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Control Systems Engineering Exam Reference Manual: A Practical Study Guide

for the NCEES Professional Engineering (PE) Licensing Examination

Bryon Lewis, CSE, P.E.

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Table of Contents

PREFACE 1

ABOUT THE AUTHOR 1

GENERAL INFORMATION 2

STATE LICENSING REQUIREMENTS 2

Eligibility 2

Examination Schedule 2

Application Procedures and Deadlines 3

DESCRIPTION OF EXAMINATION 3

Exam Format 3

Exam Content 3

Exam Scoring 7

REFERENCE MATERIALS FOR THE EXAM 7

OVERVIEW OF RECOMMENDED BOOKS 7

Recommended Books and Materials for Testing 8

Recommended Books and Courses for Additional Study 9

REVIEW OF PROCESS CONTROL SUBJECTS 10

OVERVIEW OF PROCESS MEASUREMENT AND CALIBRATION 10

Process Signal and Calibration Terminology 10

Level and Pressure Measurement 13

Density Measurement 18

Flow Measurement 19

Flowmeter Applications Chart 21

Temperature Measurement 22

Weight Measurement 25

OVERVIEW OF PROCESS CONTROL 25

Degrees of Freedom 25

Control Loops 27

Controller and Control Modes 28

Controller Tuning 33

Block Diagram Algebra 36

Nyquest Stability Criterion 38

Routh Stability Criterion 39

Laplace Transform 41

SIZING PROCESS CONTROL ELEMENTS 43
SIZING ELEMENTS AND FINAL DEVICES. 43
FLOW MEASUREMENT. 43
Fluids (and other useful equations) 43
Orifice Type Meters 44
Orifice Sizing Factors. 49
Turbine Meter 49
Control Valve Sizing 51
Control Valve Application Comparison Chart 53
Control Valve for Liquid 53
Control Valve for Gas 55
Control Valve for Steam 59
Pressure Relief Valve Sizing 62
Excerpts from ASME Unfired Pressure Vessel Code 64
Rupture Disk Sizing. 65

OVERVIEW OF DISCRETE CONTROL SUBJECTS 68
OVERVIEW OF DIGITAL LOGIC 68
Gates and Inverters 68
ISA Binary Logic 68
Relay Ladder Logic 69
Sealing Circuits. 70

ANALOG SIGNALS AND ISA SYMBOLS 71
OVERVIEW OF ANALOG SIGNALS 71
ISA P&ID (Piping and Instrumentation Diagram) 72
ISA Standard Loop Diagram 73

OVERVIEW – SAFETY INSTRUMENTED SYSTEMS 74
OVERVIEW OF PROCESS SAFTEY AND SHUTDOWN 74
SIS (Safety Instrumented Systems) 74
SIF (Safety Instrumented Function) 75
SIL (Safety Integrity Level) 76
Example SIL Evaluation 77

OVERVIEW OF INDUSTRIAL CONTROL NETWORKS 80
OVERVIEW OF NETWORKS AND COMMUNICATIONS 80
Fieldbus Networks. 80
Ethernet Networks and Communications 81
Intelligent and Smart Devices 82

OVERVIEW OF NEC and NFPA CODES 83
LIST OF NFPA CODE 83
NFPA 70 NEC - National Electric Code 83
Voltage Drop Formulas 84
Cable Sizing Formulas 84
Voltage Drop Sizing Examples 85
Comparison of NEMA Enclosures 85
Zener Diode Barrier 91
NFPA 77 Static Electricity 92
NFPA 78 Lightning Protection 92
NFPA 79 Industrial Machinery 93
NFPA 496 Purged and Pressurized Systems 94

THE FISHER CONTROL VALVE HANDBOOK 94
GUIDE TO USING THE FISHER CONTROL VALVE HANDBOOK 94
Important Sections and Pages in the FCVH 95

APPENDIX 96
Table A1. Specific Gravity for Some Common Fluids 96
Table A2. Specific Gravity and Gas Constants for Some Common Gases 99
Table A3. The Kinematic Viscosity for Some Common Fluids 100
Table A4. The Absolute Viscosity for Some Common Liquids 107
Table A5. The Absolute Viscosity for Some Common Gases 109
Table A6. Thermocouple Table (Type J) 110
Table A7. Thermocouple Table (Type K) 112
Table A8. Thermocouple Table (Type E) 115
Table A9. Thermocouple Table (Type T) 117
Table A10. Platinum 100 Ohm RTD DIN Curve Table 118
ISA SYMBOLS 119
ISA Identification Letters 119
Typical Letter Combinations 120
General Instrument or Function Symbol 121
Signal Lines 122
Worked Examples 123
Selection and Sizing of Relief Valves 123
Table A11. Typical Properties of Gases 130

REFERENCES 134

Preface

Most state licensing boards in the United States recognize the Control System Engineering (CSE) and offer the NCEES exam in this branch of engineering. There are, however, four states that do not offer the CSE exam—Alaska, Hawaii, New York, and Rhode Island. If you live in one of these states, you may choose to pursue licensing in another discipline (such as electrical, mechanical, or chemical engineering). Or you can try to arrange to take the CSE exam in a neighboring state.

The Control Systems Engineering (CSE) exam covers a broad range of subjects, from the electrical, mechanical and chemical engineering disciplines. This exam is not on systems theory, but on process control and basic control systems. Experience in engineering or designing process control systems is almost a necessity to pass this exam.

Study of this reference manual should adequately prepare the experienced engineer or designer to take the CSE exam. However, passing the exam depends on an individual applicant's demonstrated ability and cannot be guaranteed.

I have included a list of recommended books and material. The recommended books contain information, invaluable to passing the exam. Even if you could take as many books as you want into the exam site, it is better not to overwhelm yourself—too much information can become distracting. Remember you will be under pressure to beat the clock. Study your reference books and tab the tables and information you need. This will ensure you do not waste time.

The *Fisher Control Valve Handbook* is strongly recommended to obtain the full benefits of this study review guide. The pages in the second and third editions of the handbook are referenced in numerous worked examples. The *Fisher Control Valve Handbook* can be obtained free or for minimal cost from your local Fisher Valve representative. The book is also available from Brown's Technical Book Shop, **1517 San Jacinto, Houston, Texas, 77002.**

About The Author

Bryon Lewis is a Professional Engineer, licensed in Control Systems Engineering. He is also a Senior Member of ISA, an SME Certified Manufacturing Engineer, and a licensed Master Electrician. Mr. Lewis has over 20 years' experience in electrical, mechanical, instrumentation, and control systems.

General Information

STATE LICENSING REQUIREMENTS

Eligibility

Licensing of engineers is intended to protect the public health, safety, and welfare. State licensing boards have established requirements to be met by applicants for licenses which will, in their judgment, achieve this objective.

Licensing requirements vary somewhat from state to state but have some common features. In all states, candidates with a 4-year engineering degree from an ABET/EAC-accredited program and four years of acceptable experience can be licensed if they pass the Fundamentals of Engineering (FE) exam and the Principles and Practice of Engineering (PE) exam in a specific discipline. References must be supplied to document the duration and nature of the applicant's work experience.

Some state licensing boards will accept candidates with engineering technology degrees, related-science (such as physics or chemistry) degrees, or no degree, with compensating increases in the amount of work experience. Some states allow waivers of one or both of the exams for applicants with many years (6–20) of experience. Additional procedures are available for special cases, such as applicants with degrees or licenses from other countries.

Note: Recipients of waivers may encounter difficulty in becoming licensed by "reciprocity" or "comity" in another state where waivers are not available. Therefore, applicants are advised to complete an ABET accredited degree and to take and pass the FE/EIT exam. Some states require a minimum of four year experiences after passing the FE/EIT exam, before allowing one to sit for the PE (principals and practices) exam. Some states will not allow experience incurred before the passing of the FE/EIT exam.

It is necessary to contact your licensing board for the up-to-date requirements of your state. Phone numbers and addresses can be obtained by calling the information operator in your state capital, or by checking the Internet at www.ncees.org or nspe.org.

Examination Schedule

The CSE exam is offered once per year, on the last weekend in October, (typically on Friday). Application deadlines vary from state to state, but typically are about three or four months ahead of the exam date.

Application Procedures and Deadlines

Requirements and fees vary among state jurisdictions. Sufficient time must be allotted to complete the application process and assemble required data. PE references may take a month or more to be returned. The state board needs time to verify professional work history, references, and academic transcripts or other verifications of the applicant's engineering education.

After accepting an applicant to take one of the exams, the state licensing board will notify him or her where and when to appear for the exam. They will also describe any unique state requirements such as allowed calculator models or limits on the number of reference books taken into the exam site.

DESCRIPTION OF EXAMINATION

Exam Format

The NCEES Principles-and-Practice of Engineering examination (commonly called the PE examination) in Control Systems Engineering (CSE) is an eight-hour examination. The examination is administered in a four hour morning session and a four hour afternoon session.

Each session contains forty (40) questions in a multiple-choice format.

Each question has a correct or "best" answer. Questions are independent, so an answer to one question has no bearing on the following questions.

All of the questions are compulsory; applicants should try to answer all of the questions. Each correct answer receives one point. If a question is omitted or the answer is incorrect, a score of zero will be given for that question. There is no penalty for guessing.

Exam Content

The subject areas of the CSE exam are described by the exam specification and are given in six areas. ISA supports Control Systems Engineer (CSE) licensing and the examination for Professional Engineering. ISA is responsible for the content and questions in the NCEES examination. Refer to the ISA web site (http://www.isa.org) for the latest information concerning the CSE examination.

The following details what to expect on the examination and breaks down the examination into the six parts. The percentage and number of questions are given for each part of the examination at the time this guide was written.

I. MEASUREMENT

24% of Examination 19 Questions

1. Sensor technologies applicable to the desired type of measurement (e.g., flow, pressure, level, temperature, analytical, counters, motion, vision, etc.)
2. Sensor characteristics (e.g., rangeability, accuracy and precision, temperature effects, response times, reliability, repeatability, etc.)
3. Material compatibility
4. Calculations involved in: pressure drop
5. Calculations involved in: flow element sizing
6. Calculations involved in: level, differential pressure
7. Calculations involved in: unit conversions
8. Calculations involved in: velocity
9. Calculations involved in: linearization
10. Installation details (e.g., process, pneumatic, electrical, etc.

II. SIGNAL AND TRANSMISSION

12.5% of Examination 10 Questions

A. Signals - 11.5%, 9 questions

1. Pneumatic, electronic, optical, hydraulic, digital, analog
2. Transducers (e.g., analog/digital [A/D], digital/analog [D/A], current/pneumatic [I/P] conversion, etc.)
3. Intrinsically Safe (IS) barriers
4. Grounding, shielding, segregation, AC coupling
5. Basic signal circuit design (e.g., two-wire, four-wire, isolated outputs, loop powering, etc.)
6. Calculations: circuit (voltage, current, impedance)
7. Calculations: unit conversions

B. Transmission - 1.25%, 1 question

1. Different communications systems architecture and protocols (e.g., fiber optics, coaxial cable, wireless, paired conductors, fieldbus, Transmission Control Protocol/Internet Protocol [TCP/IP], OLE Process Control [OPC])
2. Distance considerations versus transmission medium

III. FINAL CONTROL ELEMENTS

20% of Examination 16 Questions

A. Valves - 12.5%, 10 questions

1. Types (e.g., globe, ball, butterfly, etc.)

2. Characteristics (e.g., linear, low noise, equal percentage, shutoff class, etc.)
3. Calculation (e.g., sizing, split range, noise, actuator, speed, pressure drop, air/gas consumption, etc.)
4. Applications of fluid dynamics (e.g., cavitation, flashing, choked flow, Joule-Thompson effects, two-phase, etc.)
5. Material selection based on process characteristics (e.g., erosion, corrosion, plugged, extreme pressure, temperature, etc.
6. Accessories (e.g., limit switches, solenoid valves, positioners, transducers, air regulators, etc.)
7. Environmental constraints (e.g., fugitive emissions, packing, special sealing, etc.)
8. Installation practices (e.g., vertical, horizontal, bypasses, troubleshooting, etc.)

B. Pressure Relieving Devices - 5%, 4 questions

1. Pressure Relieving Valves: Types (e.g., conventional spring, balanced bellows, pilot operated, etc.)
2. Pressure Relieving Valves: Characteristics (e.g., modulating, pop action, etc.)
3. Pressure Relieving Valves: Calculations (e.g., sizing considering inlet pressure drop, back pressure, multiple valves, etc.)
4. Pressure Relieving Devices: Material selection based on process characteristics
5. Pressure Relieving Valves: Installation practices (e.g., linking valves, sparing the valves, accessibility for testing, car sealing inlet valves, piping installation, etc.)
6. Rupture discs (types, characteristics, application, calculations, etc.)

C. Other Final Control Elements - 2.5%, 2 questions

1. Motor controls
2. Solenoid valves
3. On-off devices/relays
4. Self-regulating devices

IV. CONTROL SYSTEMS ANALYSIS

16% of Examination 13 Questions

A. Documentation - 7.5%, 6 questions

1. Drawings (e.g., PFD, P&ID, Loop Diagrams, Ladder Diagrams, Logic Drawings, Cause and Effects Drawings, SAFE Charts, etc.)

B. Theory - 6%, 5 questions

1. Basic processes (e.g., compression, combustion, distillation, hydraulics, etc.)
2. Process dynamics (e.g., loop response, P-V-T relationships, simulations, etc.)

3. Basic control (e.g., regulatory control, feedback, feed forward, cascade, ratio, PID, split-range, etc.)
4. Discrete control (e.g., relay logic, Boolean algebra)
5. Sequential control (e.g., batch)

C. Safety - 2.5%, 2 questions

1. Safety system design (e.g., Safety Instrumented System [SIS], Safety Requirements Specification [SRS], application of OSHA 1910, etc.)

V. CONTROL SYSTEMS IMPLEMENTATION

16% of Examination 13 Questions

1. HMI (e.g., graphics, alarm management, trending, historical data)
2. Ergonomics (e.g., human factors engineering, physical control room arrangement, panel layout)
3. Configuration and programming (e.g., PLC, DCS, Hybrid systems, SQL, Ladder logic, sequential function chart, structured text, function block programming, data base management, specialized controllers, etc.)
4. System comparisons and compatibilities (e.g., advantages and disadvantages of system architecture)
5. Installation requirements (e.g., shielding, constructability, input/output termination, environmental, heat load calculations, power load requirements, purging, lighting, etc.)
6. Commissioning (e.g., performance tuning, loop checkout, etc.)
7. Safety Instrumented System [SIS] model validation calculations (e.g., Safety Integrity Level [SIL], reliability, availability, etc.)
8. Troubleshooting (e.g., root cause failure analysis and correction)

VI. CODES, STANDARDS, REGULATIONS

7.5% of Examination 6 Questions

1. Working knowledge of applicable Codes, Standards, and Regulations: American National Standards Institute (ANSI)
2. Working knowledge of applicable Codes, Standards, and Regulations: Institute of Electrical & Electronics Engineers (IEEE)
3. Working knowledge of applicable Codes, Standards, and Regulations: ISA
4. Working knowledge of applicable Codes, Standards, and Regulations: National Electrical Code (NEC)
5. Working knowledge of applicable Codes, Standards, and Regulations: National Electrical Manufacturers Association (NEMA)

6. Working knowledge of applicable Codes, Standards, and Regulations: National Fire Protection Association (NFPA)
7. Working knowledge of applicable Codes, Standards, and Regulations: Occupational Safety and Health Administration (OSHA)

Exam Scoring

NCEES exams are scored independently. There are no pre-specified percentages of candidates that must pass or fail.

Assisted by a testing consultant, a panel of licensed CSEs uses recognized psychometric procedures to determine a passing score corresponding to the knowledge level needed for minimally-competent practice in the discipline.

The passing score is expressed as the number of questions out of 80 that must be answered correctly. The method used for pass-point determination assures that the passing score is adjusted for variations in the level of exam difficulty and that the standard is consistent from year to year.

Starting in October 2005, candidates have received results expressed either as "Pass" or "Fail"; failing candidates no longer receive a numerical score. Published passing rates are based on first-time takers only, omitting the results for repeat takers.

Reference Materials for the Exam

Overview of Recommended Books

The list of recommended books and materials for testing will be necessary to help you pass the CSE examination. Use a book you are comfortable with. A substitution with the same material and information may be used.

The list of recommended books and materials for additional study can be helpful in the review of subjects and preparation for the examination.

One of the books, Fisher's *Control Valve Handbook*, is necessary to work many of the examples in this book. The information and tables in the *Control Valve Handbook* will be constantly referenced. See the Preface for information in obtaining the *Control Valve Handbook*. The book

may also be downloaded in PDF format from Fisher: http://www.documentation.emersonprocess.com/groups/public/documents/book/cvh99.pdf

Remember to keep the review simple. The test is not on control systems theory studies, but rather on simple general functional design. Again keep your studies simple; control systems theory will only encompass about 3% of the examination.

Recommended Books and Materials for Testing

- NCEES APPROVED CALCULATOR (Have a spare with new batteries installed). I recommend the TI-36X Solar (any light). Practice with the calculator you will be using. (See http://www.ncees.org for a current list of approved calculators.)

- CONTROL VALVE HANDBOOK (3rd Ed.), Fisher Controls, Marshalltown, IA, 1989.

- Norman A. Anderson, INSTRUMENTATION FOR PROCESS MEASUREMENT AND CONTROL (3rd Ed.), CRC Press LLC, Boca Raton, FL, 1997. [Measurement; instrument calibration; orifice sizing; valve sizing; process characteristics; charts; thermocouple tables; RTD tables; general flow and pipe data tables; nomographs; formulas; typical installation details; typical calculations.]

- NFPA No. 70 - NATIONAL ELECTRICAL CODE [Hazardous location classification; group classifications and autoignition temperatures of gases; hazardous installation codes; intrinsically safe systems installation]

- ISA-5.1-1984 (R1992) - INSTRUMENTATION SYMBOLS AND IDENTIFICATION

- ISA-5.2-1976 (R1992) - BINARY LOGIC DIAGRAMS FOR PROCESS OPERATIONS

- ISA-5.3-1983 - GRAPHIC SYMBOLS FOR DISTRIBUTED CONTROL/ SHARED DISPLAY INSTRUMENTATION, LOGIC, AND COMPUTER SYSTEMS

- ISA-5.4-1991 - STANDARD INSTRUMENT LOOP DIAGRAMS

Recommended Books and Courses for Additional Study

- B. G. Lipták, INSTRUMENT ENGINEERS' HANDBOOK - PROCESS MEASUREMENT, 3rd Ed., ISA, 2002 [Instrument symbols, performance, and terminology; measurement of flow, level, temperature, pressure and density; safety, weight and miscellaneous sensors; analytical instrumentation]

- B. G. Lipták, INSTRUMENT ENGINEERS' HANDBOOK - PROCESS CONTROL, 3rd Ed., ISA, 2002 [Control theory; controller, transmitters, converters and relays; control centers, panels and displays; control valves, on-off and throttling; regulators; process control systems]

- Robert N. Bateson, INTRODUCTION TO CONTROL SYSTEM TECHNOLOGY (6th Ed.), Prentice-Hall, Upper Saddle River, NJ, 1999. [Block diagram algebra; servomechanisms; electrical, mechanical, thermal and gas flow elements; bode plots; laplace transforms; digital signal conditioning; ac and dc motors; control valves; discrete process control and PLCs; modes of control; process characteristics; analysis and design.]

- H. D. Baumann, CONTROL VALVE PRIMER (3rd Ed.), ISA, 1998. [Control valves and control loops; selection and sizing; fail safety; flow characteristics; positioners; actuators; stem forces; installation; materials; environmental concerns; electric vs. pneumatic actuators]

- Bernard Grob, Grob: Basic Electronics (any Ed.), McGraw-Hill Science. [AC-DC circuit theory; network theorems; network analysis]

- Bissell C.C., Control Engineering (2nd Ed.), Chapman and Hall. [A simple easy to follow book on control systems engineering. Approximately 200 pages and less than $25.00. A very practical book.]

- ISA offers a 3-1/2 day instructor led Control Systems Engineer (CSE) PE exam review course at different locations across the nation. The cost of the course is approximately $1,299.

- ISA offers an Automation and Control Curriculum - 44 Courses. The cost for all 44 courses is approximately $750.

Review of Process Control Subjects

Overview of Process Measurement and Calibration

The process control industry covers a wide variety of applications: petrochemical; pharmaceutical; pulp and paper; food processing; material handling; even commercial applications.

Process control in a plant can include discrete logic, such as relay logic or a PLC; analog control, such as single loop control or a DCS (distributed control system); pneumatic; hydraulic and electrical systems as well. The Control Systems Engineer must be versatile and have a broad range of understanding of the engineering sciences.

The Control Systems Engineer (CSE) examination encompasses a broad range of subject to ensure minimum competency. This book will review the foundations of process control and demonstrate the breadth and width of the CSE examination.

Process Signal and Calibration Terminology

The most important terms in process measurement and calibration are range, span, zero, accuracy and repeatability. Let us start by defining Span; Range; Lower Range Value (LRV); Upper Range Value (URV); Zero; Elevated Zero; Suppressed Zero.

range: The region in which a quantity can be measured, received, or transmitted, by an element, controller or final control device. The range can usually be adjusted and is expressed by stating the *lower* and *upper range-values.*

NOTE 1: For example:

	Full Range	Adjusted Range	LRV	URV
a)	0 to 150°F	None	0°F	150°F
b)	–20 to +200°F	–10 to +180°F	–10°F	+180°F
c)	20 to 150°C	50 to 100°C	50°C	100°C

NOTE 2: Unless otherwise modified, input *range* is implied.

NOTE 3: The following compound terms are used with suitable modifications in the units: *measured variable range, measured signal range,* indicating scale range, chart scale range, etc. See Tables 1 and 2.

NOTE 4: For multi-range *device*s, this definition applies to the particular range that the *device* is set to measure.

Table 1 — Illustrations of the use of range and span terminology

TYPICAL RANGES		NAME	RANGE	LOWER RANGE-VALUE	UPPER RANGE-VALUE	SPAN	SUPPLEMENTARY DATA
0	+100	—	0 to 100	0	+100	100	—
20	+100	SUPPRESSED ZERO RANGE	20 to 100	20	+100	80	SUPPRESSION RATIO = 0.25
-25	+100	ELEVATED ZERO RANGE	–25 to +100	–25	+100	125	—
–100	0	ELEVATED ZERO RANGE	–100 to 0	-100	0	100	—
–100	–20	ELEVATED ZERO RANGE	–100 to –20	-100	-20	80	—

range-limit, lower: The lowest value of the *measured variable* that a *device* is adjusted to measure.

NOTE: The following compound terms are used with suitable modifications to the units: *measured variable* lower range-limit, *measured signal* lower range-limit, etc. See *Tables 1 and 2.*

range-limit, upper: The highest value of the *measured variable* that a *device* is adjusted to measure.

NOTE: The following compound terms are used with suitable modifications to the units: *measured variable* upper range-limit, *measured signal* upper range-limit, etc. See *Tables 1 and 2.*

span: The algebraic difference between the *upper* and *lower range-values.*

NOTE 1: For example:

Range 0 to 150°F, *Span* 150°F

Range –10 to 180°F, *Span* 190°F

Range 50 to 100°C, *Span* 50°C

NOTE 2: The following compound terms are used with suitable modifications to the units: *measured variable range, measured signal range,* etc.

NOTE 3: For multi-range *device*s, this definition applies to the particular *range* that the *device* is set to measure. See *Tables 1 and 2.*

live-zero: The lower range value (LRV) is said to be set to zero, as a reference point, whether it is at zero or not. This LRV can be 0%; -40°F; 4mA; 1V; 3 PSI. All LRVs are an example of the *ZERO* (Live Zero), in process control signals or elements.

elevated-zero: The *lower range-value of the range* is below the value of zero. The LRV of the range must be raised to *Live Zero*, for the instrument to function properly. The output signal of the measured value will always be 0 to 100%. If the LRV of the range is too low, the instrument may not be able to reach 100% output.

NOTE 1: For example: input signal = (-100 in H2O to 25 in H2O)
output signal = (4mA to 20mA)

Table 2 —Illustrations of the use of the terms measured variable, measured signal, range and span

TYPICAL RANGES	TYPE OF RANGE	RANGE	LOWER RANGE-VALUE	UPPER RANGE-VALUE	SPAN
(1) THERMOCOUPLE 0 2000°F TYPE K T/C	MEASURED VARIABLE	0 to 2000°F	0°F	2000°F	2000°F
–0.68 +44.91 mV	MEASURED SIGNAL	–0.68 to +44.91 mV	–0.68 mV	+44.91 mV	45.59 mV
0 20 x100=°F	SCALE AND/OR CHART	0 to 2000°F	0°F	2000°F	2000°F
(2) FLOWMETER 0 10 000 lb/h	MEASURED VARIABLE	0 to 10 000 lb/h	0 lb/h	10,000 lb/h	10,000 lb/h
0 100 in H_2O	MEASURED SIGNAL	0 to 100 in H_2O	0 in H_2O	100 in H_2O	100 in H_2O
0 10 x1000=lb/h	SCALE AND/OR CHART	0 to 10,000 lb/h	0 lb/h	10,000 lb/h	10,000 lb/h
4 20 mA	MEASURED SIGNAL	4 to 20 mA	4 mA	20 mA	16 mA
1 5 Volts	MEASURED SIGNAL	1 to 5V	1V	5V	4V

The output signal may only reach 12mA for 25 in H_2O (100%) input, due to limitation in the electronics or pneumatics. Therefore the *Elevate* jumper must be set in the transmitter or an elevation kit must be installed in a pneumatic transmitter. See *Table 1.*

suppressed-zero: The *lower range-value of the span* is above the value of zero. The LRV of the range must be lowered to *Live Zero*, for the instrument to function properly. The output signal of the measured value will always be 0 to 100%. If the LRV of the range is too high, the instrument may not be able to reach 0% output.

NOTE 1: For example: input signal = (50 in H_2O to 200 in H_2O)
output signal = (4mA to 20mA)

The output signal may only reach 6mA for 50 in H_2O (0%) input, due to limitation in the electronics or pneumatics. Therefore the *Suppress* jumper must be set in the transmitter or a suppression kit must be installed in a pneumatic transmitter. See *Tab1e 1*.

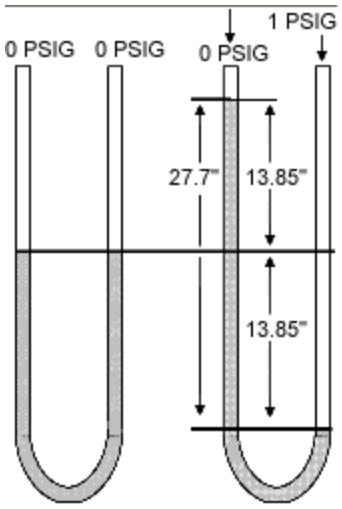
Level and Pressure Measurement

The level in a vessel or tank can be measured by a number of methods: differential pressure; displacement of volume; bubbler tube; capacitance; sonar; radar; weight, to name a few. This book will focus on differential pressure, displacement of volume, and bubbler tube for the examination.

Head pressure measurement
Head pressure is independent of the tank's height or area. The transmitter measures head pressure. Head pressure is the measure of the potential energy in the system. The transmitter measurement is from how high is the fluid falling. The distance the fluid falls dictates the force generated (F=ma). This is why the density of the fluid must be known to calibrate a pressure transmitter for a process. The calibration process uses specific gravity (S.G.), the ratio of a known density of a fluid divided by the density of water (H_2O).

To illustrate these facts we will start with one gallon of water. The gallon of water equals 231 cubic inches and weighs approximately 8.324 pounds. Pressure is measured in PSI (pounds per square inch). Only one (1) square inch of area is needed to calculate the height of the water and the force it is excerpting. Remember force divided by area = pressure.

Stack 231 cubic inches of water on top of each other, to form a tall column of water, with a base of one (1) square inch. The column of water will be 231 inches tall. Divide the height of the column of water, 231 inches, by the weight of one (1) gallon of water, 8.324 pounds. The result will be 27.691 or 27.7 inches of water per pound of water over a one square inch area. Therefore 27.7 inches H_2O, of head pressure, equals one (1) PSI.



By knowing the specific gravity of the fluid to be measured, multiplied by the height of the tank in inches, an equivalent value in inches of water can be made. The transmitter can now be calibrated in inches of water, regardless of the fluid. If the tank's fluid has a S.G. equal to 0.8 and is 100 inches tall, then the height in H_2O will be (100″ x 0.8 = 80″).

Calibration procedure

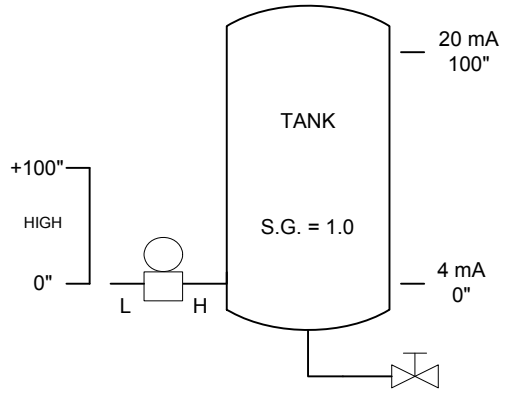
Differential pressure or differential head pressure is used to calibrate transmitters for pressure, level, flow and density. The transmitter has a high side, marked with an H, and a low side, marked with a L. The low side will typically go to atmospheric pressure or to the fixed height wet leg measurement. The high side will typically go to the tank, where the varying height of fluid is to be measured. When calibrating an instrument remember: The low side is the negative scale, below zero, and the high side is the positive scale, above zero. The transmitter's sensor element is static in position or elevation and therefore the transmitter itself is always equal to zero elevation.

The formula for calibration is:
(high side inches x S.G.) – (low side inches x S.G.) = lower or upper range value.
Note: lower range value when empty and upper range value when full.

The calibration procedure below is as follows.
See Example 1. The low side is open to atmosphere. The atmosphere adds zero inches of water to the low side. The high side is connected to the tank. The first line of math will be the LRV. The second line of math will be the URV. The tank has 100 inches of fluid with a S.G. of 1.0. The calibrated *Range* of the instrument will be 0″ to 100″ of water or H_2O. The *Span* of the transmitter is (100″ x 1.0 = 100″).

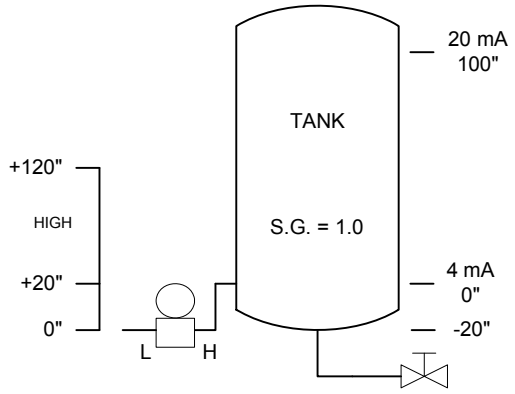
See Example 2. The low side is open to atmosphere. The atmosphere adds zero inches of water to the low side. The high side is connected to the tank. The first line of math will be the LRV. The second line of math will be the URV. The tank has 100 inches + the tube adds 20″ of fluid with a S.G. of 1.0. The calibrated *Range* of the instrument will be 20″ to 120″ of water or H_2O. Remember the minimum measurement can not be lower than 20″, the fixed tube height. Suppress the zero and make 20″ a live zero to the instrument.

Example 1: Open Tank
Zero-Based Level Application



Tank Level = 0 to 100 inches
S.G. = 1.0
(switch jumper to normal zero)
(0″ x 1.0) – (0″ x 1.0) = 0″ = 4 mA
(100″ x 1.0) – (0″ x 1.0) = 100″ = 20 mA
Calibrate range from 0″ to 100″ H_2O

Example 2: Open Tank
Suppress the Zero

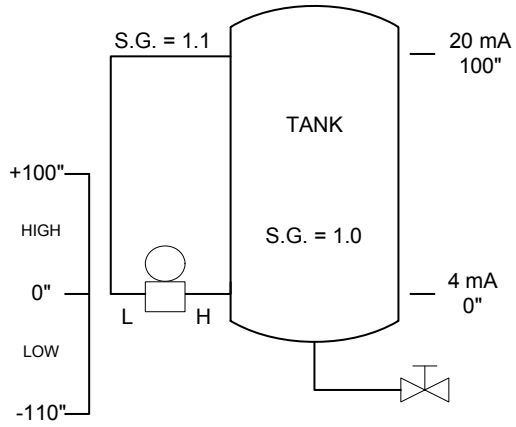


Tank Level = 0 to 100 inches
S.G. = 1.0
(switch jumper to suppress zero)
(20″ x 1.0) – (0″ x 1.0) = 20″ = 4 mA
(120″ x 1.0) – (0″ x 1.0) = 120″ = 20 mA
Calibrate range from 20″ to 120″ H_2O

See Example 3. The low side is connected to the top of the closed tank. The high side is connected to the bottom of the closed tank. The tank's pressure does not matter, because the low and high line cancels each other out. The wet leg has 100 inches of fluid with a S.G. of 1.1. The first line of math will be the LRV. The second line of math will be the URV. The tank has 100 inches of fluid with a S.G. of 1.0. The calibrated *Range* of the instrument will be -110″ to -10″ of water or H_2O. Elevate the zero and make -10″ a live zero to the instrument. The *Span* of the transmitter is (100″ x 1.0 = 100″).

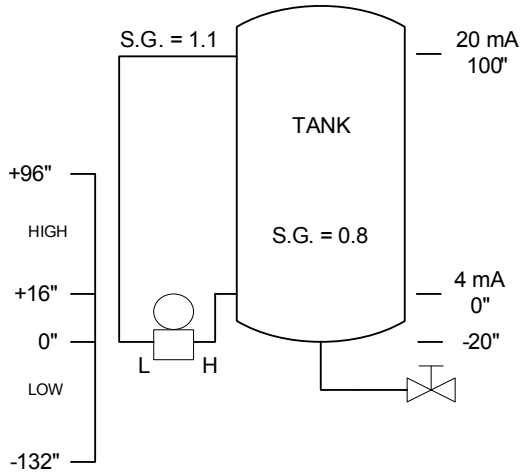
See Example 4. The low side is connected to the top of the closed tank. The high side is connected to the bottom of the closed tank. The tank's pressure does not matter, because the low and high line cancels each other out. The wet leg has 120 inches of fluid with a S.G. of 1.1. The first line of math will be the LRV. The second line of math will be the URV. The tank has 100 inches + the tube adds 20″ of fluid with a S.G. of 0.8. The calibrated *Range* of the instrument will be -116″ to -36″ of water or H_2O. Remember the minimum measurement can not be lower than 20″ on the high side, the fixed height tube. Elevate the zero and make -116″ a live zero. The *Span* of the transmitter is (100″ x 0.8 = 80″).

Example 3: Closed Tank
Elevate the Zero



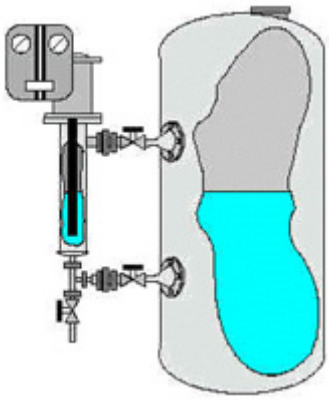
Tank Level = 0 to 100 inches
S.G. = 1.0 Wet Leg S.G. = 1.1, Height = 100″
(switch jumper to elevate zero)
(0″ x 1.0) – (100″ x 1.1) = -110″ = 4 mA
(100″ x 1.0) – (100″ x 1.1) = -10″ = 20 mA
Calibrate range from -110″ to -10″ H_2O

Example 4: Closed Tank
Elevate the Zero (transmitter below tank)



Tank Level = 0 to 100 inches
S.G. = 0.8 Wet Leg S.G. = 1.1, Height = 120″
(switch jumper to elevate zero)
(20″ x 0.8) – (120″ x 1.1) = -116″ = 4 mA
(120″ x 0.8) – (120″ x 1.1) = -36″ = 20 mA
Calibrate range from -116″ to -36″ H_2O

Level Displacer (buoyancy)



The displacer tube for liquid level measurement is based on Archimedes principle that, the buoyancy force exerted on a sealed body immersed in a liquid is equal to the weight of the liquid displaced.

There are two types of displacer transmitters in common use today: torque tube and spring operated.

$$f = \frac{V_{df}}{231}(8.33)G_f$$

where,

f = buoyancy force in lbf

V_{df} = total volume of displaced process fluid in cubic inches

Ls = the submerged length of the displacer in process fluid

231 = cubic inches in one gallon of water

8.33 = weight of one gallon of water in pounds

G_f = specific gravity of displaced process fluid

Sample problem: A. What is the force upward on the 30" displacer, if the displacer is 4" in diameter and submerged 10" in a fluid with a specific gravity of 0.72? B. What is the mA output and percent output?

A. Find displaced volume

$$V_{df} = \left(\frac{\pi * D^2}{4}\right) * L_s = \left(\frac{\pi * 16}{4}\right) * 10 = 125.66\ in^3$$

Find displacement force upward

$$f = \frac{V_{df}}{231}(8.33)G_f = \frac{125.66}{231}(8.33)(0.72) = 3.26\ lbf$$

B. Find displacement force upward for 30 inches and then the percent output and mA

$$V_{df} = \left(\frac{\pi * D^2}{4}\right) * L_s = \left(\frac{\pi * 16}{4}\right) * 30 = 376.99\ in^3$$

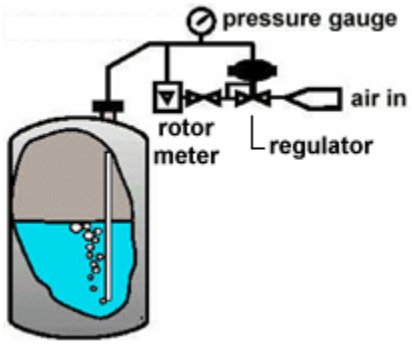
$$f = \frac{V_{df}}{231}(8.33)G_f = \frac{376.99}{231}(8.33)(0.72) = 9.79\ lbf$$

$$\% = \frac{3.26}{9.79} = 0.333 * 100 = 33.3\%\ \text{output}$$

$$(0.333 * 16mA) + 4mA = 9.328mA\ \text{output}$$

Level – Bubble tube method

The bubble tube measures the level of the process fluid by measuring the back pressure. This simple level measurement has a dip tube installed with the open end close to the bottom of the process vessel.



A flow of gas, usually air or nitrogen, passes through the tube and the resultant air pressure in the tube corresponds to the hydraulic head of the liquid in the vessel. The air pressure in the bubble tube varies proportionally with the change in head pressure.

$$h = L_{TS}G_f$$

where,

h = head pressure in inches of water

L_{TS} = length of tube submerged in process fluid

G_f = specific gravity of process fluid

Sample problem: A. What is the head pressure measurement of a bubbler tube submerged 24″ in a fluid with a specific gravity of 0.85? B. What is the mA output and percent output if the transmitter is calibrated for a tube 100″ long?
A. Find head pressure of the process fluid $h = L_{TS}G_f = 24 * 0.85 = 20.4$ inches H_2O B. Find percent and mA output $\% = \frac{24}{100} = 0.24 * 100 = 24\%$ output $(0.24 * 16mA) + 4mA = 7.84mA$ output

Density measurement

Head pressure and volume displacement can be used to measure density. By using a differential head pressure transmitter, calibrated in inches of water, with the high and low lines connected to a tank at a fixed distance of separation, such as 12″, and both taps completely submerged below the lowest fluid level, the height measured in inches of water divided by 12″ is the S.G. of the unknown fluid. If the fluid height measurement was divided into the fixed 12″ of displacement, density would be measured.

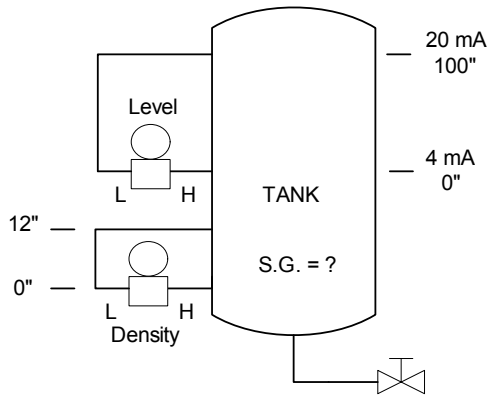


Figure 1

Note the upper level measurement can be any height and the fluid to be measured of any density.

With the specific gravity (S.G.) known from the lower density transmitter, a second upper transmitter, calibrated in inches of water for level, can be added. The level measurement can be divided by the S.G. measurement from the lower density transmitter, to show the true height of the fluid in the tank (see Figure 1).

Flow measurement

Like level measurement, flow measurement is also head pressure and zero elevation based. Head pressure is the measure of the endowed potential energy in the system. The transmitter measurement is from how high the fluid falls to its velocity squared. The velocity is squared, due to the fact that the fluid is constantly being accelerated through the pipe, as potential energy is endowed into the fluid by the pump's head pressure.

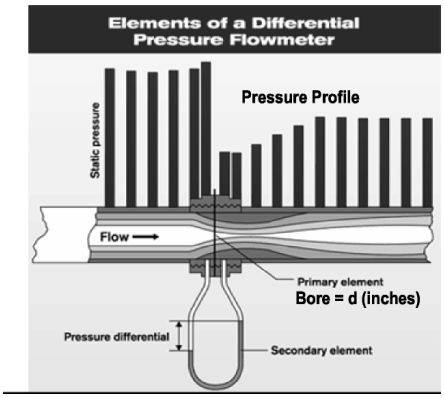
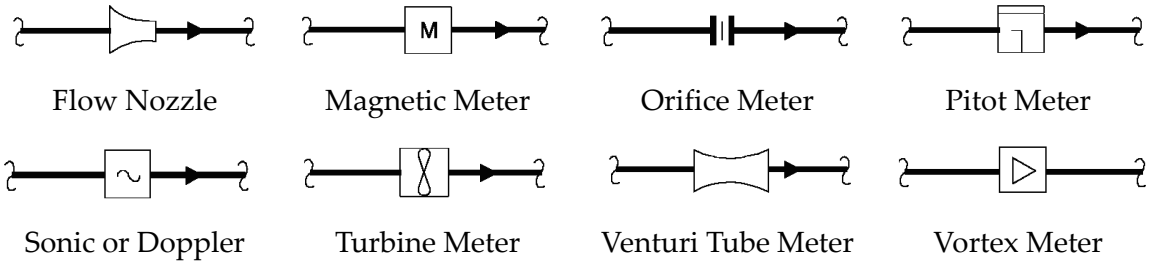


Figure 2

Head pressure is lost across the orifice element due to the fact that, energy loss is the product of energy flow multiplied by the resistance thought which it flows (see Figure 2).

Sizing of the orifice will be discussed in detail in the section on Sizing Process Control Elements and Final Devices. You should familiarize yourself with the different types of flowmeters, their applications, and their ISA symbols.

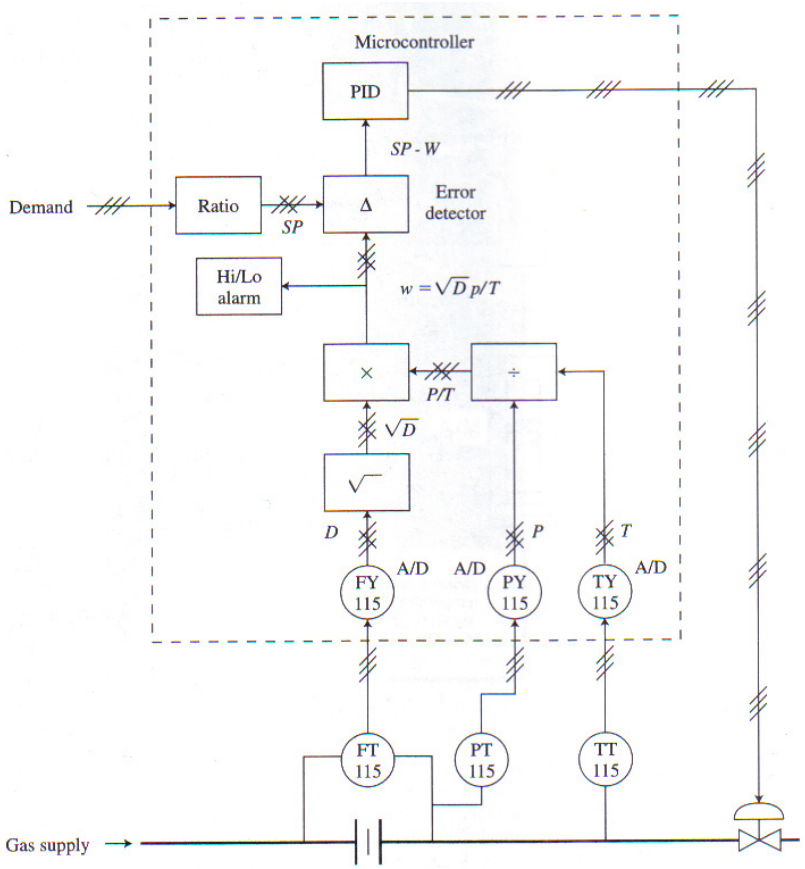
ISA Meter Symbols



Flow Nozzle | Magnetic Meter | Orifice Meter | Pitot Meter

Sonic or Doppler | Turbine Meter | Venturi Tube Meter | Vortex Meter

Mass Flow Metering



From Bulletin C-404A, Courtesy of the Foxboro Company

Mass flow of gas:

$$w = \frac{m}{t} = \frac{M}{10^3 R}\left[\frac{V}{t}\right]\left[\frac{p}{T}\right]$$

Substituting Q for V/t:

$$w = \frac{MQ}{10^3 R}\left[\frac{p}{T}\right]$$

Substituting for Q:

$$Q = k\sqrt{D};\ \ k = \frac{Mk_f}{10^3 R}$$

Finally the simplified mass flow equation:

$$w = k\sqrt{D}\left[\frac{p}{T}\right]$$

where,

w = mass flow rate, kilogram/second

Q = volume flow rate, cubic meters per second

p = absolute pressure, Pascal's

T = absolute temperature, Kelvin

M = gram molecular weight of gas (g/mol)

R = universal gas constant = 8.314 J/K*mol

D = flowmeter differential pressure in Pascals

k = mass flow proportionality constant

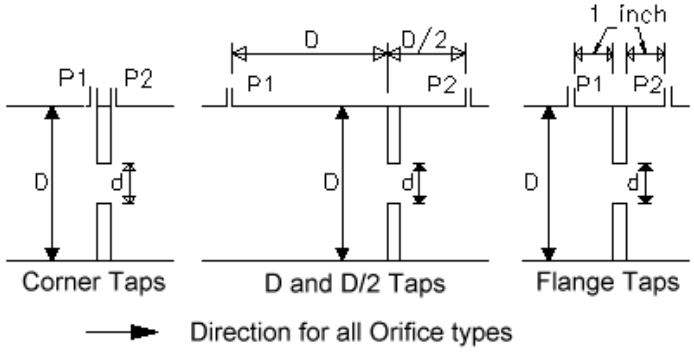
k_f = flowmeter proportionality constant

V = volume of gas

Table 3 - Flowmeter applications chart

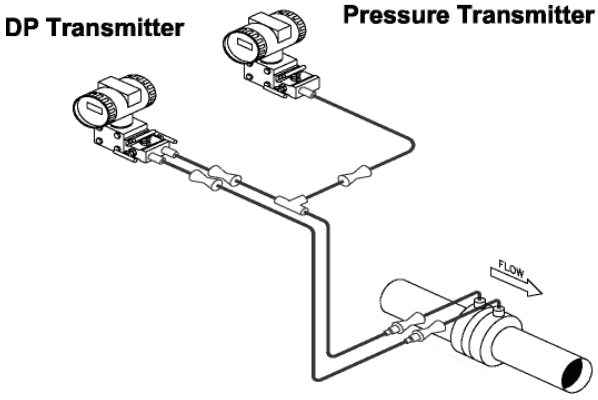
Sensor	Rangeability	Accuracy	Advantages	Disadvantages
orifice	3.5:1	2-4% of full span	-low cost -extensive industrial practice	-high pressure loss -plugging with slurries
venturi	3.5:1	1% of full span	-lower pressure loss than orifice -slurries do not plug	-high cost -line under 15 cm
flow nozzle	3.5:1	2% full span	-good for slurry service -intermediate pressure loss	-higher cost than orifice plate -limited pipe sizes
elbow meter	3:1	5-10% of full span	-low pressure loss	-very poor accuracy
annubar	3:1	0.5-1.5% of full span	-low pressure loss -large pipe diameters	-poor performance with dirty or sticky fluids
turbine	20:1	0.25% of measurement	-wide rangeability -good accuracy	-high cost -strainer needed, especially for slurries
vortex shedding	10:1	1% of measurement	-wide rangeability -insensitive to variations in density, temperature, pressure, and viscosity	-expensive
positive displacement	10:1 or greater	0.5% of measurement	-high rangeability -good accuracy	-high pressure drop -damaged by flow surge or solids
Coriolis mass flow	100:1	0.05-0.15% of measurement	-good accuracy	-expensive

Orifice tap dimensions for head type meters

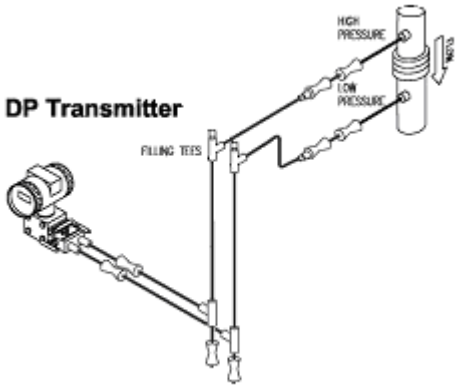


Meter Connection Orientation

Gas or Air Installation

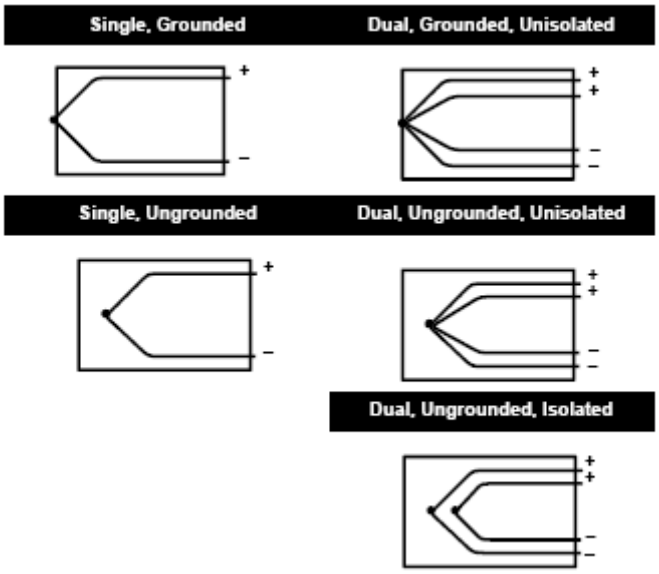


Steam or Liquid Installation



Temperature measurement

In the process industry, temperature measurements are typically made with thermocouples, RTDs (Resistance Temperature Detector) and industrial thermometers. Industrial thermometers are typically of the liquid (class I), vapor (class II), and gas (class III) type.

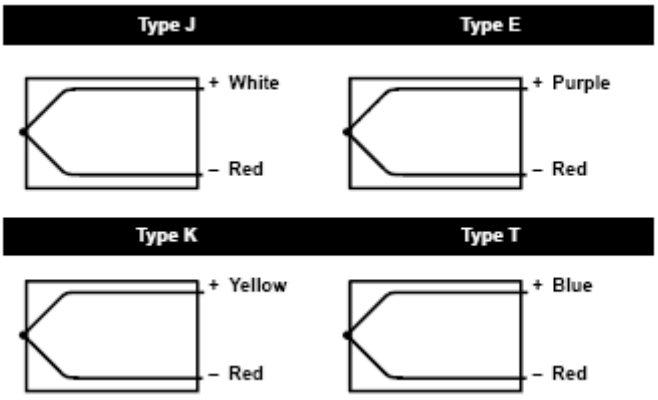


The five major thermocouple configurations are shown to the left.

The first two thermocouples are welded or grounded, as shown, to the outside metal protective sheathing.

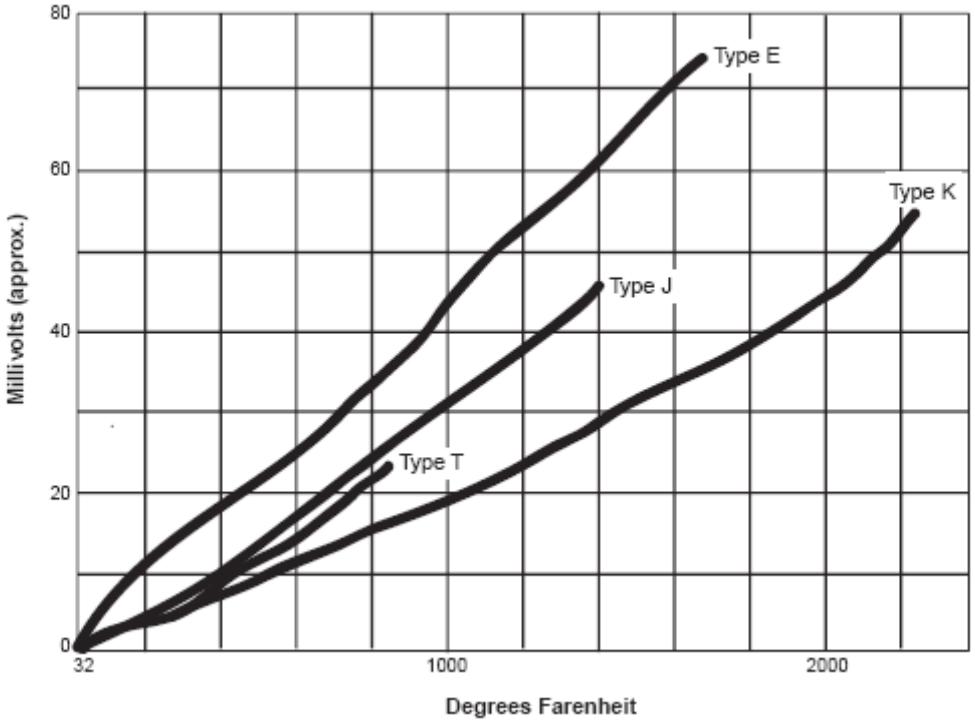
The bottom three thermocouples are ungrounded and should never touch the metal protective sheathing; otherwise they are shorted to ground.

Thermocouples should be extended with thermocouple extension wire and thermocouple termination blocks, but can be extended with standard copper wire and standard terminal blocks. This is due to the fact that the voltages generated at the extension junctions cancel each other out. One side is positive and the other side is negative.



The four major thermocouples used in the process industry are Type “J”; Type “E”; Type “K”; Type “T”. The red wire is always negative with thermocouples.

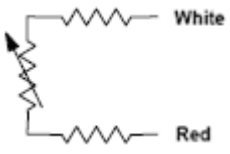
The color diagrams are shown to the left. The millivolt output and temperature ranges for the four thermocouple types are shown in the following graph.



The process control industry also uses RTDs (Resistance Temperature Detectors) for many applications, for example, when precise temperature measurement is needed, such as mass flow measurements or critical temperature measurements of motor bearings.

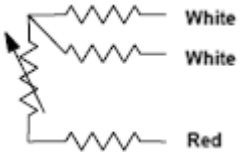
RTDs typically come in 10 ohm copper and 100 ohm platinum elements. Their resistance is typically very linear over the scale.

Resistance and millivolt tables for the examination can be found at Omega.com or in the **Tables Used In The Examination** section of this guide.



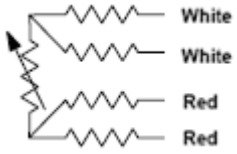
2-wire RTD

Good for close applications, at the transmitter.



3-wire RTD

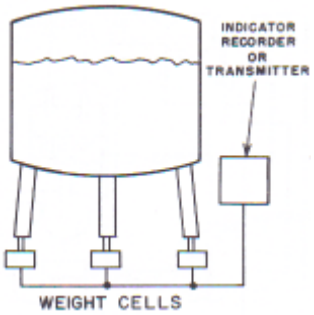
Good for further distance applications. Remote from the transmitter.



4-wire RTD

Best application and usually uses 20 mA driving current and voltage measurement.

Weight measurement



Weight measurements are typically made with strain gauges attached to metal bars. The bending moment of the bar causes the strain gauge to elongate, resulting in an increase of resistance in the strain gauge. This variable resistance is connected to a bridge circuit and a voltage is measured across the bridge. The voltage is proportional to the weight applied to the measuring bar.

This strain gauge technology is used in measuring the weight in tanks and weight on conveyor belts. The tare weight (tank weight) is nulled out and the voltage is set to zero or 0%, in the bridge circuit. Then the maximum weight to be measured is applied. These weights are NIST (National Institute of Standards and Technology) certified. The span voltage is then calibrated to maximum or 100%. This measurement is the net weight. (Remember all calibration processes should be repeated at least three times.)

Overview of Process Control

The process control industry covers a wide variety of applications: petrochemical; pharmaceutical; pulp and paper; food processing; material handling; even commercial applications.

Process control in a plant can include discrete logic, such as relay logic or a PLC; analog control, such as single loop control or a DCS (distributed control system); pneumatic; hydraulic and electrical systems as well. The Control Systems Engineer must be versatile and have a broad range of understanding of applied sciences.

The Control Systems Engineer (CSE) examination encompasses a broad range of subjects to ensure minimum competency. This book will review the foundations of process control and demonstrate the breadth and width of the CSE examination.

Degrees of Freedom

In an unconstrained dynamic or other system, the number of independent variables required to specify completely the state of the system at a given moment must be defined. If the system has constraints, that is, kinematic or geometric relations between the variables, each such relation reduces by one the number of degrees of freedom (DOF) of the system.

Process Variables - (Equations + Constants) = Degrees of Freedom

Degrees of Freedom = The Minimum Number of Process Controllers required

Example 1: An Airplane

Variables

Altitude	1
Latitude	1
Longitude	1
	3
Minus Constants	0
Minus Equations	0
Degrees of freedom =	3



DOF = 3 – (0+0) = 3

Three (3) controllers are needed. One (1) for each variable.

Example 2: A Train

Variables

Altitude	1
Latitude	1
Longitude	1
	3
Minus Constants	
Altitude	1
Latitude	1
Minus Equations	0
Degrees of freedom =	1



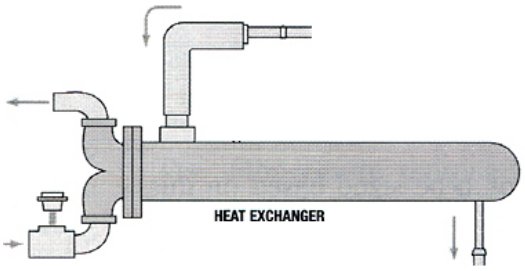
DOF = 3 – (2+0) = 1

One (1) controller is needed. One (1) for Longitude only.

Example 3: A Hot Water Heat Exchanger

Variables

Ws (flow rate of steam)	1
Wcw (flow rate of cold water)	1
Whw (flow rate of hot water)	1
Q (quantity of steam)	1
Ps (supply pressure of steam)	1
Tcw (temperature of cold water)	1
Thw (temperature of hot water)	1
	7



Minus Constants

Q (quantity of steam)	1
Ps (supply pressure of steam)	1
Tcw (temperature of cold water)	1
	3

Minus Equations

Material Balance (conservation of mass)	1
Energy Balance (conservation of energy)	1
	2

DOF = 7 – (3+2) = 2
Two (2) controllers are needed.

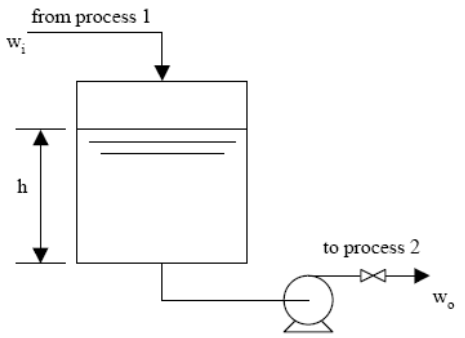
a) One (1) to controller for steam flow.

b) One (1) to controller for the energy equation (mass*Cp*deltaT). The controller will be a temperature controller, and on the outlet water temperature. It will provide a remote setpoint to the steam flow controller.

Control Loops

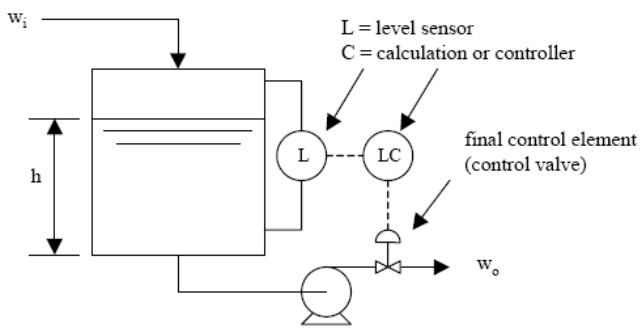
In general terms, a *control loop* is a group of components working together as a system to achieve and maintain the desired value of a system variable by manipulating the value of another variable in the control loop. Each control loop has at least one input and one output. There are two types of control loops: open loop and closed loop.

In an *open loop* system, the controller **does not** have a feedback signal from the system. The controller has a setpoint and a fixed output signal. The output signal does not vary regardless of the system disturbances.



An example of an open loop system would be a car, when using the accelerator pedal only. The accelerator pedal is held in fixed position. When the car goes up a hill, the car will tend to slow down. The decrease in speed is inversely proportional to the increase in slope.

In a *closed loop* system, the controller **does** have a feedback signal from the system. The controller has a setpoint, a feedback input signal and a varying output signal. The output signal increases or decreases proportionally to the error of the setpoint compared to the input signal. The input signal varies proportionally to the system disturbances and the gain of the measurement sensor.



An example of a closed loop system would be a car, when using the speed control only. When the car goes up a hill, the car will tend to speed up to maintain the setpoint speed, regardless of increase in slope. The increase in slope is a systems disturbance, but there can be more than one disturbance on a system. A head wind would add to the error of increasing slope, giving the car even more power to increase the speed to setpoint, say 55 mph.

All control systems have their limitations of control. Either the ability to respond to fast changing systems disturbances, frequency response, or the limitation to add or remove energy to the system, i.e. the valve is at 0% or 100%. When referring to system response, the valve or servo mechanism has limited speed of movement due to mechanical design, a slew rate of movement. The valve or servo mechanism can only move so many inches or degrees in a time period. Frequency is the reciprocal of time.

The process variable or feedback input signal is always measured in 0% to 100% and is typically evenly divisible by 4 or measured at 25% increments.

Examples:

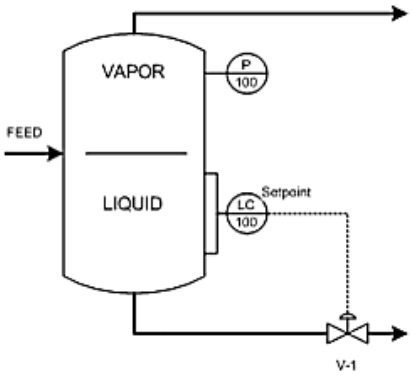
3 to 15 PSI	12 PSI span
4 to 20 mA	16 mA span
1 to 5 Volts	4 Volts span

Controller and Control Modes

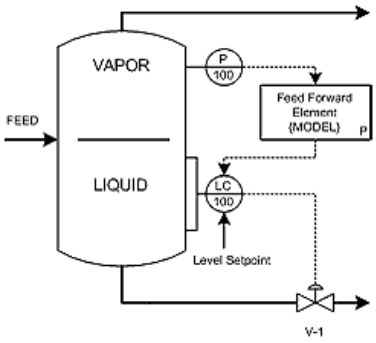
Familiarize yourself with the different control modes and the ISA Standards and symbols for representing the modes on a P&ID (Piping & Instrumentation Drawing).

The most common types of closed loop control modes are: feedback, feedforward, cascade, and ratio.

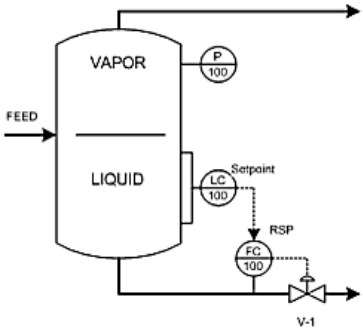
Feedback Control Loop:



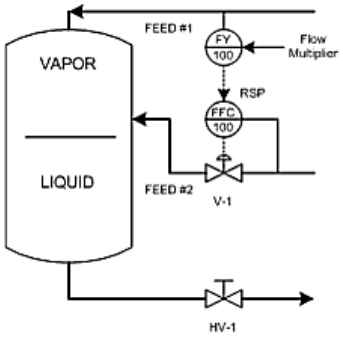
Feedforward Control Loop:

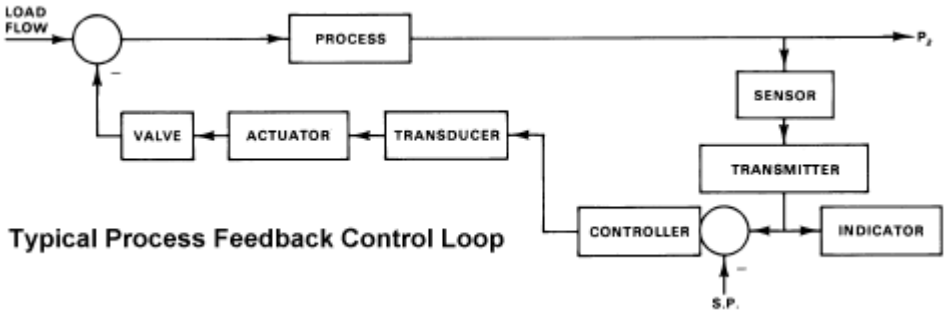


Cascade Control Loop:



Ratio Control Loop:



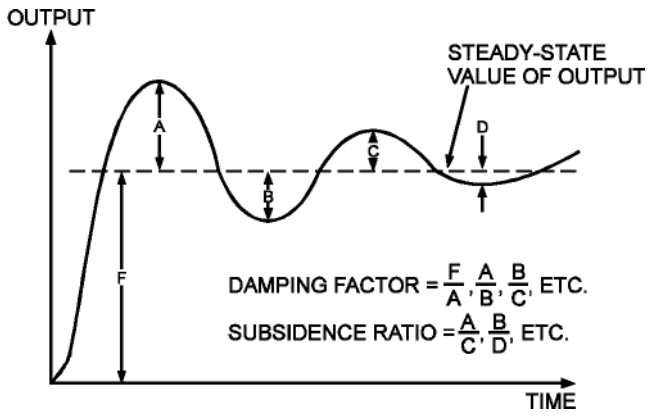


Typical Process Feedback Control Loop

We will now look at controller and control loop characteristics. Mathematically we will describe the response of a control loop and calculate the overshoot and damping of a typical control loop.

To the right side is a graph showing a typical controller response to a setpoint change. Most engineers use 0.25 amplitude damping for control of loops in the process industry.

Let us find out how to solve for the above-mentioned criteria.



Find Damping

F=50 PSI; A= 8.15 PSI

Find the damping from overshoot:

$$A\%os = 100e^{-\pi\xi/\sqrt{1-\xi^2}}$$

OR

USE SIMPLE METHOD BELOW

$$\xi = \sqrt{\frac{\left[\ln(OS)\right]^2}{\pi^2 + \left[\ln(OS)\right]^2}}$$

The damping from overshoot is:

$$\frac{8.15}{50} * 100 = 16.3\%$$

$$16.3 = 100e^{-\pi\xi/\sqrt{1-\xi^2}}$$

$$e^{\pi\xi/\sqrt{1-\xi^2}} = \frac{100}{16.3}$$

$$\pi\xi/\sqrt{1-\xi^2} = \ln\frac{100}{16.3} = 1.814$$

$$\pi^2\xi^2 = 1.814^2\left(1-\xi^2\right)$$

$$\pi^2\xi^2 = 3.29\left(1-\xi^2\right)$$

$$\pi^2\xi^2 = 3.29 - 3.29\xi^2$$

$$9.869\xi^2 = 3.29 - 3.29\xi^2$$

$$9.869\xi^2 + 3.29\xi^2 = 3.29$$

$$\left(9.869 + 3.29\right)\xi^2 = 3.29$$

$$\xi^2 = \frac{3.29}{9.869 + 3.29}$$

$$\xi = \sqrt{0.25}$$

$$\xi = 0.5$$

Find Overshoot and Peak Value

F=50 PSI; $\xi = 0.5$

The first overshoot is:

$$A\% = 100e^{-\pi\xi/\sqrt{1-\xi^2}}$$

The second overshoot is:

$$C\% = 100e^{-3\pi\xi/\sqrt{1-\xi^2}}$$

The percent overshoot and peak is:

$$A\% = 100e^{-\pi 0.5/\sqrt{1-0.5^2}}$$

$$A\% = 100e^{-1.57/\sqrt{0.75}}$$

$$A\% = 100e^{-1.812}$$

$$A\% = 100(0.163)$$

$$A\% = 16.3\%$$

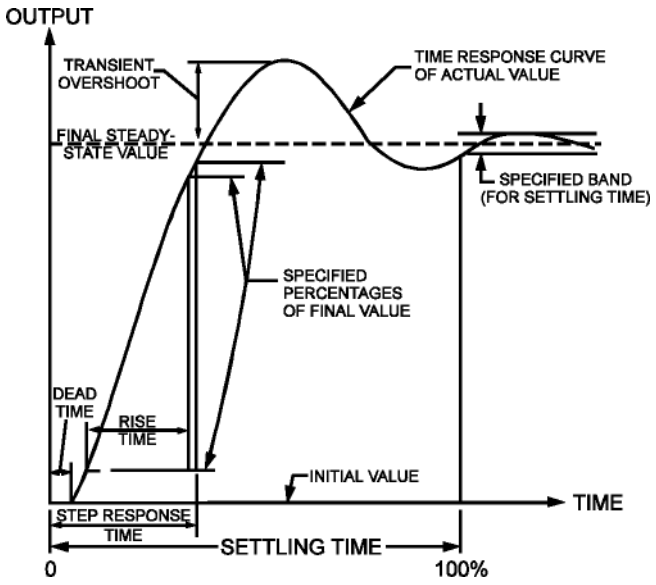
$$50\,psi\,(0.163) = 8.17\,psi\ overshoot$$

$$50\,psi + 8.17\,psi = 58.17\,psi\ \ peak$$

We will now calculate the rise time, period, natural frequency and the settling time. We will refer to the graph to the right and the previously used graph for the peak amplitude designations.

Notice rise time in the graph on the right. It rises in a vertical line from 10% to 90% of steady state value. This is the definition of rise time.

Notice step response in the graph on the right. It rises in a vertical line from 0% to 63.2% of peak value. This is the definition of step response time.



The time constant will be step response time minus the dead time or lag time.

Find the Time Constant

Step response time: 6 seconds Dead time: 1 second	$\tau = T_{sr} - T_d$ $\tau = 6 - 1$ $\tau = 5\ seconds$

Find the Period

Step response time: 6 seconds Dead time: 1 second Time Constant: 5 seconds Damping: 0.5	$P = \frac{2\pi\tau}{\sqrt{1-\xi^2}}$ $P = \frac{6.28(5)}{\sqrt{1-0.5^2}}$ $P = 36.26\ seconds$

Find the Time Constant from the Period

Period: 36.26 seconds Damping: 0.5	$\tau = \frac{\sqrt{1-\xi^2}}{2\pi} P$ $\tau = \frac{\sqrt{1-0.5^2}}{6.28} 36.26$ $t = 5\ seconds$

Using Transfer Functions

Find the Damping from the Function

G(s)= $\frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$ G(s)= $\frac{25}{s^2 + 5s + 25}$ $\xi =$ Damping Ratio Damping: 0.5	$s^2 + 5s + 25$ $s^2 + 2\xi\omega_n s + \omega_n^2;\ \ \omega_n = \sqrt{25}$ $2\xi\omega_n s = 5s$ $\xi = \frac{5}{2\omega_n}; = \frac{5}{2\sqrt{25}} = \frac{5}{10} = 0.5$

Find the Poles from the Function

$G(s)=\frac{\omega_n^2}{s^2+2\xi\omega_n s+\omega_n^2}$ $G(s)=\frac{25}{s^2+5s+25}$ Pole1: -2.5+j4.33 Pole2: -2.5-j4.33	Poles: $p_1; p_2 = \frac{-b \pm \sqrt{b^2-4ac}}{2}$ $p_1; p_2 = \frac{-5 \pm \sqrt{25-4(25)}}{2}$ $p_1; p_2 = \frac{-5 \pm \sqrt{\lvert 25-100 \rvert}}{2} = -2.5 \pm j\ 4.33$

Controller Tuning

We will now look at two different methods for tuning a controller, the Ultimate Gain (Continuous Cycling), and Process Reaction Curve (Step Response) methods.

Tuning based on the ultimate gain method

Essentially, the tuning method works by oscillating the process. Turn off the Integral mode or set time to zero (0) and turn off the derivative mode. Increase the gain of the controller and make a slight setpoint change. Repeat the process and gradually increase the gain of the controller each time, until a sustained oscillation is achieved as shown in the following figure. This is called the ultimate gain. The proportional band is the reciprocal of the gain.

Tune the controller by entering the new values from the calculations in table 4 below.

The table values are to be entered as ***gain***. If you need to convert gain to proportional band, then Pu=1/Ku and Ku=1/Pu. Convert ***after*** applying the table calculations.

The period or time contain equals Tu in minutes. The time calculation will be entered as ***minutes per repeat*** for Integral time and Derivative time as ***minutes***.

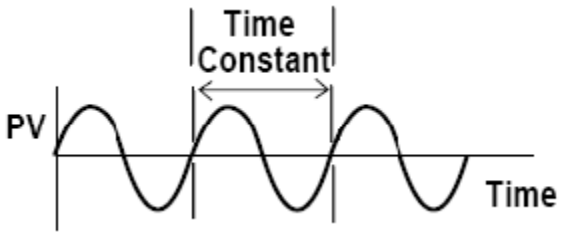


Table 4. Tuning parameters for the closed loop Ziegler-Nichols method

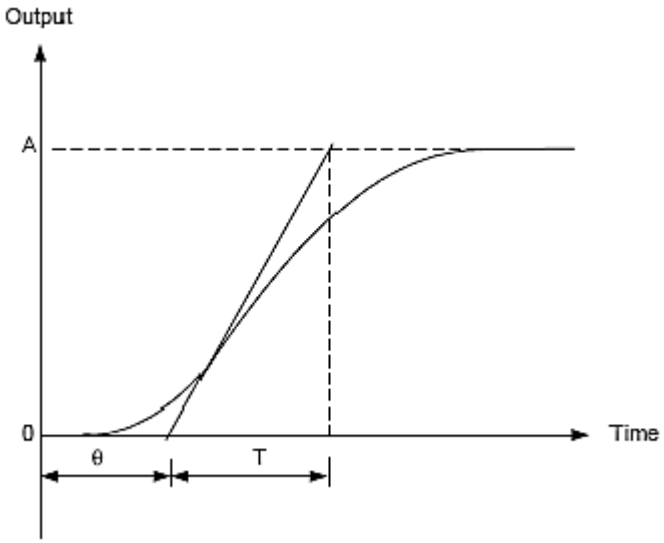
Controller type	Gain, Kc	Integral time, TI	Derivative time, TD
P	$0.5K_u$		
PI	$0.45K_u$	$\frac{T_u}{1.2}$	
PID	$0.6K_u$	$\frac{T_u}{2}$	$\frac{T_u}{8}$

Example: Tune using Ultimate Gain (continuous cycling)

Time Constant: 12 minutes Gain Ku: 2.2	$K_c = 0.6K_u = (0.6)(2.2) = 1.32$
Note: T_I = minutes per repeat	$T_I = \frac{T_U}{2} = \frac{12}{2} = 6 \text{ min}$
T_I^{-1} = repeats per minute	$T_D = \frac{T_u}{8} = \frac{12}{8} = 1.5 \text{ min}$

Tuning based on the process reaction curve

In process control, the term 'reaction curve' is sometimes used as a synonym for a step response curve. Many chemical processes are stable and well damped, and for such systems the step response curve can be approximated by a first-order-plus-deadtime model and it is relatively straightforward to fit the model parameters to the observed step response. Look at the reaction curve to the right.



Essentially, the tuning method works by manually causing a step change in the process. This is accomplished by putting the controller in manual and forcing an output change of the controller. Record the step change process reaction curve on the chart recorder and follow the setup instructions below.

1. Locate the inflection point, i.e., the point where the curve stops curving upwards and starts to curve downwards.
2. Draw a straight line through the inflection point, with the same gradient as the gradient of the reaction curve at that point. (see the graph above)
3. The point where this line crosses the initial value of the output is assumed to be zero, (zero may equal 50 psi or 500 degrees, but set it to a live zero), gives the apparent time delay or dead time θ.
4. The straight line reaches the steady state value "A", (Δ PV), of the output at time T + θ. Draw a line straight down. T is time constant.
5. The gain slope K is given by A/T.

Table 5. Tuning parameters for the open loop Ziegler-Nichols method

Controller type	Gain, Kc	Integral time, TI	Derivative time, TD
P	$\frac{T}{K\theta}$		
PI	$\frac{0.9T}{K\theta}$	$\frac{\theta}{0.3}$	
PID	$\frac{4T}{3K\theta}$	$\frac{\theta}{0.5}$	0.5θ

Example: Tune using Process Reaction Curve (step response)

Time Constant T: 10 minutes
Dead Time : 5 minutes
A = Delta PV: 8 psi

Note: T_I = minutes per repeat

T_I^{-1} = repeats per minute

$$K = Slope = \frac{A}{T}; K = \frac{8}{10} = 0.8$$

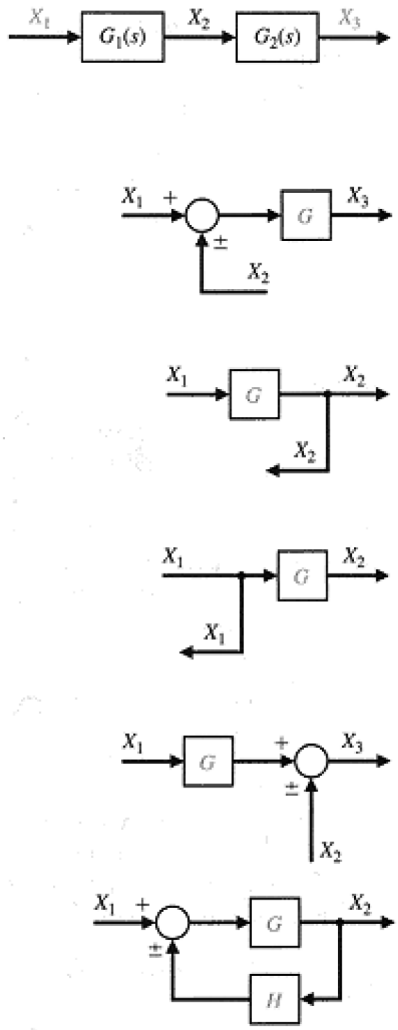
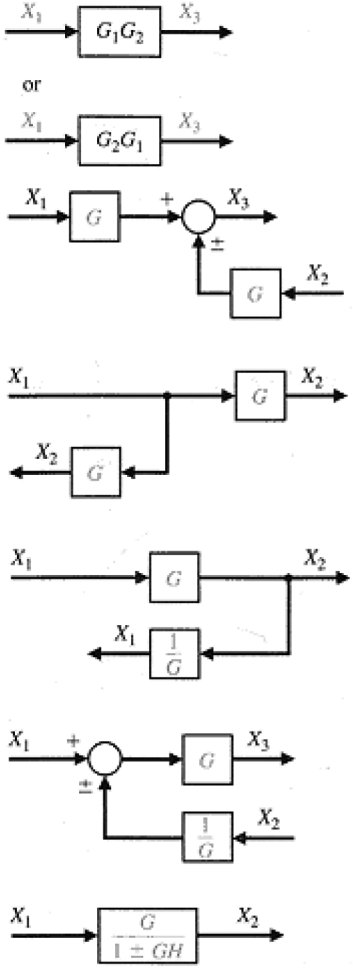
$$K_c = \frac{4T}{3K\theta} = \frac{(4)(10)}{(3)(0.8)(5)} = 3.33$$

$$T_I = \frac{\theta}{0.5} = \frac{5}{0.5} = 10 \text{ min}$$

$$T_D = 0.5\theta = (0.5)(5) = 2.5 \text{ min}$$

Block Diagram Algebra

(Simplification Methods)

Original Block Diagram	Equivalent Block Diagram
	

Block Diagram Algebra Reduction (Example)

Start at figure (a), the original multivariable diagram and simplify.

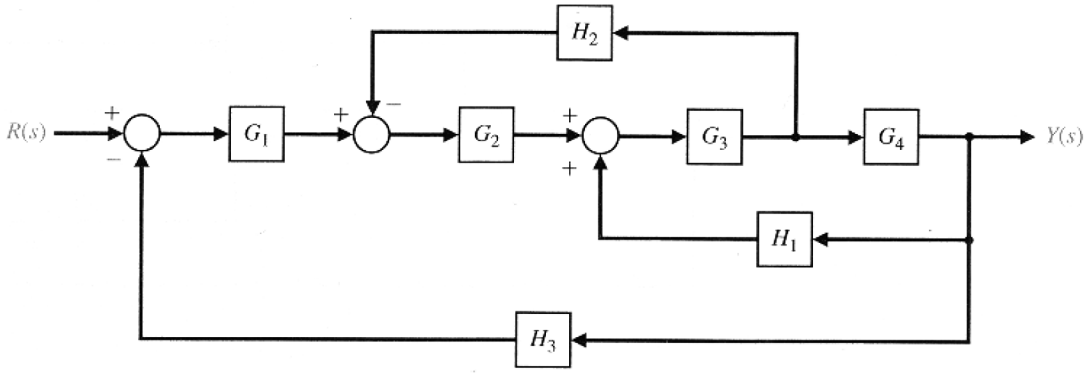


Figure (a)

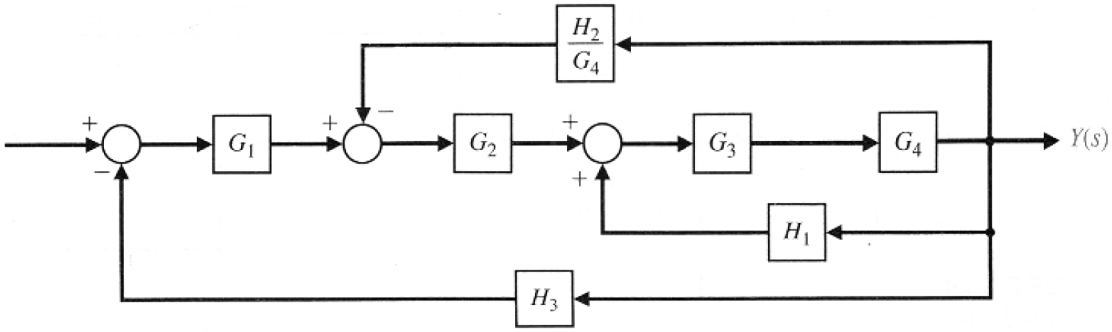


Figure (b)

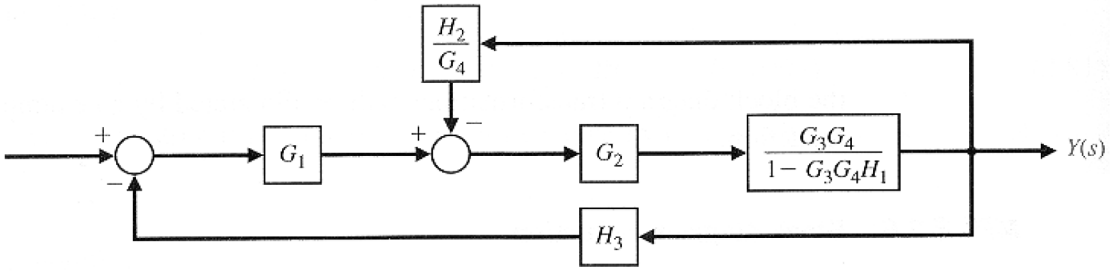


Figure (c)

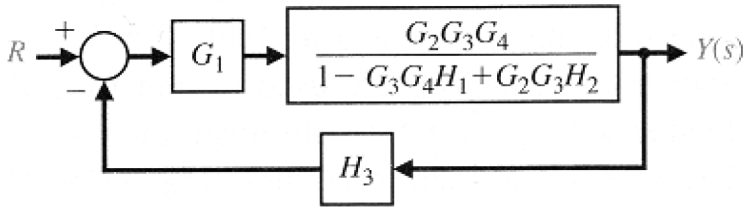


Figure (d)

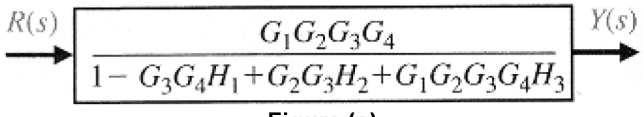
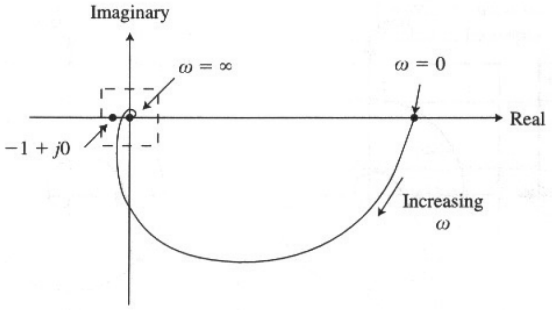


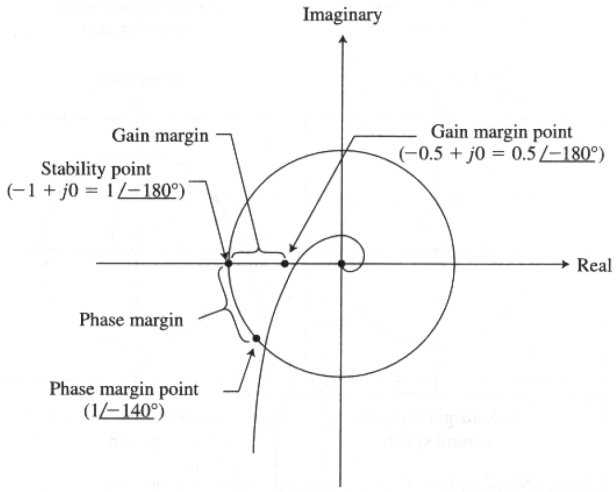
Figure (e)

Nyquist Stability Criterion

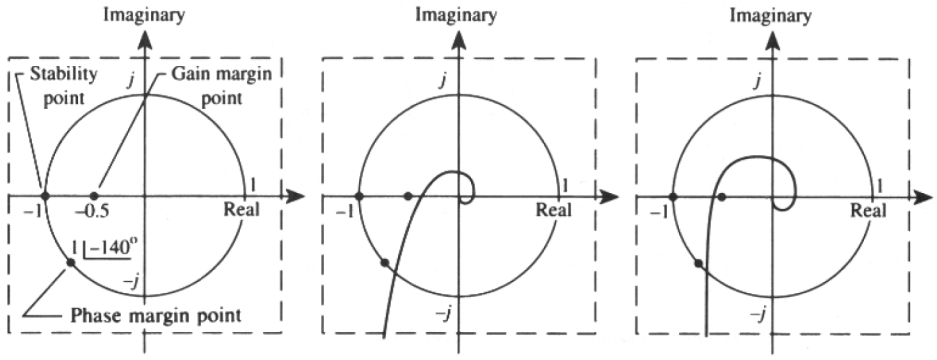
Most closed-loop systems are open-loop stable and do not have any pole (open-loop pole) in the right half of the s plane. Closed-loop systems that are stable will not have any root in the right half plane. The Nyquist diagram [Ref. 3] of an open-loop stable system does not encircle the (–1, j0) point.



a) Nyquist diagram of a stable control system



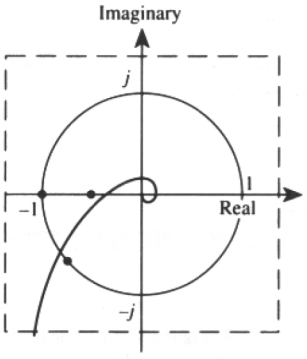
b) Close-up view of the boxed region in part a



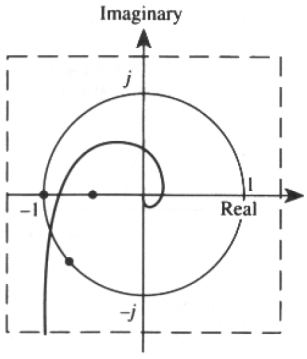
a) Close-up of the unit circle.

b) A stable control system

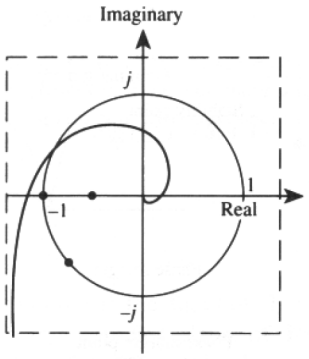
c) A marginally stable control system



d) A marginally stable control system



e) A marginally stable control system



f) An unstable control system

Routh Stability Criterion

For given coefficients a_i of the *characteristic equation* the method of Routh, which is an alternative to the method of Hurwitz, can be applied [Ref. 20]. Here the coefficients $a_i\,(i=0,1,...,n)$ will be arranged in the first two rows of the Routh schema, which contains $n+1$ rows:

Row n	s^n	a_o	a_2	a_4	a_6	…	…	0
Row n-1	s^{n-1}	a_1	a_3	a_5	a_7	…	…	0
Row n-2	s^{n-2}	b_1	b_2	b_3	b_4	…	0	
Row n-3	s^{n-3}	c_1	c_2	c_3	c_4	…	0	
:	:	:	:	:				
Row 3	s^3	d_1	d_2	0				
Row 2	s^2	e_1	e_2	0				
Row 1	s^1	f_1						
Row 0	s^0	g_1						

The coefficients $b_1, b_2.b_3, ...$ in the third row are the results from cross multiplication the first two rows according to

$$b_1 = \frac{a_1 a_2 - a_0 a_3}{a_1}$$

$$b_2 = \frac{a_1 a_4 - a_0 a_5}{a_1}$$

$$b_3 = \frac{a_1 a_6 - a_0 a_7}{a_1}$$

$$\vdots$$

Building the cross products you start with the elements of the first row. The calculation of these *b* values will be continued until all remaining elements become zero. The calculation of the *c* values are performed accordingly from the two rows above as follows:

$$c_1 = \frac{b_1 a_3 - a_1 b_2}{b_1}$$

$$c_2 = \frac{b_1 a_5 - a_1 b_3}{b_1}$$

$$c_3 = \frac{b_1 a_7 - a_1 b_4}{b_1}$$

$$\vdots$$

From these new rows further rows will be built in the same way, where for the last two rows finally

$$f_1 = \frac{e_1 d_2 - d_1 e_2}{e_1}$$

and

$$g_1 = e_2$$

follows.

Note: For our example, the last two rows are:

$$d_1 = \frac{c_1 b_2 - b_1 c_2}{c_1}$$

and

$$e_1 = c_2$$

Note: If there are only four polynomials, decrement the last two rows by one letter again and do not use the c_1, c_2, c_3, … determinates. A pattern should be emerging now.

Now the *Routh criterion* includes the following:

A polynomial $P(s)$ is Hurwitzian, if and only if the following three conditions are valid:

a) all coefficients $a_i \left(i = 0,1,...,n\right)$ are positive.
b) all coefficients $b_1, c_1, ...$ in the first column of the Routh schema are positive.
c) all coefficients $b_1, c_1, ...$ in the first column of the Routh schema are not zero.

As in the first row of the Routh schema, a coefficient is negative the system is unstable.

For proving instability, it is sufficient to build the Routh schema only until negative or zero value occurs in the first column. In the example, given the schema could have been stopped at the fifth row.

Another interesting property of the Routh scheme says that the number of roots with positive real parts is equal to the number of changes of sign of the values in the first column.

Example

$$P(s) = s^5 + 2s^4 + 30s^3 + 50s^2 + 110s + 240$$

$$Note: \left\{P(s) = a_0 + a_1 + a_2 + a_3 + a_4 + a_5\right\}$$

The Routh schema is:

s^5	a_0	a_2	a_4	0
s^4	a_1	a_3	a_5	0
s^3	b_1	b_2	0	
s^2	c_1	c_2		
s^1	d_1	0		
s^0	e_1			

5	1	30	110	0
4	2	50	240	0
3	5	-10	0	
2	54	240		
1	-32.22	0		
0	240			

Laplace transform

Corresponding elements of the Laplace transform		
Nr.	time response $f(t), f(t)=0$, for $t<0$	Laplace transformed $F(s)$
1	pulse $\delta(t)$	1
2	unit step $\sigma(t)$	$\frac{1}{s}$
3	t	$\frac{1}{s^2}$
4	t^2	$\frac{2}{s^3}$
5	$\frac{t^n}{n!}$	$\frac{1}{s^{n+1}}$
6	e^{-at}	$\frac{1}{s+a}$
7	te^{-at}	$\frac{1}{(s+a)^2}$
8	t^2e^{-at}	$\frac{2}{(s+a)^3}$

9	$t^n e^{-at} t^2$	$\frac{n!}{(s+a)^{n+1}}$
10	$1-e^{-at}$	$\frac{a}{s(s+a)}$
11	$\frac{1}{a^2}\left(e^{-at}-1+at\right)$	$\frac{1}{s^2(s+a)}$
12	$(1-at)e^{-at}$	$\frac{s}{(s+a)^2}$
13	$\sin \omega_o t$	$\frac{\omega_o}{s^2+\omega_o{}^2}$
14	$\cos \omega_o t$	$\frac{s}{s^2+\omega_o{}^2}$
15	$e^{-at} \sin \omega_o t$	$\frac{\omega_o}{(s+a)^2+\omega_o{}^2}$
16	$e^{-at} \cos \omega_o t$	$\frac{s+a}{(s+a)^2+\omega_o{}^2}$
17	$\frac{1}{a} f\left(\frac{t}{a}\right)$	$F(as)(a>0)$
18	$e^{at} f(t)$	$F(s-a)$
19	$f(t-a) \quad for \quad t>a\geq 0$ $0 \quad for \quad t<a$	$e^{-as} F(s)$
20	$-t\, f(t)$	$\frac{dF(s)}{ds}$
21	$(-t)^n\, f(t)$	$\frac{d^n F(s)}{ds^n}$
22	$f_1(t)\, f_2(t)$	$\frac{1}{2\pi j}\int_{c-j\infty}^{c+j\infty} F_1(p) F_2(s-p)\, dp$

Sizing Process Control Elements

Sizing Elements and Final Devices

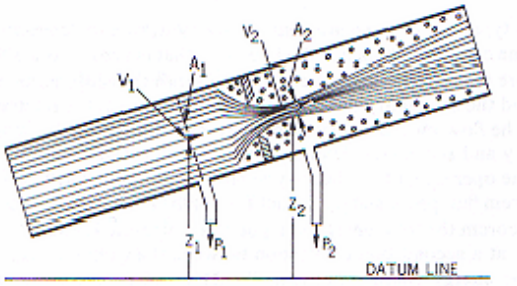
The process control industry covers a wide variety of applications of elements and final correction devices.

The Control Systems Engineer (CSE) examination encompasses a broad range of valve applications and sizing for different services, possibly an orifice meter; a turbine meter; pressure relief valve or safety rupture disk. This book will cover essential basics for the CSE examination.

Flow Measurement

Fluids (and other useful equations)

$$Z_1 + \frac{V_1^2}{2g} + \frac{p_1}{\gamma} = Z_2 + \frac{V_2^2}{2g} + \frac{p_2}{\gamma}$$



$$A_1 V_1 = A_2 V_2$$

$\frac{P_2}{P_1} = \left(\frac{F_1}{F_2}\right)^2$; very useful in the examination

$$R_e = \frac{3160 * flow\,rate(gpm) * Specific\,Gravity}{Pipe\,ID(inches) * Viscosity(cp)} ; for\ liquids$$

$$R_e = \frac{6.316 * Flow\,Rate(LB/Hr)}{Pipe\,ID(inches) * Viscosity(cp)} ; for\ gases\ and\ steam$$

$$R_e = 1000 \frac{v(m/s)\,D(mm)}{\upsilon(cSt)}$$

Orifice Type Meters

The basic equation for liquid flow through an orifice plate is:

$$Q = 5.667 SD^2 \sqrt{\frac{h}{G_f}}$$

We will reference Norman Anderson's book: *Instrumentation for Process Measurement and Control* and reference Table 4-2, or see this guide's Table 6 - Sizing Factors.

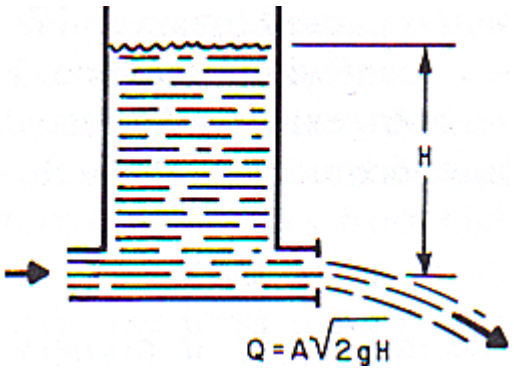
Let us review the math that derives this volumetric flow equation.

$$V^2 = 2gH$$

$$V = \sqrt{2gH}$$

$$Q = AV$$

$$Q = A\sqrt{2gH}$$



$$H = \frac{h}{12G_f}; \text{ Note: h is in inches, put it in feet}$$

$$Q = A\sqrt{2g\frac{h}{12G_f}}$$

$$Q(gpm) = [time\ scaling * volume\ scaling] * \left\{ \sqrt{\frac{2g}{1}} * \frac{A}{144} * \sqrt{\frac{h}{12G_f}} \right\}; \text{ Note: scale inches to feet}$$

$$Q(gpm) = \left[\frac{60\,\text{sec}}{1\,\text{min}} * \frac{1728\,\text{in}^3}{231\,\text{in}^3} \right] * \sqrt{\frac{2(g)}{12}} * \frac{\pi d^2}{4*144} * \sqrt{\frac{h}{G_f}}$$

$$Q(gpm) = \left[\frac{60\,\text{sec}}{1\,\text{min}} * \frac{1728\,\text{in}^3}{231\,\text{in}^3} \right] * \sqrt{\frac{64.34\,\text{ft}}{12\,\text{in sec}^2}} * \frac{\pi}{4*144\,\text{in}^2} * (d\ \text{in})^2 * \sqrt{\frac{h\ \text{in}}{G_f}}$$

$$Q(gpm) = \frac{60\,\text{sec}}{\text{min}} * \frac{7.4805\,\text{gal}}{\text{ft}^3} * \frac{2.3155\,\text{ft}}{\text{in sec}} * \left[\frac{0.00545}{\text{in}^2} * (d\ \text{in})^2 \right] \text{ft}^2 * \sqrt{\frac{h\ \text{in}}{G_f}}$$

$$Q(gpm) = \frac{60}{\min} * 7.4805\ \text{gal} * 2.3155 * 0.00545 * d^2 * \sqrt{\frac{h}{G_f}} = 5.667 * d^2 * \sqrt{\frac{h}{G_f}} * \frac{\text{gal}}{\min}$$

$$Q(gpm) = 5.667 * d^2 * \sqrt{\frac{h}{G_f}}$$

Add factor for coefficients of friction, viscosity, convergence, and divergence.

$$Q(gpm) = 5.667 * Kd^2 * \sqrt{\frac{h}{G_f}}$$

Since *K* and *d* (orifice diameter) are unknowns:

$S = K\left(\frac{d^2}{D^2}\right)$ … So, cancel the pipe diameter (D^2) and…

The basic equation for liquid through an orifice type device is:

$$Q(gpm) = 5.667SD^2\sqrt{\frac{h}{G_f}}$$

Using the sizing equation and the sizing factor table, we accurately size orifices taps; pipe taps; nozzle and venture; lo-loss tube; and dall tube.

The basic equation for gas through an orifice type device is:

$$Q(scfh) = 218.4SD^2\frac{T_{abs}}{P_{abs}}\sqrt{\frac{hp_f}{T_f G_f}}$$

If conditions are 60°F and 14.7psia then the formula can be reduced to:

$Q(scfh) = 7,727SD^2\sqrt{\frac{hp_f}{T_f G_f}}$; ONLY at 60°F and 14.7 psia conditions

The basic equation for steam through an orifice type device is:

$$W(\textit{pounds per hour}) = 359SD^2\sqrt{h\gamma_f}$$

where,

$G_f = Specific\ gravity,\ for\ gas \dfrac{(molecular\ weight\ of\ gas)}{28.97(is\ the\ \text{M.W.}\ of\ air)}$

$h = Head\ in\ inches$

$P_{abs} = Reference\ pressure\ (psi\ absolute)$

$P_f = Fluid\ operating\ pressure\ (psi\ absolute)$

$T_{abs} = Reference\ temperature\ (psi\ absolute); (Reference\ temp\ in\ {}^{o}\text{F}+460)$

$T_f = Fluid\ operating\ temperature\ (psi\ absolute); (Reference\ temp\ in\ {}^{o}\text{F}+460)$

$\gamma_f = Specific\ weight\ of\ the\ steam\ or\ vapor\ in\ pounds\ per\ cubic\ foot\ (operating\ cond.)$

Liquid Sample Problem: Gasoline is carried in a 3-inch schedule 40 pipe (ID=3.068). A concentric sharp-edged orifice plate, with corner taps, is used to measure the flow. If the Beta Ratio is 0.500, maximum flow rate is 100 gpm, and s.g. = 0.75, what is the differential head and span of the flowmeter transmitter?

$$Q(gpm) = 5.667SD^2\sqrt{\frac{h}{G_f}}$$

From Table 6: $S = 0.1568$

$$100(gpm) = 5.667(0.1568)(3.068)^2\sqrt{\frac{h}{0.75}}$$

$$\frac{100(gpm)}{5.667(0.1568)(3.068)^2} = \sqrt{\frac{h}{0.75}}$$

$$\left(\frac{100(gpm)}{8.3639}\right)^2 = \left(\sqrt{\frac{h}{0.75}}\right)^2$$

$$11.9561^2 = \frac{h}{0.75}$$

$$142.95(0.75) = h$$

$$107.21 = h \text{ (span)}$$

Calibrate transmitters 0 to 100% and 4mA to 20mA.
The calibrated range of the transmitter will be 0 to 107.21 inches H_2O.

Gas Sample problem: natural gas is carried in a 6-inch schedule 40 pipe (ID=6.065). Flowing temperature is 60F at 30 psig pressure. A concentric sharp-edged orifice plate, with flange taps, is used to measure the flow. If maximum flow rate is 4,000,000 scf per day; s.g. = 0.60, and the differential head of the flow meter transmitter is 50 inches H_2O. What is the orifice hole bore diameter?

$$Q(scfh) = 218.4SD^2 \frac{T_{abs}}{P_{abs}} \sqrt{\frac{hp_f}{T_f G_f}}$$

Change flow from per day to per hour and temperature and pressure to absolute:

$$\frac{4,000,000}{24hrs} = 218.4S(6.065)^2 \frac{520}{14.7} \sqrt{\frac{(50)(44.7)}{(520)(0.60)}}$$

$$\frac{4,000,000}{24} = 218.4S(6.065)^2 \frac{520}{14.7} \sqrt{\frac{(50)(44.7)}{(520)(0.60)}}$$

Find the "S" sizing factor:

$$\frac{4,000,000}{(24)760,609.46} = S = 0.2191$$

From Table 6 data:
Beta = 0.575 *S* = 0.2144
Beta = 0.600 *S* = 0.2369

This will require interpolation:

$$Beta = \left[\left(\frac{0.2191 - 0.2144}{0.2369 - 0.2144}\right)(0.600 - 0.575)\right] + 0.575 = 0.5802$$

Find the orifice hole diameter:

$$\text{d} = \text{Beta} * \text{pipe ID} = \text{hole size}$$
$$\text{d} = 0.5802 * 6.065 = 3.519 \,\text{inches}$$

For the calibrated range of the transmitter 0 to 50 inches H_2O, and a flow rate of 166,666.7 scfh or 4,000,000 scfd, the orifice hole bore diameter = 3.519 inches

Steam Sample Problem: Dry saturated steam is carried in an 8-inch schedule 80 pipe (ID=7.625). A flow nozzle is used to measure the flow. If the Beta Ratio is 0.450, and the static pressure is 345 psig, what is the flow rate with a differential head pressure of 200 inches H_2O across the meter?

$$W(\textit{pounds per hour}) = 359SD^2\sqrt{h\gamma_f}$$

Find the density from the saturated steam tables in the FCVH (Chapter 10). A gauge pressure of 345.3 gives a specific volume of 1.2895.

$$\text{Density in } \frac{\text{lb}}{\text{ft}^3} = \frac{1}{\text{specific volume in } \frac{\text{ft}^3}{\text{lb}}}$$

$$\gamma_f = \frac{1}{1.2895} = 0.7755$$

From Table 6: $S = 0.2026$

$$W(\textit{pounds per hour}) = 359(0.2026)(7.625)^2\sqrt{(200)(0.7755)} = 52,664.68\ lb/hr$$

I suggest using Norman Anderson's book, *Instrumentation for Process Measurement and Control*, and working all examples for liquid, steam, and vapor.

Table 6 - Orifice Sizing Factors

Beta Or d/D Ratio	Square Edged Orifice; Flange Corner or Radius Taps	Full-Flow (Pipe) 2 ½D & 8D Taps	Nozzle and Venturi	Lo-Loss Tube	Dall Tube	Quadrant-Edged Orifice
0.100	0.005990	0.006100				
0.125	0.009364	0.009591				
0.150	0.01349	0.01389				
0.175	0.01839	0.01902				
0.200	0.02402	0.02499				0.0305
0.225	0.03044	0.03183				0.0390
0.250	0.03760	0.03957				0.0484
0.275	0.04558	0.04826				0.0587
0.300	0.05432	0.05796	0.08858			0.0700
0.325	0.06390	0.06874	0.1041			0.0824
0.350	0.07429	0.08086	0.1210	0.1048		0.0959
0.375	0.08559	0.09390	0.1392	0.1198		0.1106
0.400	0.09776	0.1085	0.1588	0.1356	0.1170	0.1267
0.425	0.1977	0.1247	0.1800	0.1527	0.1335	0.1443
0.450	0.1251	0.1426	0.2026	0.1705	0.1500	0.1635
0.475	0.1404	0.1625	0.2270	0.1900	0.1665	0.1844
0.500	0.1568	0.1845	0.2530	0.2098	0.1830	0.207
0.525	0.1745	0.2090	0.2810	0.2312	0.2044	0.232
0.550	0.1937	0.2362	0.3110	0.2539	0.2258	0.260
0.575	0.2144	0.2664	0.3433	0.2783	0.2472	0.292
0.600	0.2369	0.3002	0.3781	0.3041	0.2685	0.326
0.625	0.2614	0.3377	0.4159	0.3318	0.2956	0.364
0.650	0.2879	0.3796	0.4568	0.3617	0.3228	
0.675	0.3171	0.4262	0.5016	0.3939	0.3499	
0.700	0.3488	0.4782	0.5509	0.4289	0.3770	
0.725	0.3838		0.6054	0.4846	0.4100	
0.750	0.4222		0.6667	0.5111	0.4430	
0.775	0.4646			0.5598	0.4840	
0.800	0.5113			0.6153	0.5250	
0.820				0.6666	0.5635	

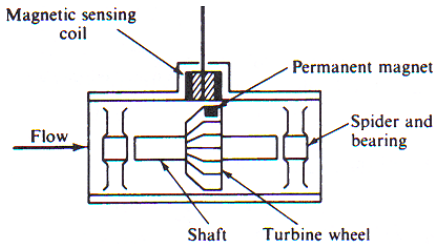
Turbine Meter

The basic equation for flow through a turbine meter is:

$$V = KN; V = Volume; K = Volume\ per\ pulse; N = number\ of\ pulses$$

The average flow rate is equal to the total volume divided by the time interval.

$$Q_{avg} = \frac{V}{\Delta t} = K\frac{N}{\Delta t}$$



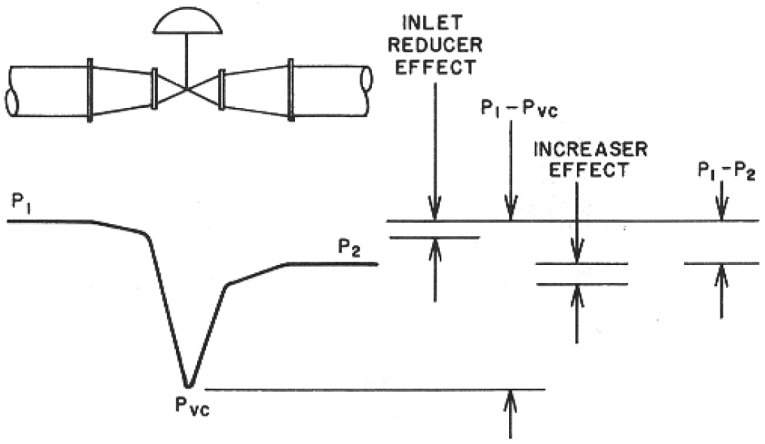
But $f = \frac{N}{\Delta t}$ is the number of pulses per unit time…

$$Q_{avg} = Kf$$

Sample problem: A turbine meter has a K value of 1.22 in^3 per pulse. A: Determine the liquid volume transferred for a pulse count of 6400. B: Determine the flow rate, if each pulse has a duration of 40 seconds. C: What is the totalized flow after 15 minutes?
A: Liquid volume $V = KN$ $V = \left(1.22in^3\right)\left(6400\right) = 7808in^3$ $Gallons = 7808in^3 * \frac{1gal}{231in^3} = 33.8\ gal$ B: Flow Rate $Q = \frac{V}{\Delta t}$ $Q = \frac{7808in^3}{40\sec} = \frac{195.2in^3}{\sec}$ $Q = \frac{195.2in^3}{\sec} * \frac{60\sec}{1\min} * \frac{1gal}{231in^3} = 50.7\frac{gal}{\min}$ C: Totalized flow after 15 minutes $Q = 50.7\frac{gal}{\min} * 15\min = 760.5gal$

CONTROL VALVE SIZING

Note: The* Fisher Control Valve Handbook *is needed for C_v reference.
Read Chapter 5 on Control Valve Selection. All variables are discussed in detail. For sizing, we will keep the equations simple and to the point. Later in this guide, we will discuss how to use the *Fisher Control Valve Handbook* (FCVH) 4th Edition as a reference for different portions of the CSE examination. ISA also offers video tape training in detail on control valve sizing and selection, Control Valves and Actuators Series. The manual to accompany the videos is *Control Valves and Actuators - Manual*, ISBN: 978-1-55617-183-3.



Basic equation for liquid flow

$$q = \left(N_1 F_p C_v\right)\sqrt{\frac{\Delta p}{G_f}}\ ;\ \textit{Note: } N_1 \textit{ = always equal to 1 for psia}$$

Basic equation for gas flow

$$q = \left(N_1 N_7 F_p C_v P_1 Y\right)\sqrt{\frac{x}{G_f T_1 Z}};\ \textit{Note: } N_1 \textit{ = alwys equal to 1 for psia, } N_7 = 1360$$

Basic equation for steam flow

$$w = \left(N_1 N_6 F_p C_v Y\right)\sqrt{x P_1 \gamma_1};\ \textit{Note: } N_1 \textit{ = alwys equal to 1 for psia, } N_6 = 63.3$$

where,

$G_f = Specific\ gravity,\ for\ gas \dfrac{(molecular\ weight\ of\ gas)}{28.97(is\ the\ \text{M.W.}\ of\ air)}$

$C_v = Valve\ sizing\ coefficient$

$F_k = Ratio\ of\ specific\ heat\ factors$

$F_p = Piping\ geometric\ factor$

$K_1 = Inlet\ velocity\ head\ loss\ coefficient$

$K_2 = Outlet\ velocity\ head\ loss\ coefficient$

$K_i = Inlet\ head\ loss\ coefficient;\ K_1 + K_{B1}$

$K_{B1} = Inlet\ Bernoulli\ coefficient$

$K_{B2} = Outlet\ Bernoulli\ coefficient$

$\Sigma K = K_1 + K_2 + K_{B1} - K_{B2}$

$N_1 = 1.00\ (for\ psia;\ equation\ constant\ see\ the\ FCVH,\ Chapter\ 5)$

$N_6 = 63.3\ (for\ lb/h;\ equation\ constant\ see\ the\ FCVH,\ Chapter\ 5)$

$N_7 = 1360\ (for\ scfh;\ equation\ constant\ see\ the\ FCVH,\ Chapter\ 5)$

$N_9 = 7320\ (for\ scfh;\ equation\ constant\ see\ the\ FCVH,\ Chapter\ 5)$

$\Delta p = Pressure\ in\ psid\ across\ the\ valve$

$P_1 = Inlet\ pressure\ (psi\ absolute)$

$q = Volumetric\ Flow\ (in\ gpm\ for\ liquid\ or\ scfh\ for\ gas)$

$T_1 = Fluid\ operating\ temperature\ (psi\ absolute);\ (reference\ temp\ in\ {}^{o}\text{F}+460)$

$w = Volumetric\ flow\ (in\ pounds\ per\ hour)$

$x = Ratio\ of\ delta\ pressure\ to\ inlet\ pressure\ absolute$

$Z = Fluid\ compressibility$

$\gamma_f = Specific\ weight\ of\ the\ steam\ or\ vapor\ in\ pounds\ per\ cubic\ foot\ (operating\ cond.)$

Control Valve Application Comparison Chart

Valve Type	Characteristic and Rangeability	Uses on slurries, Dirty solid bearing fluids	Relative Cost	Rating as Control Valve
Globe body with characterized plug or cage Sizes from needle up to 24 inches	Equal percentage or linear Max 50:1 Approx. 35:1 for needle	Very poor, can be constructed of corrosion resistant materials	High, very high in larger sizes	Excellent; any desired characteristic can be designed into this type valve
Ball valve availability up to 42 inches	Equal percentage Approx. 50:1 Ball can be characterized	Reasonably good, can be constructed of corrosion resistance materials	Medium	Excellent, if characteristic is suitable
Butterfly valve availability up to 150 inches	Equal percentage or linear Approx. 30:1 (some can characterized for quick opening)	Poor, a variety of material for construction available	Lowest cost for large size valves	Good, if characteristic is suitable
Saunders valve availability up to 20 inches	Approx. Linear 3:1 conventional 15:1 dual range	Very good, available with liner to resist corrosion	Medium	Conventional is poor; dual range is fair. Use only when ability is needed to handle dirty flow
Pinch valve availability up to 24 inches	Approx. Linear 3:1 to 15:1, depending on type	Excellent, several materials available to resist corrosion	Low	Poor to fair. Use only when ability is needed to handle dirty flow

Control Valve for Liquid

The basic equation for liquid flow through a control valve is:

$$q = \left(N_1 F_p C_v\right)\sqrt{\frac{\Delta p}{G_f}} \; ; \; \textit{Note: } N_1 \textit{ = always equal to 1 for psia}$$

Solving for C_v we get:

$$C_v = \frac{q}{\left(N_1 F_p\right)\sqrt{\frac{\Delta p}{G_f}}} \; ; \; \textit{Note: } N_1 \textit{ = always equal to 1 for psia}$$

$$F_p = \left[1 + \frac{\Sigma K}{890}\left(\frac{C_v}{d^2}\right)^2\right]^{-1/2} \; ; \; \textit{Note: } F_p \textit{ = piping geometry factor}$$

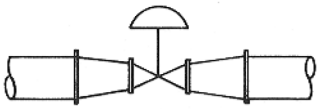
The piping geometry factor covers elbows and reducing fittings attached to each side of the valve body. See the *Fisher Control Valve Handbook* Chapter 5 use of F_p.

Now the equation becomes:

$$C_v = \frac{q}{F_p\sqrt{\frac{\Delta p}{G_f}}}$$

IMPORTANT NOTE:
If you work the Fisher globe valve example in the FCVH, you will find they are using data from the second edition, not the third edition. The example is not incorrect for a class 300 valve. It only appears incorrect with the third edition data at hand. ***Here is an example.***

Sample problem: We will now assume an 8-inch pipe connected to a Globe Valve, with the following service, Liquid Propane. Size the equal percentage valve for the following criteria.



q = 800 gpm	T_1 = 70°F	G_f = 0.5	ΔP = 25 psi
P_1 = 300 psig; 314.7 psia	P_2 = 300 psig; 314.7 psia	Pv = 124.3	Pv = 616.3 psia

A: Find the approximate C_v, this needed to find F_p (for now set to $F_p = 1$).

$$C_v = \frac{q}{F_p\sqrt{\frac{\Delta p}{G_f}}}$$

$$C_v = \frac{800}{\sqrt{\frac{25}{0.5}}} = 113.13$$ *Note: If piping were the same size as the valve, we're done.*

From FCVH chapter 5, we find a 3" Globe Valve (equal percentage) has a maximum C_v of 136 at full open. Now we will plug this C_v into the piping geometry equation to get the installed valve C_v.

$$\Sigma K = K_1(\textit{the entry factor}) + K_2(\textit{the exit factor})$$

$$\sum K = K_{1+2} = 1.5\left(1 - \frac{d^2}{D^2}\right)^2$$ *Note:* $K_1 + K_2 = (0.5 + 1)\left(1 - \frac{d^2}{D^2}\right)^2$ *same size piping*

$$\sum K = K_{1+2} = 1.5\left(1 - \frac{3^2}{8^2}\right)^2 = 1.11$$

$$F_p = \left[1 + \frac{\sum K}{890}\left(\frac{C_v}{d^2}\right)^2\right]^{-1/2}$$ *Note: F_p = piping geometry factor.*

$$F_p = \left[1 + \frac{1.11}{890}\left(\frac{136}{3^2}\right)^2\right]^{-1/2} = 1.2848^{-1/2} = 0.8822$$

Find the corrected C_v for the installed valve.

$$C_v = \frac{q}{F_p\sqrt{\frac{\Delta p}{G_f}}}$$

$$C_v = \frac{800}{(0.8822)\sqrt{\frac{25}{0.5}}} = \frac{800}{6.238} = 128.24 \text{ or } 129$$

This shows a 3″ valve is too small; it will require the 4″ with the maximum $C_v = 224$.

$$\% = \frac{129}{224} = 57.6\% \text{ of maximum } C_v \text{ and about } 76\% \text{ open}$$

In the FCVH fourth edition, a type ED valve is used in the table and a 3″ would be correct with a C_v of 136, but it is too small. The valve would be (129/136) or 95% of maximum C_v, and you might not get the required flow through the valve for throttling.

Remember, valves start choking at about 75% throttle, so size your C_v to fit at about 50% maximum C_v. ***Valves in throttling services should be sized for 200% operational flow, this allows the valve to open up further and correct for process upset rapidly.***

Control Valve for Gas

The basic equation for gas flow through a control valve is:

$$q = \left(N_1 N_7 F_p C_v P_1 Y\right)\sqrt{\frac{x}{G_f T_1 Z}}; \text{ Note}: N_1 = \text{always equal to } 1 \text{ for psia}, N_7 = 1360$$

$$C_v = \frac{q \text{ (in scfh)}}{1360 F_p P_1 Y\sqrt{\frac{x}{G_f T_1 Z}}}; \text{ Note}: \text{for volumetric flow units}$$

where,

$Y = 1 - \left(\frac{x}{3F_k x_{TP}}\right)$; expansion factor, velocity down stream will be greater than upstream

$F_k = \frac{k}{1.4}$; ratio of specific heats factor

k = ratio of specific heats

$x = \frac{\Delta P}{P_1}$; pressure drop ratio of ΔP to inlet pressure P_1

$Y = 1 - \left(\frac{x}{3F_k x_{TP}}\right)$; expansion factor, must be between 1.0 and 0.667

x_T = pressure drop ratio required to produce maximum flow through the valve when $F_k = 1.0$.(x_T can be found in valve coefficients table)

$x_{TP} = \frac{x_T}{F_p^2}\left[1 + \frac{x_T K_i}{N_5}\left(\frac{C_v}{d^2}\right)^2\right]^{-1}$; pressure drop ratio factor with installed fitting attached

where,

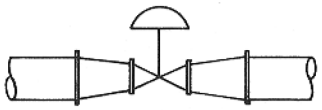
$F_p = \left[1 + \frac{\sum K}{890}\left(\frac{C_v}{d^2}\right)^2\right]^{-1/2}$; piping geometry factor

$K_i = K_1 + K_{b1}$; inlet head loss coefficient

$K_1 = 0.5 * \left(1 - \frac{d^2}{D^2}\right)^2$ *and* $K_2 = 1 * \left(1 - \frac{d^2}{D^2}\right)^2$

$K_{B1} = 1 - \left(\frac{d^2}{D^2}\right)^2$ and $K_{B2} = 1 - \left(\frac{d^2}{D^2}\right)^2$ OR $K_{B1} = 1 - \left(\frac{d}{D}\right)^4$ and $K_{B2} = 1 - \left(\frac{d}{D}\right)^4$; Bernoulli coefficients

Sample problem: We will now assume 6″ inch pipe connected to a Globe Valve, with the following service, Natural Gas. Size the equal percentage valve for the following criteria.



q = 800,000 scfh	T_1 = 60°F = 520°R	G_g= 0.60	ΔP = 150 psi
P_1 = 400 psig; 414.7 psia	P_2 = 250 psig; 264.7 psia	k = 1.31	M = 17.38

A: For use of molecular weight substitute M for G_f and N_9 for N_7 and set N_9 to 7320. We will use specific gravity and N_7 = 1360.

$$C_v = \frac{q\ (in\ scfh)}{1360 F_p P_1 Y \sqrt{\frac{x}{G_f T_1 Z}}}$$; *Note:* for volumetric flow units

B: First find the approximate valve size and C_v for formulas. Set Fp = 1, Y = 1, Z = 1.

$$x = \frac{\Delta P}{P_1} = \frac{150}{414.7} = 0.362;\ F_k x_T = \left(\frac{1.31}{1.4}\right)0.68 = 0.636$$

$$C_v = \frac{q\ (in\ scfh)}{1360 P_1 \sqrt{\frac{x}{G_f T_1}}} = \frac{800,000}{1360(414.7)\sqrt{\frac{0.362}{(0.60)(520)}}} = 41.64\ or\ 42$$

When the pressure differential ratio x reaches a value of F_K x_T. The limiting value of x is defined as the critical differential pressure ratio. The value of x used in any of the sizing equations and in the relationship for Y shall be held to this limit even if the actual pressure differential ratio is greater. Thus, the numerical value of Y may range from 0.667, when $x = F_K$ x_T, to 1.0 for very low differential pressures. The x_T comes from the valve coefficient tables in the FCVH. (recalculate if done, if not move forward for now).

From the FCVH valve coefficients table we see a 3″ valve with the C_v = 136.

C: Calculate for piping geometric factors. Inlet = 6″ and Outlet=6″ schedule 40 pipe.

$$\Sigma K = K_1 + K_2 + K_{B1} - K_{B2}$$

$$K_1 = 0.5 * \left(1 - \frac{d^2}{D^2}\right)^2 = 0.5 * \left(1 - \frac{2^2}{6^2}\right)^2 = 0.395$$

$$K_2 = 1 * \left(1 - \frac{d^2}{D^2}\right)^2 = 1 * \left(1 - \frac{2^2}{6^2}\right)^2 = 0.79$$

$$K_{B1} = 1 - \left(\frac{d^2}{D^2}\right)^2 = 1 - \left(\frac{2}{6}\right)^4 = 0.8888$$

$$K_{B2} = 1 - \left(\frac{d^2}{D^2}\right)^2 = 1 - \left(\frac{2}{6}\right)^4 = 0.8888$$

Sum resistance coefficients and Bernoulli coefficients and get piping geometry factor:

$$\Sigma K = 0.395 + 0.790 + 0.0123 - 0.0123 = 1.185$$

$$F_p = \left[1 + \frac{\Sigma K}{890}\left(\frac{C_v}{d^2}\right)^2\right]^{-1/2} = \frac{1}{\sqrt{1 + \frac{1.185}{890}\left(\frac{59.7}{2^2}\right)^2}} = 0.878$$

Find the pressure drop ratio for the installed fitting attached to the valve.

$$K_i = K_1 + K_{B1} = 0.395 + 0.8888 = 1.2838$$

From the tables in the FCVH: N_5=1000 and x_T=0.69

$$x_{TP} = \frac{x_T}{F_p^2}\left[1 + \frac{x_T K_i}{N_5}\left(\frac{C_v}{d^2}\right)^2\right]^{-1} = \frac{x_T}{F_p^2\left[1 + \frac{x_T K_i}{N_5}\left(\frac{C_v}{d^2}\right)^2\right]}$$

$$x_{TP} = \frac{0.69}{0.878^2\left[1 + \frac{(0.69)(1.2838)}{1000}\left(\frac{59.7}{2^2}\right)^2\right]} = 0.747$$

D: Find the expansion factor Y, it must be between 1.0 and 0.667

$$F_k = \frac{1.31}{1.4} = 0.936; \text{ratio of specific heats factor}$$

$$Y = 1 - \left(\frac{x}{3F_k x_{TP}} \right) = 1 - \left(\frac{0.362}{3(0.936)(0.747)} \right) = 0.827$$

$$C_v = \frac{q\ (in\ scfh)}{1360 F_p P_1 Y \sqrt{\frac{x}{G_f T_1}}} = \frac{800,000}{1360(0.878)(414.7)(0.827)\sqrt{\frac{0.362}{(0.60)(520)}}} = 57.35\ or\ 57$$

Since a 2 inch valve has a C_v of 59.7 the valve needs to be a 3" valve.

$$\% = \frac{57}{136} = 42\%\ \ of\ maximum\ C_v\ and\ about\ 64\%\ open$$

$$C_g = 40\, C_v \sqrt{x_T};\ \ if\ needed\ to\ convert\ C_v\ to\ C_g\ (\text{as in the FCVH})$$

Control Valve for Steam

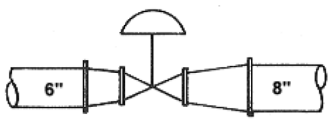
The basic equation for vapor or steam flow through a control valve is:

$w = \left(N_1 N_6 F_p C_v Y \right) \sqrt{x P_1 \gamma_1}$; Note: N_1 = always equal to 1 for psia, N_6 = 63.3

$C_v = \frac{w(inlb/h)}{63.3 F_p Y \sqrt{x P_1 \gamma_1}}$; Note: for mass flow units in pounds per hour

Sample problem: We will now assume 6 inch pipe in and 8 inch pipe out, connected to a type ED Globe Valve, with the following service, Process Steam. Size a linear valve for the following criteria.

Note: γ_1 and k can be found in the steam tables in the FCVH.



q = 125,000 lb/h	T_1 = 500°F = 960°R	G_g = 0.60	ΔP = 250 psi
P_1 = 500 psig; 514.7 psia	P_2 = 250 psig; 264.7 psia	k = 1.28	γ_1 = 1.0434

A: First find the approximate valve size and C_v for formulas. Set Fp = 1, Y = 1.

$$C_v = \frac{w\,(in\ lb/h)}{63.3F_pY\sqrt{xP_1\gamma_1}} = \frac{125,000}{63.3(1)(1)\sqrt{(0.49)(514.7)(1.0434)}} = 121.7\ or\ 122$$

When the pressure differential ratio x reaches a value of $F_k x_T$, The limiting value of x is defined as the critical differential pressure ratio. The value of x used in any of the sizing equations, and in the relationship for Y, shall be held to this limit, even if the actual pressure differential ratio is greater. Thus, the numerical value of Y may range from 0.667, when $x = F_k x_T$, to 1.0 for very low differential pressures. The x_T comes from the valve coefficient tables in the FCVH. (Recalculate if done; if not, move forward for now.)

The FCVH shows a 3" with a C_v = 136, but we want to throttle around 50%, so a 4" with the C_v of 236 will be selected.

B: Calculate for piping geometric factors. Inlet = 6" and Outlet = 8" schedule 40 pipe.

$$K_1 = 0.5*\left(1-\frac{d^2}{D^2}\right)^2 = 0.5*\left(1-\frac{4^2}{6^2}\right)^2 = 0.154$$

$$K_2 = 1*\left(1-\frac{d^2}{D^2}\right)^2 = 1*\left(1-\frac{4^2}{8^2}\right)^2 = 0.5625$$

$$K_{B1} = 1-\left(\frac{d^2}{D^2}\right)^2 = 1-\left(\frac{4}{6}\right)^4 = 0.8025$$

$$K_{B2} = 1-\left(\frac{d^2}{D^2}\right)^2 = 1-\left(\frac{4}{8}\right)^4 = 0.9375$$

Sum resistance coefficients and Bernoulli coefficients and get piping geometry factor:

$$\Sigma K = K_1 + K_2 + K_{B1} - K_{B2}$$

$$\Sigma K = 0.154 + 0.5625 + 0.8025 - 0.9375 = 0.7355$$

$$F_p = \left[1+\frac{\Sigma K}{890}\left(\frac{C_v}{d^2}\right)^2\right]^{-1/2} = \frac{1}{\sqrt{1+\frac{0.7355}{890}\left(\frac{236}{4^2}\right)^2}} = 0.920$$

C: Find the pressure drop ratio for the installed fitting attached to the valve.

$$K_i = K_1 + K_{b1} = 0.154 + 0.8025 = 0.9565$$

From the tables in the FCVH: $N_5 = 1000$ and $x_T = 0.69$

(NOTE: Because a 4-inch valve is to be installed in a 6-inch inlet pipe and 8-inch outlet pipe respectfully, the x_T term must be replaced by x_{TP} in the Y expansion factor formula below. This is not necessary if the pipe inlet and the pipe outlet are the same size; in other words, there are no reducer fittings are being used.)

$$x_{TP} = \frac{x_T}{F_p^2}\left[1 + \frac{x_T K_i}{N_5}\left(\frac{C_v}{d^2}\right)^2\right]^{-1} = \frac{x_T}{F_p^2\left[1 + \frac{x_T K_i}{N_5}\left(\frac{C_v}{d^2}\right)^2\right]}$$

$$x_{TP} = \frac{0.69}{0.9097^2\left[1 + \frac{(0.69)(0.9565)}{1000}\left(\frac{236}{4^2}\right)^2\right]} = 0.729$$

D: Find the expansion factor Y, it must be between 1.0 and 0.667

$$F_k = \frac{1.28}{1.4} = 0.914; \text{ratio of specific heats factor}$$

$$[x = \Delta P/P_1 = 0.49] < [x = F_k x_T]; \textit{subsitute } x_{TP} \textit{ for } x_T$$

$$[x = \Delta P/P_1 = 0.49] < [x = F_k x_{TP} = (0.914)(0.729) = 0.6663]$$

Pressure ratio is smaller than critical limits, so we will use $x = 0.486$.

$$Y = 1 - \left(\frac{x}{3F_k x_{TP}}\right) = 1 - \left(\frac{0.486}{3(0.914)(0.729)}\right) = 0.757$$

Note: Replace x_{TP} with x_T if pipe size in and out are the same size as valve

$$C_v = \frac{w\,(in\ lb/h)}{63.3F_p Y\sqrt{xP_1\gamma_1}} = \frac{125{,}000}{63.3(0.920)(0.757)\sqrt{(0.486)(514.7)(1.0434)}} = 175.5 \textit{ or } 175$$

This shows a 4″ valve is the correct size.

$$\% = \frac{175}{224} = 78.1 \ \ of \ \ maximum \ C_v \ \ and \ \ about \ 88\% \ open$$

Note: this valve is borderline of being at maximum flow, valves start choking at about 75%. It is correct for the test, but in real life I would go with a 6 inch valve for more capacity.

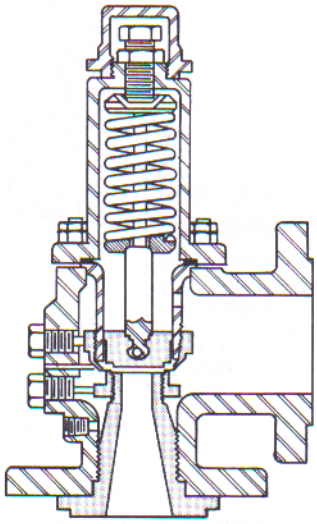
$$C_g = 40\, C_v \sqrt{x_T}; \ \ if \ \ needed \ to \ convert \ C_v \ to \ C_g \ (\text{as in the FCVH})$$

$$\gamma_1 = \text{Reciprocal of specific volume } \frac{1}{\left(ft^3 / lb\right)}; \left(\frac{ft^3}{lb} \text{ as in the FCVH steam tables}\right)$$

Pressure Relief Valve Sizing

Gas and Vapor

The basic equation for gas and vapor flow through a pressure relief valve is:



ASME VIII Code equation

$$W = K_b CKAP\sqrt{\frac{M}{TZ}}$$

Solving for area gives:

$$A = \frac{W}{K_b CKP\sqrt{\frac{M}{TZ}}}$$

where,
W = required relieving rate, mass flow
T = relieving temperature, absolute
Z = compressibility factor
M = molecular weight
C = gas constant = a function of ratio of specific heats

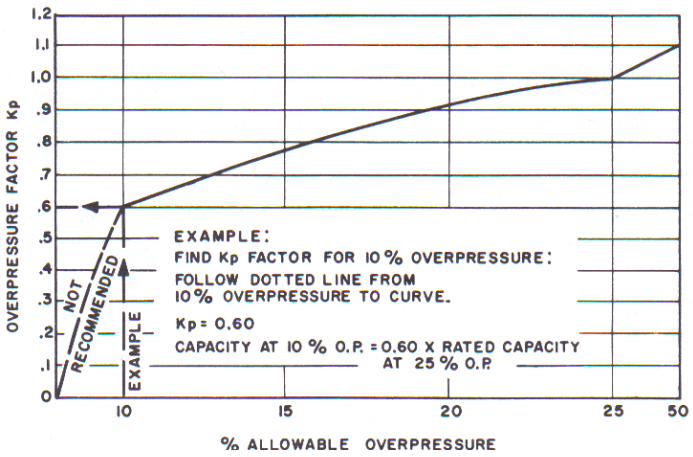
$$C = 520\sqrt{k\left(\frac{2}{k+1}\right)^{\frac{k+1}{k-1}}}$$

k = specific heats ratio
K_b = back-pressure correction factor, dimensionless (0.62)
P = (Set pressure x 1.10) plus atmospheric pressure (psia)

Liquid

The basic equation for liquid flow through a pressure relief valve is:

$$Q_g = 27.2A\sqrt{\frac{P_d}{G}}K_p K_u K_w$$



Solving for area gives:

$$A = \frac{Q_g}{27.2\sqrt{\frac{P_d}{G}}K_p K_u K_w}$$

where,
Q_g = required relieving rate of liquid, gpm
A = actual nozzle area of valve, square inches
P_d = inlet (relieving) pressure, less any constant back pressure, PSID
G = specific gravity of liquid flowing conditions relative to water at 60°F
K_p = overpressure correction for liquid (0.60)
K_w = variable or constant back-pressure factor for bellows sealed valves only
K_u = viscosity correction factor

Steam

The basic equation for liquid flow through a pressure relief valve is:

$$W_s = 51.5AKPK_bK_{sh}$$

Solving for area gives:

$$A = \frac{W_s}{51.5KPK_bK_{sh}}$$

where,
W_s = required steam capacity, lbm/hr
A = required nozzle area of valve, square inches
P = relieving pressure = [set pressure x (1 + accumulation allowed by code having jurisdiction)] + atmospheric pressure (PSIA)
K_b = vapor gas correction for constant back pressure above critical pressure
K_{sh} = superheat correction factor (see manufacturer's tables)

Note: See Appendix A for worked examples using ASME code. Refer to manufacturer's examples for using manufacturer-specific material and formulas.

Excerpts from ASME Unfired Pressure Vessel Code

UG-125(c)—All unfired pressure vessels other than unfired steam boilers shall be protected by pressure-relieving devices that will prevent the pressure from rising more than 10% above the maximum allowable working pressure, except when the excess pressure is caused by exposure to fire or other unexpected sources of heat.

UG-125(d)—Where an additional hazard can be created by exposure of a pressure vessel to fire or other unexpected sources of external heat (for example, vessels used to store liquefied flammable gases), supplemental pressure-relieving devices shall be installed to protect against excessive pressure. Such supplemental pressure-relieving devices shall be capable of preventing the pressure from rising more than 20% above the maximum allowable working pressure of the vessel. A single pressure-relieving device may be used to satisfy the requirements of this paragraph and (c), provided it meets the requirements of both paragraphs.

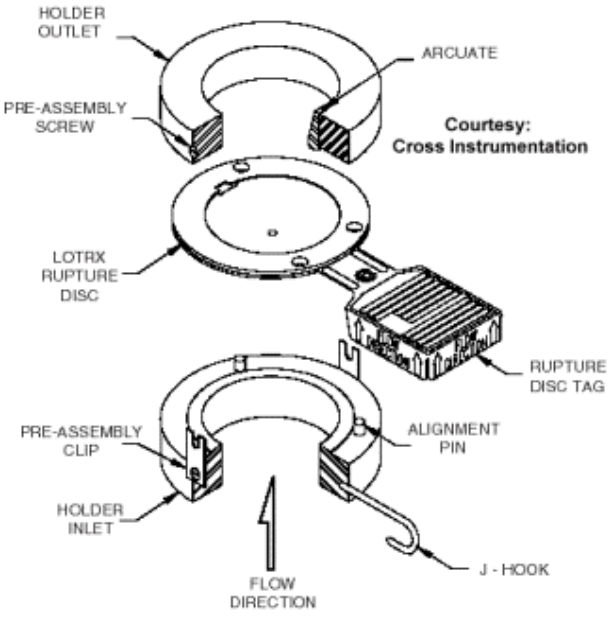
UG-133(a)—When safety or relief valves are provided, they shall be set to blow at a pressure not exceeding the maximum allowable working pressure of the vessel at the operating temperature, except as permitted in (b). If the capacity is supplied in more than one safety or relief valve, only one valve need be set to open at a pressure not exceeding the maximum allowable working pressure of the vessel; the additional valves may be set to open at a higher pressure, but not to exceed 105% of the maximum allowable working pressure of the vessel. See Paragraph UG-125(c).

UG-133(b)—Protective devices permitted in Paragraph UG-125(d) as protection against excessive pressure caused by exposure to fire or other sources of external heat shall be set to operate at a pressure not in excess of 110% of the maximum allowable working pressure of the vessel. If such a device is used to meet the requirements of both Paragraphs UG-125(c) and UG-125(d), it shall be set to operate at not over the maximum allowable working pressure.

UG-133(f)—The set pressure tolerances, plus or minus, of safety or relief valves, shall not exceed 2 PSI (13.8 kPA) for pressures up to and including 70 PSIG (483 kPa), and 3% for pressures above 70 PSIG (483 kPa).

RUPTURE DISK SIZING

The ASME Code requires that when a rupture disk is used as the primary relief device, it must be sized to prevent the pressure from rising above 110% of the MAWP (UG-125(c)). If used as a secondary relief device or as multiple relief devices, the size must prevent the pressure from rising above 116% of the MAWP (UG-125(c)(1)). If used as a supplementary relief device for hazards external to the protected vessel or system, the size must prevent the pressure from rising above 121% of the MAWP (UG-125(c)(2)).



Note: Where rupture disks are installed upstream of a relief valve, the rupture disc is normally the same size as the relief valve inlet nozzle.

Rupture Disk Sizing Example 1

Sample problem: We will size a rupture disk for the following service, **LIQUID**. Size the rupture disk for the following criteria. Application: (Primary Relief).			
Q = 1500 gpm (required)	Vessel MAWP = 45 psig	$G_f(\rho)$= 0.85	
Back Pressure = 5 psig			
Use 10% over-pressure as permitted by ASME code. Volumetric flow rate, conventional U.S. units: $A = \frac{Q}{186}\sqrt{\frac{\rho}{P}} = \frac{1500}{186}\sqrt{\frac{(.85)(62.37)}{(1.1)(45-5)}} = 8.85$ sq. inch. This should be all you need for the CSE Examination. Use manufacturer's catalog for the actual disk to order your application.			

where,
A = disk area.
C = gas constant = a function of ratio of specific heats
k = specific heats ratio
M = molecular weight
P = inlet pressure, gauge (psig)

P_1 = inlet pressure, absolute (psia).
P_2 = outlet pressure, absolute (psia)
T = relieving temperature, absolute
V_A = volumetric flow rate, actual conditions
V_S = volumetric flow rate, standard conditions
Z = compressibility factor
ρ = specific gravity

Rupture Disk Sizing Example 2

Sample problem: We will size a rupture disk for the following service, **GAS (Air)**. Size the rupture disk for the following criteria. Application: (Primary Relief).			
Q = 5000 scfm (required)	Vessel MAWP = 150 psig	$G_f(\rho)$= 0.85	K = 1.4
Back Pressure = 20 psig	Flow temperature = 250°F	M = 29	Z = 1

Use 10% over-pressure as permitted by ASME code. Flow pressure ratio:

$$\frac{P_2}{P_1} = \frac{20 + 14.7}{(1.1)(150) + 14.7} = 0.193$$

Critical pressure ratio:

$$r_c = \left[\frac{2}{k+1}\right]^{\frac{k}{k-1}} = \left[\frac{2}{1.4+1}\right]^{\frac{1.4}{1.4-1}} = 0.528$$

P2/P1 is less than r_c, therefore the flow will be sonic. For conventional U.S. units:

$$C = 520\sqrt{k\left[\frac{2}{k+1}\right]^{\frac{k+1}{k-1}}} = 520\sqrt{1.4\left[\frac{2}{1.4+1}\right]^{\frac{1.4+1}{1.4-1}}} = 356$$

Given the required flow in actual cubic feet per minute:

$$A = 9.02\frac{V_A}{C}\sqrt{\frac{M}{ZT}} = 9.02\left(\frac{5000}{356}\right)\sqrt{\frac{29}{(1)(250+460)}} = 25.6 \text{ sq. inch}$$

This should be all you need for the CSE Examination.

Use manufacturer's catalog for the actual disk to order your application.

Rupture Disk Sizing Example 3

Sample problem: We will size a rupture disk for the following service, **GAS (some process)**. Size the rupture disk for the following criteria. Application: (Primary Relief).			
Q = 2000 scfm (required)	Vessel MAWP = 15 psig	$G_f(\rho)$= 0.72	K = 1.26
Back Pressure = 5 psig	Flow temperature = -40°F	M = 29	Z = 0.95

In this case 10% of gauge pressure is less than 3 psi, therefore 3 psi over-pressure is permitted by ASME code.

Flow pressure ratio:

$$\frac{P_2}{P_1} = \frac{5+14.7}{3+15+14.7} = 0.602$$

Critical pressure ratio:

$$r_c = \left[\frac{2}{k+1}\right]^{\frac{k}{k-1}} = \left[\frac{2}{1.26+1}\right]^{\frac{1.26}{1.26-1}} = 0.553$$

P2/P1 is greater than r_c, therefore the flow will be subsonic. For conventional U.S. units:

$$C = 735\sqrt{\frac{k}{k+1}\left[\left(\frac{P_2}{P_1}\right)^{\frac{2}{k}} - \left(\frac{P_2}{P_1}\right)^{\frac{k+1}{k}}\right]} = 735\sqrt{\frac{1.26}{1.26+1}\left[(0.602)^{\frac{2}{1.26}} - (0.602)^{\frac{1.26+1}{1.26}}\right]} = 115.67$$

Given the required flow in standard cubic feet per minute:

$$A = \frac{V_S\sqrt{ZMT}}{3.92CP_1} = \frac{2000\sqrt{(.95)(29)(.72)(460-40)}}{(3.92)(115.67)(3+15+14.7)} = 12.31 \text{ sq. inch.}$$

Use manufacturer's catalog for the actual disk to order your application.

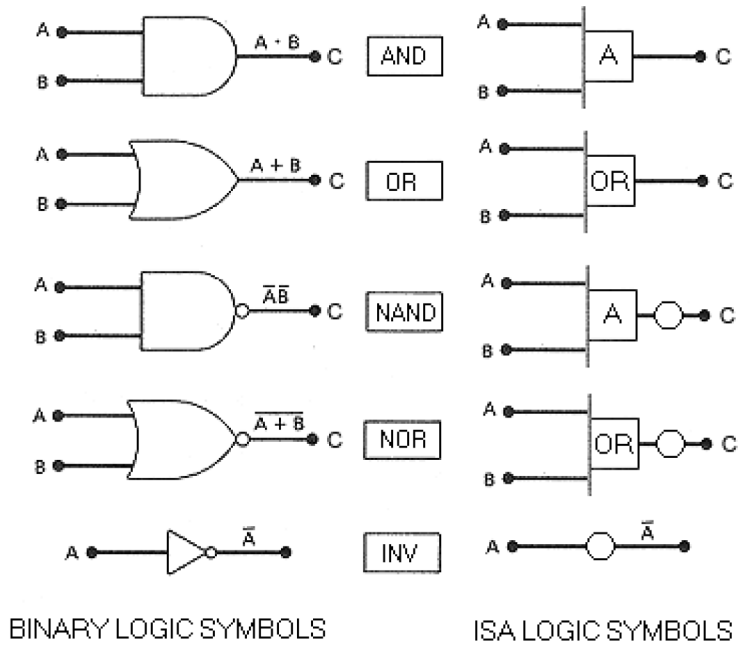
Overview of Discrete Control Subjects

Overview of Digital Logic

Discrete control plays a vital role in the process control industry. Discrete control is used for material handling, lockouts and safeties, indicators, alarms and switching applications. Discrete control usually takes the form of RLL (Relay Ladder Logic) or digital logic combined with some type of mechanical apparatus. The PLC (Programmable Logic Controller) is the workhorse of the industry today and is covered on the CSE Exam with ISA binary logic and Relay Ladder Logic.

Gates and Inverters

Familiarize yourself with the following binary logic table and its functions. The ISA logic is the same in function, although the symbols are slightly different.



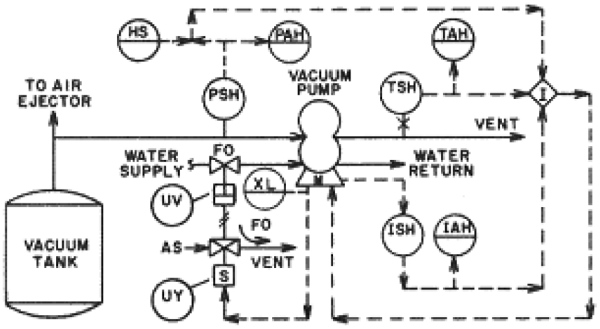
ISA Binary Logic

Familiarize yourself with the previous binary logic table and its functions. The ISA logic is used in the examination. Look at some examples of its use such as in the ISA's "Control Systems Engineer Study Guide" and "ISA-5.2-1976 (R1992) Binary logic Diagrams for Process Operations".

ISA Usage of Binary Logic

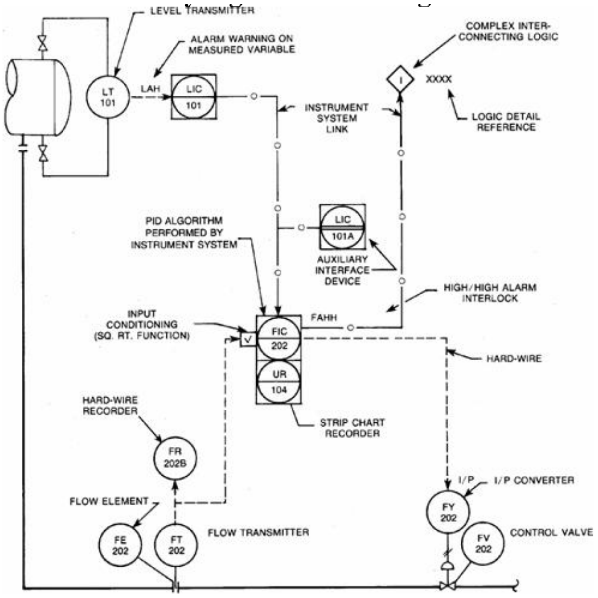
The CSE exam may have a diagram similar to below. Questions will be asked as to the state or outcome of the logic, if certain states occur in the process. Familiarize yourself with this type of logic and control diagram.

Process and Instrumentation Diagram



Control system for standby vacuum pump

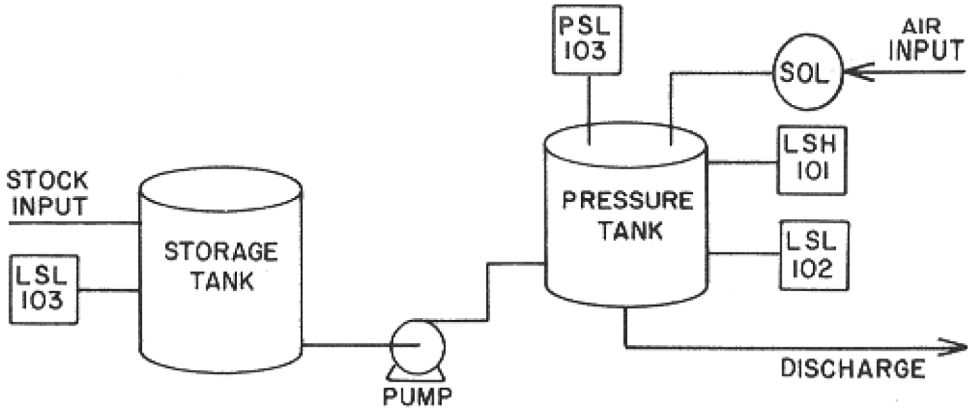
Binary Logic Control Diagram

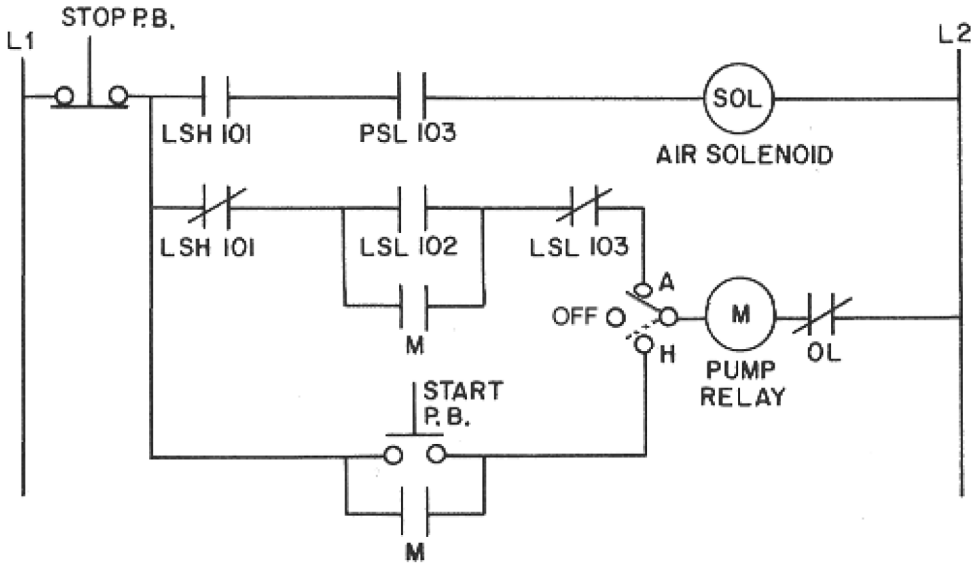


Logic diagram for standby vacuum pump

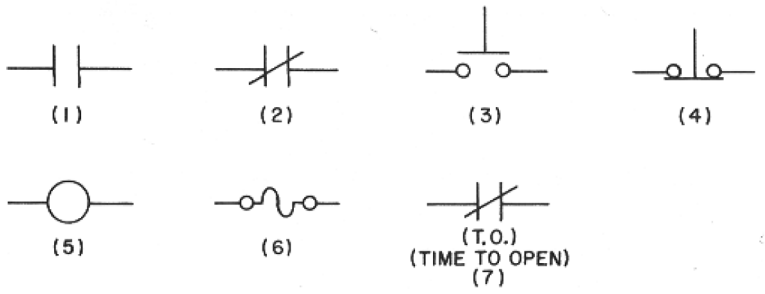
Relay Ladder Logic (RLL)

The CSE exam may have a diagram similar to below. Questions will be asked as to the state or outcome of the logic, if certain states occur in the process. Familiarize yourself with this type of logic and control diagram.





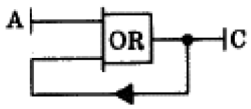
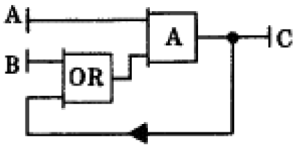
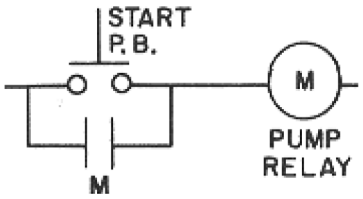
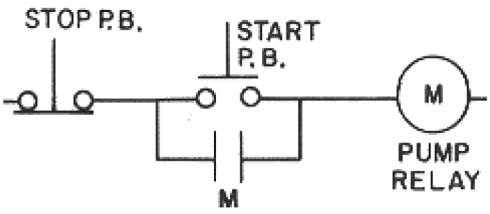
The basic RLL symbols listed below are (1) NO or examine on; (2) NC or examine off; (3) NO button, function such as energize; (4) NC button, function such as de-energize; (5) Coil such as on a relay, solenoid, motor starter; (6) OL, over current protection; (7) timing contact shown in standard contact form.



Sealing Circuits

Two types of sealing circuits can be seen below. The first is an OR gate. Once a signal is applied to the gate's "A" input, the gate seals and stays on until the system power is removed. This would be like a relay being energized and the contact held closed until the relay's power is removed.

The second is like the sealing circuit on a motor control starter. The gate's input "A" is the stop button and the gate's input "B" is the start button. Once input "B" is set to "1" or pushed on, the output "C" stays on until input "A", the stop button, is pressed open and set to "0" or off.

	
	
Equivalent Sealing Circuit	Equivalent Stop/Start Sealing Circuit

Analog Signals and ISA Symbols

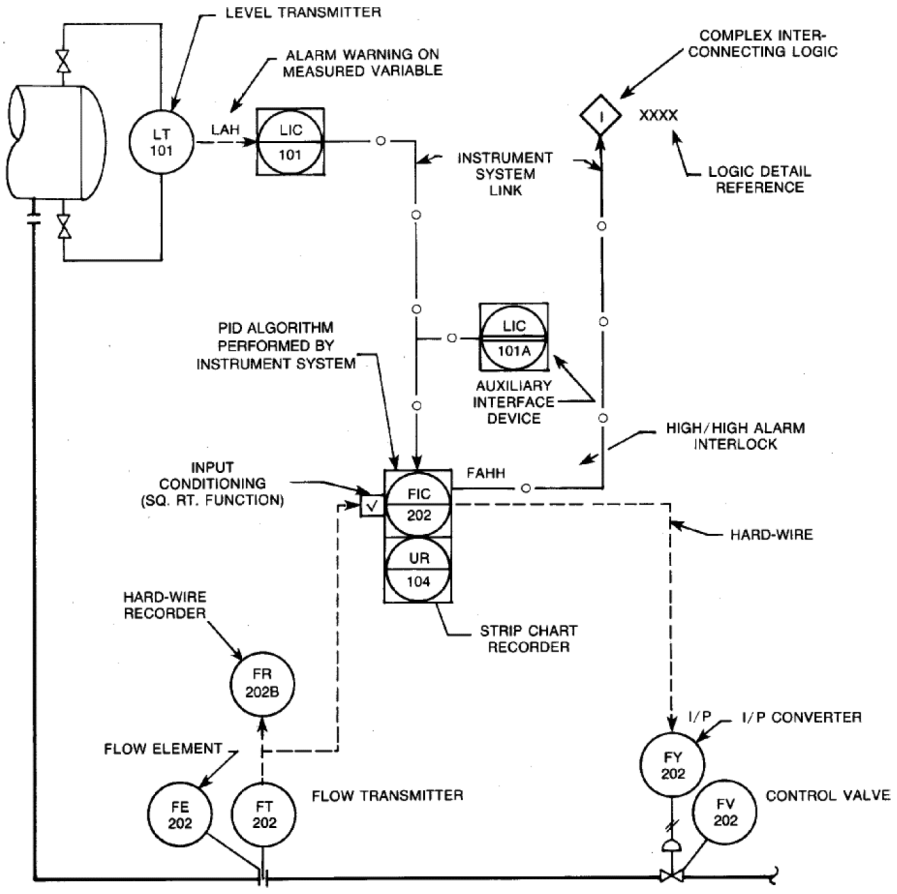
Overview of Analog Signals

On the CSE Exam, there may be a few questions on ISA symbols for electrical and pneumatic circuits. Study the following ISA standards publications:

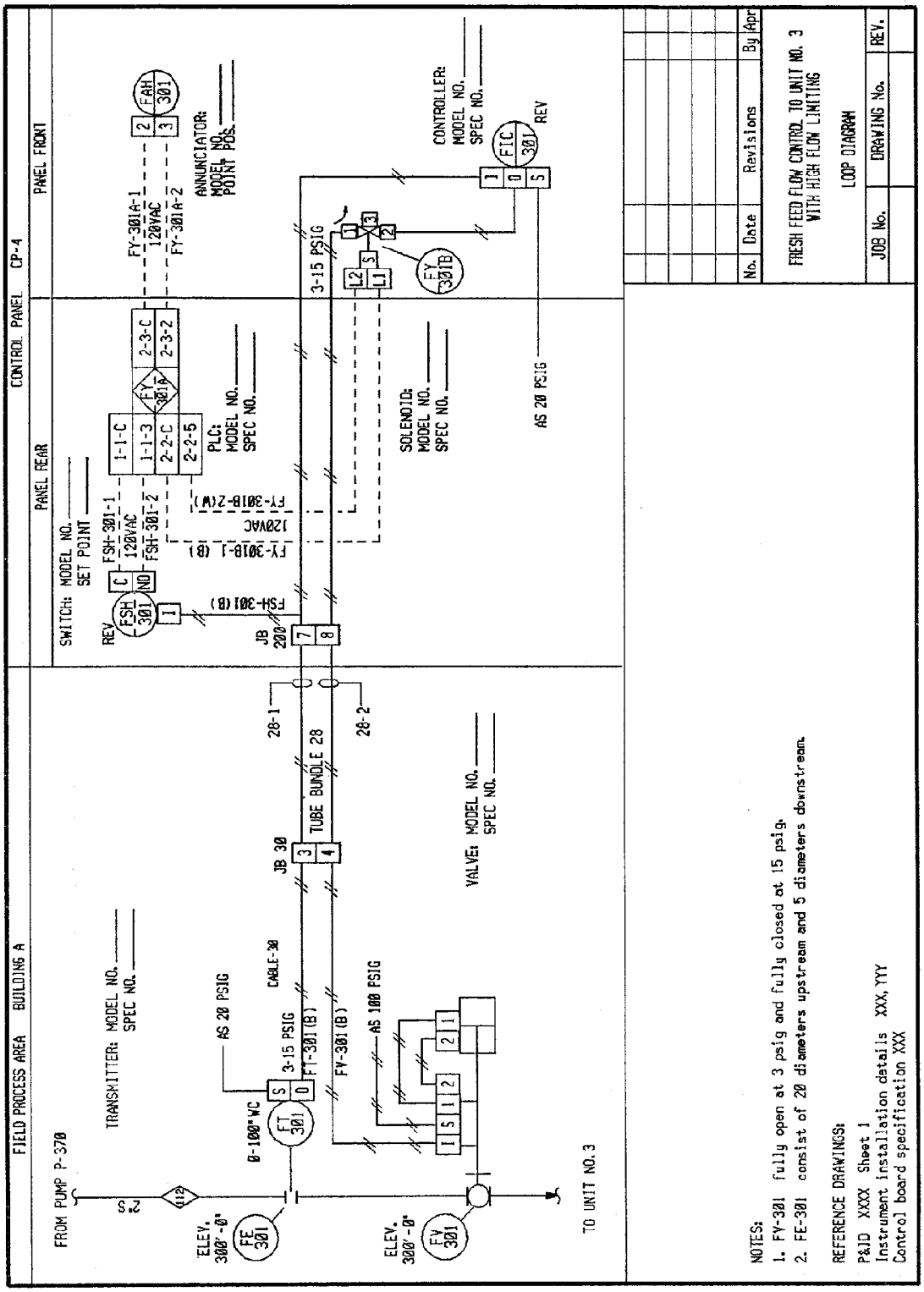
ISA-5.1-1984 (R1992)	Instrumentation Symbols and Identification
ISA-5.2-1976 (R1992)	Binary Logic Diagrams for Process Operations
ISA-5.3-1983	Graphic Symbols for Distributed Control/ Shared Display
ISA-5.4-1991	Standard Instrument Loop Diagrams

I consider these required elements. There are numerous problems dealing with all the above standards. You will be tested on details, so do not feel comfortable with your company's standards. Only the exact ISA Standard is correct. There may be questions from the documentation text, not just symbols.

This is a standard **ISA P&ID** (Piping and Instrumentation Diagram) as might be seen on the CSE Examination.



This is a standard **ISA Instrument Loop Diagram** as might be seen on the CSE Examination. The exam may ask questions related to terminals, symbols and connections.



Overview – Safety Instrumented Systems

Overview of Process Safety and Shutdown

On the CSE Exam there will be a few questions on SIS (Safety Instrumented Systems) and SIL (Safety Integrity Levels). We will discuss some of the calculations and data you may encounter on the test.

SIS (Safety Instrumented Systems)

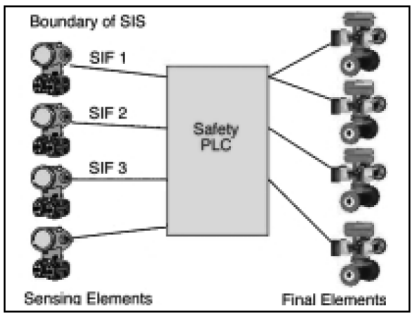
OSHA law incorporates as the guideline that "good engineering practice" will be used in evaluating and engineering safety instrumented systems (SIS). This means that the program follows the codes and standards published by such organizations as the American Society of Mechanical Engineers, American Petroleum Institute, American National Standards Institute, National Fire Protection Association, American Society for Testing and Materials, National Board of Boiler and Pressure Vessel Inspectors, and ISA. Other countries have similar requirements.

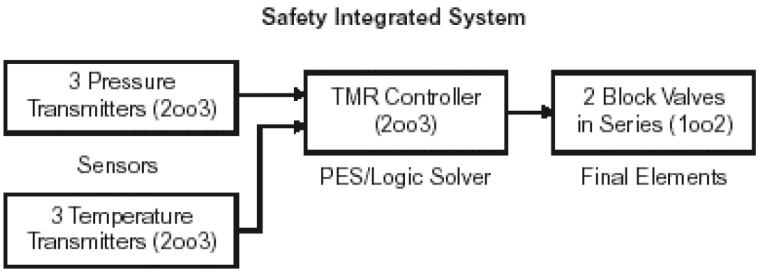
The OSHA approved code standards for the implementation of SIS are ANSI/ISA-84.00.01 (IEC 61511 modified): [For Safety Integrated System Designers, Integrators and Users], and IEC 61508: [For Manufacturers and Suppliers of Devices and Equipment].

IEC-61508 is currently divided into seven parts:
1. General Requirements
2. Requirements for Electrical/Electronic/Programmable Electronic Safety Systems
3. Software Requirements
4. Definitions and abbreviations of terms
5. Guidelines for application of part 1
6. Guidelines for application of parts 2 and 3
7. Bibliography of techniques

IEC-61508 also defines a SIL 4, which is discussed in the Safety Integrity Level section.

NOTE: There is no code required by law, only suggested guidelines to follow.





Voting or (Polling of the System)

It is also important to understand the voting systems, (polling systems), of SIS/SIL rated PLC controllers (Logic Solvers). The following is read X out of X.

1oo1 = one out of one	1oo1D = one out of one with diagnostics
1oo2 = one out of two	1oo2D = one out of two with diagnostics
2oo2 = two out of two	2oo2D = two out of two with diagnostics
2oo3 = two out of three	2oo3D = two out of three with diagnostics

SIF (Safety Instrumented Function)

The safety instrumented function sheet includes the following information:

- Input
  - Type
  - Redundancy
  - Voting Architecture
  - Testing Interval
- Logic Solver Type
- Actuator
  - Type
  - Redundancy
  - Voting Architecture
  - Test Interval
- Final Element
  - Type
  - Redundancy

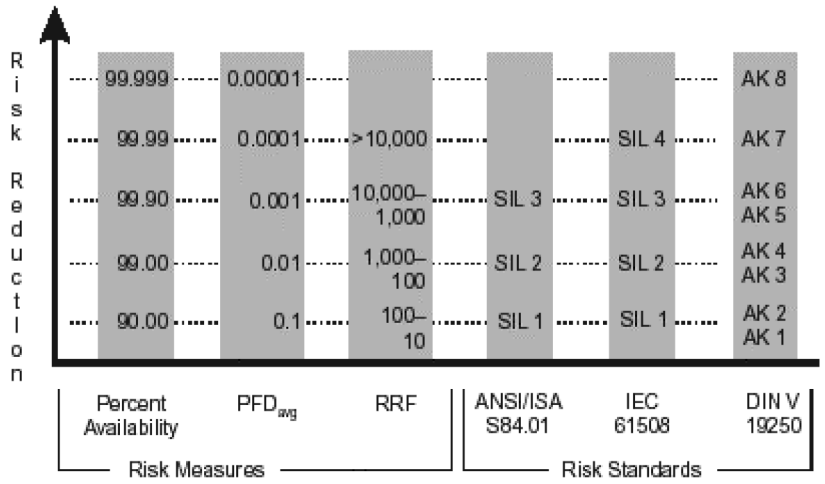
- Voting Architecture
- Testing Interval
- Diagnostic Requirements For All Devices

- Alarms
- Maintenance Provisions
- Bypass Requirements
- Manual ESD Requirements
- SIL Verification
- Predicted Spurious Trip Rate

SIL (SAFETY INTEGRITY LEVEL)

If concluded that an SIS is required, ANSI/ISA-84.00.01 (IEC 61511 modified) and IEC 61508 require that a target SIL be assigned. The assignment of a SIL is a corporate decision based on risk management and risk tolerance philosophy. Safety regulations require that the assignment of SILs should be carefully performed and documented. A qualitative view of SIL has slowly developed over the last few years as the concept of SIL has been adopted at many chemical and petrochemical plants. This qualitative view can be expressed in terms of the impact of the SIS failure on plant personnel and the public or community.

- **"4"** - Catastrophic Community Impact.
- **"3"** - Employee and Community Protection.
- **"2"** - Major Property and Production Protection. Possible injury to employee.
- **"1"** - Minor Property and Production Protection.



Safety Integrity Level (SIL) and Availability
Safety Integrity Level (SIL) is a statistical representation of the safety availability of an SIS at the time of process demand. It is at the heart of acceptable SIS design and includes the following factors:

- Device integrity
- Diagnostics
- Systematic and common cause failures
- Testing
- Operation
- Maintenance

Acronyms
EUC = Equipment Under Control

Ft = Tolerable Risk level

Fnp = present risk level

MTBF = Mean Time Between Failures

PFDavg = Probability of Failure on Demand

RRF = Risk Reduction Factor

RRF = Fnp/Ft

PFDavg = 1/ RRFSIS

Example SIL Evaluation

IEC 61508 contains guidance on using both qualitative and quantitative methods to determine the SIL for a system based on risk frequency and consequence tables and graphs. The following steps illustrate application of the general guidelines contained in IEC 61508:

1. Set the target Tolerable Risk level (Ft), where Ft is the risk frequency, often determined as hazardous event frequency x consequence of hazardous event expressed numerically
2. Calculate the present risk level (Fnp) for the EUC, which is the risk frequency with no protective functions present (or unprotected risk)
3. The ratio Fnp/Ft gives the Risk Reduction Factor (RRF) required to achieve the target tolerable risk

4. Determine the amount of RRF to be assigned to the SIS (RRF_{SIS}). The reciprocal of RRF_{SIS} gives the target average Probability of Failure on Demand (PFDavg) the SIS must achieve.
5. Translate the PFDavg value into a SIL value (using guidance tables)

Consider a system with EUC that has an unprotected risk frequency (Fnp) of 1 hazardous event per 5 years (Fnp = 0.2/year), [0.2 = 1/5], with a consequence classified as "Critical". Tables 3 and 4 show examples of guidance tables used for risk classification and class interpretation of accidents from IEC 61508-5.

Risk Classification of Accidents: Table B1 of IEC 61508-5

Frequency	Catastrophic	Critical	Marginal	Negligible
	> 1 death	1 death or injuries	Minor injury	Production Loss
1 per year	I	I	I	II
1 per 5 years	I	I	II	III
1 per 50 years	I	II	III	III
1 per 500 years	II	III	III	IV
1 per 5000 years	III	III	IV	IV
1 per 50000 years	IV	IV	IV	IV

Risk Classification of Accidents: Table B2 of IEC 61508-5

Risk Class	Interpretation
I	Intolerable risk
II	Undesirable risk, tolerable only if risk reduction is impracticable or if cost are grossly disproportionate to the improvement gained
III	Tolerable risk if the cost of risk reduction would exceed the improvement gained
IV	Negligible risk

Using tables B1 and B2, the unprotected risk is determined as class I. The target is to reduce this risk to a tolerable risk of class III, i.e., 1 hazardous event per 500 to 5000 years.

If we consider the safest target, Ft = 1 hazardous event in 5000 years, this represents a frequency of 0.0002 events/year.

This gives a target risk reduction factor RRF of Fnp/Ft = 0.2/0.0002 = 1000

If there are no non-SIS protective layers assigned to the system, the SIS must fulfill the total RRF of 1000. So, in this case the total RRF = RRF_{SIS}.

Now PFDavg = 1/ RRF_{SIS} = 1/1000 = 0.001 = 1 x 10-3

SIL	Availability	PDF (avg)	MTBF
4	> 99.99%	10^{-5} to < 10^{-4}	100000 to 10000
3	99.9%	10^{-4} to < 10^{-3}	10000 to 1000
2	99-99.9%	10^{-3} to < 10^{-2}	1000 to 100
1	90-99%	10^{-2} to < 10^{-1}	100 to 10

Using the SIL assignments in the above table, this gives a SIL target 2.

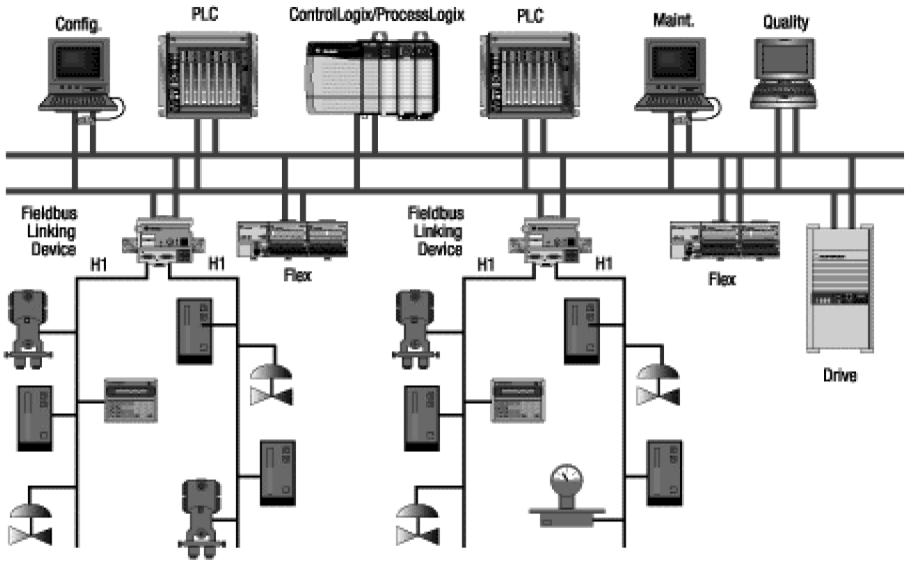
Note: Calculating MTBF based on failure rates.

$$MTBF = \frac{1}{FR_1 + FR_2 + FR_3 + ...FRn}$$

Overview of Industrial Control Networks

Overview of Networks and Communications

On the CSE Exam there may be a few questions on Fieldbus, Intelligent Devices and networks. We will briefly review the highlights of these subjects. For more information on fieldbus, contact your local distributor or the web sites of Fieldbus.org and ProfiBus.org.



Fieldbus Networks

Fieldbus is a digital, two-way, multi-drop communication link among intelligent control devices that replace the 4-20 mA analog standard devices. *The key to fieldbus is that the device is digital not analog.* There are numerous protocols on the international market: Foundation Fieldbus, ProfiBus, Asi, ControlNet, DeviceNet, Modbus, and Hart are the most popular in the process industry.

The most popular types of Fieldbus typically use EIA-485 protocol with token passing and 31.25kbps on a single twisted pair wire that can be run up to 1900 Meters. They can have 32 segments and 1024 intelligent devices per network.

The connected intelligent devices are not calibrated; the data is scaled in software. Intelligent devices may deliver from one (1) up to twelve (12) or more data variables of information from one instrument. The data is delivered in data packets to the intelligent control device or master. Possibly the valve may be the intelligent controller.

Intelligent devices need to be configured when first installed. This is done through EDDL (Electronic Device Description Language) or FDT (Field Device Tools). Most of the intelligent devices are plug and play. ProfiBus devices can even be changed out without reconfiguring the device once configured. The configuration data is stored by the Master controller.

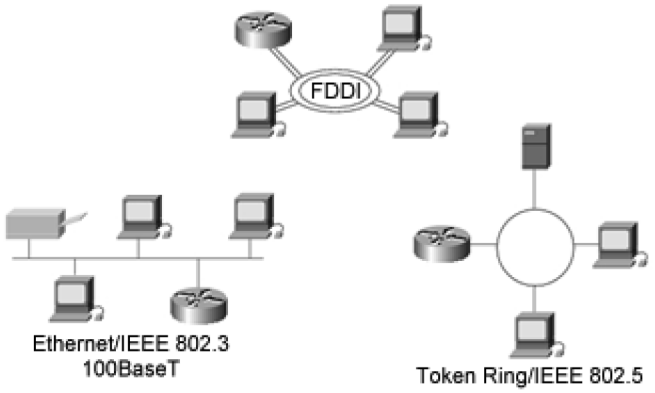
Ethernet Networks and Communications

Networks can be connected by wire, fiber optic cable, or can be wireless. There are three major categories of networks: LAN (Local Area Network), WAN (Wide Area Network) and MAN (Metropolitan Area Network). The LAN is typically limited to 100 meters (or 330′ per segment) and 1024 nodes.

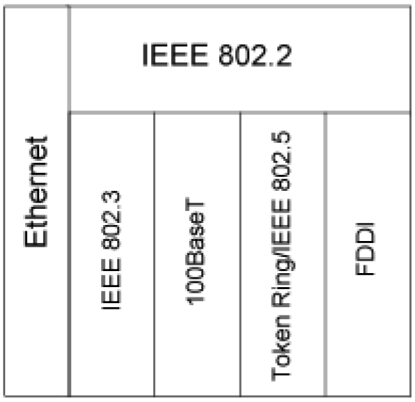
Industrial instruments typically communicate through a version of one of three communication network protocols below.

If a ***Serial Network*** they use: EIA/RS-232; EIA/RS-485; EIA/RS-488.

If an ***Ethernet Network*** they use: Ethernet/IEEE 802.3, Token Ring/IEEE 802.5, and Fiber Distributed Data Interface (FDDI).



Three typical Ethernet networks



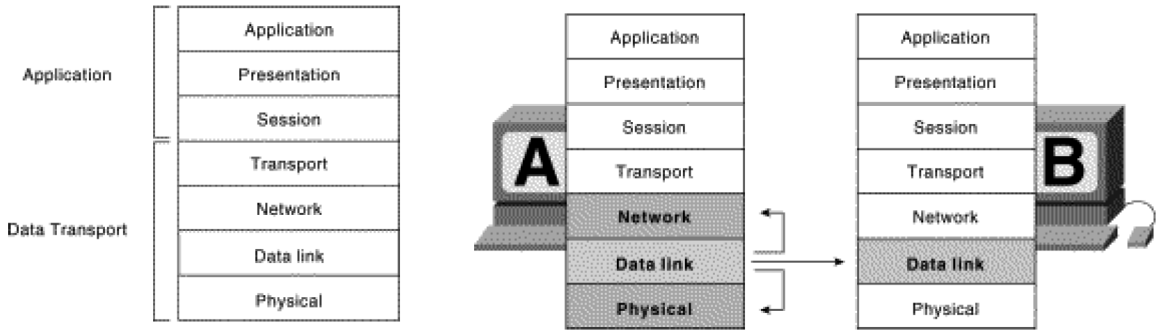
Ethernet Protocols

If the device communicates through Ethernet protocol, it typically has, *but not always*, a **MAC (Media Access Control) address**. Like a social security number this number is unique to every device. For a device on one network to talk to a device on another network using a different protocol, a **Protocol Converter or Gateway** is needed.

For more information on Ethernet, Cisco Systems offers the following at no charge:

The Cisco Systems Internetworking Technology Handbook. It is online at this address: http://www.cisco.com/univercd/cc/td/doc/cisintwk/ito_doc/ (note: there is an underscore in (ito_doc).

Layers Make Up the OSI Layers



OSI Layers **OSI Layer Services**

The above diagram shows computers communicating through the data link layer, between MAC addresses, but they are connected on the **physical layer** by media (cable, fiber optic, etc.).

Intelligent and Smart Devices

An Intelligent Device ***IS NOT*** a Smart Device. Smart Devices, such as level transmitters, are capable of being programmed or calibrated with a communicator or software over the network. A device which is neither smart nor intelligent must be calibrated and commissioned by hand.

An Intelligent Device is not calibrated in the field or shop. It is calibrated at the factory and left alone. Standard Devices and Smart Devices deliver only one variable: e.g., temperature; pressure; mass flow rate. But an Intelligent Device can deliver: e.g., temperature, pressure, delta pressure, mass flow rate, and viscosity, etc., all in one data stream (digital signal).

The information is sent in framed data packets to the controller or host, which then extracts the multiple data variables for use from the data packet. The information is typically delivered in one byte per data variable. The data packet itself may be 8 to 40 plus bytes long. A frame can be from 64 to 1,518 bytes long, in total.

This should be adequate information for the CSE examination. There are many books on the subject of Fieldbus and Intelligent Devices.

Overview of NEC and NFPA Codes

List of NFPA Codes

The CSE exam will cover code questions. We have covered ASME codes in the section on pressure relief valves and safety disks. We will now talk about codes for the installation, maintenance and operation of control systems in process plants. Here are the major codes the CSE exam may cover:

NFPA 70 NEC – National Electrical Code

NFPA 77 Static Electricity

NFPA 78 Lightning Protection

NFPA 79 Industrial Machinery

NFPA 496 Purged and Pressurized Systems

NFPA 70 NEC – National Electrical Code

Having the NEC – National Electrical Code (NFPA 70) Handbook, or a book of equal information, is required. The book contains information needed for motors, hazardous locations, NEMA classifications, and temperature group ratings. The handbook contains information about group classifications and autoignition temperature ratings of flammable gases and vapors (reprints from NFPA 497M).

- Table 310-16 – Conductor ampacities in raceways, cable or earth
- Table 430-147– Motor currents for single phase motors
- Table 430-150– Motor currents for three phase motors
- 500-2 Handbook, list of TYPE X,Y, Z purging of enclosures
- 500-3 Special precautions, group classifications of gases and vapors
- 500-3 Handbook, list of gases and vapors, with their group ratings
- 504-X Intrinsically Safe Systems (review this section)
- 504-50 Handbook, diagrams of intrinsically safe barriers
- Chapter 9-Table 8 Conductor properties and DC resistance
- Chapter 9-Table 9 AC resistance for 600 volt cables

Voltage drop will also probably be on the test. Voltage drop is just Ohm's Law.

Voltage Drop formulas:

$V_d = \left(\frac{2 * L}{1000}\right) * I * R$; for dc

$V_d = \left(\frac{2 * L}{1000}\right) * I * Z_e$; for ac $\{Z_e$ with P.F. = 100 is equal to dc=R$\}$ [For single phase]

$V_d = \left(\frac{\sqrt{3} * L}{1000}\right) * I * Z_e$; for ac $\{Z_e$ with P.F. = 100 is equal to dc= R$\}$ [For three phase]

Note: $\left(\frac{2 * L}{1000}\right) * I * Z_e * Cos30 = \left(\frac{2 * L}{1000}\right) * I * Z_e * \frac{\sqrt{3}}{2} = \left(\frac{\sqrt{3} * L}{1000}\right) * I * Z_e$

Use Specific Resistance (k) for Resistance (R)

k = 10.8; the specific resistance of copper for, 1 cm of one foot in length

$k = \left(\frac{R * cm}{1000}\right)$; specific resistance of copper = 12 (for 75°C)

cm = circular mils of copper

Next Substitute for: $\frac{R}{1000}; \frac{k}{cm}$

Cable Sizing formulas:

$V_d = \left(\frac{2 * L * I * k}{cm}\right);\ cm = \left(\frac{2 * L * I * k}{V_d}\right)$; [For single phase]

$V_d = \left(\frac{\sqrt{3} * L * I * k}{cm}\right);\ cm = \left(\frac{\sqrt{3} * L * I * k}{V_d}\right)$; [For three phase]

Voltage Drop Sizing Example 1

Sample problem: (A): What is the voltage drop for AWG 18 stranded wire 565 feet one way in coated cable? Wire carries 20 mA of dc current. *Note: Coated (wires are jacketed); uncoated (wires are not jacketed).*
Find the resistance for AWG 18 stranded wire (coated) in NEC Table 8. *R* per 1000ft = 8.45 ohms $V_d = \left(\frac{2*L}{1000}\right)*I*R;\ \left(\frac{2*565}{1000}\right)*0.020*8.45 = 0.191$ or 191mV

Voltage Drop Sizing Example 2

Sample problem: A 480 volt three phase 50 HP motor draws 65 amps and is 600 feet away. What is the voltage drop, and what size wire should we use for a 5% voltage drop?
Find the voltage drop first. $\%_{drop} = \frac{V_d}{V_{source}};\ V_d = V_{source} * \%_{drop} = 480 * 0.05 = 24$volts drop maximum Find the wire size from the maximum allowable voltage drop. $cm = \left(\frac{\sqrt{3} * L * I * k}{V_d}\right) = \left(\frac{\sqrt{3} * 600 * 65 * 12}{24}\right) = 33,774 cm$ Find the cm (area circular mils) of stranded wire (uncoated) in NEC Table 8. AWG 6 = 26,240cm < 33,774cm < AWG 4 = 41,740cm … so use AWG 4 ***Proof of voltage drop***, resistance for AWG 4 stranded wire (uncoated) in NEC Table 8. *R* per 1000ft = 0.321 ohms $V_d = \left(\frac{\sqrt{3} * L}{1000}\right) * I * Z_e = \left(\frac{\sqrt{3} * 600}{1000}\right) * 65 * 0.321 = 21.68$ or 22 Volts dropped along the wire. The wire size gives less than the required maximum of 5% voltage drop.

Comparison of NEMA Enclosures

Definitions [*from NEMA 250-2003*]

In Non-Hazardous Locations, the specific enclosure Types, their applications, and the environmental conditions they are designed to protect against, **when completely and properly installed**, are as follows:

Type 1 Enclosures constructed for indoor use to provide a degree of protection to personnel against access to hazardous parts and to provide a degree of protection of the equipment inside the enclosure against ingress of solid foreign objects (falling dirt).

Type 2 Enclosures constructed for indoor use to provide a degree of protection to personnel against access to hazardous parts; to provide a degree of protection of the equipment inside the enclosure against ingress of solid foreign objects (falling dirt); and to provide a degree of protection with respect to harmful effects on the equipment due to the ingress of water (dripping and light splashing).

Type 3 Enclosures constructed for either indoor or outdoor use to provide a degree of protection to personnel against access to hazardous parts; to provide a degree of protection of the equipment inside the enclosure against ingress of solid foreign objects (falling dirt and windblown dust); to provide a degree of protection with respect to harmful effects on the equipment due to the ingress of water (rain, sleet, snow); and that will be undamaged by the external formation of ice on the enclosure.

Type 3R Enclosures constructed for either indoor or outdoor use to provide a degree of protection to personnel against access to hazardous parts; to provide a degree of protection of the equipment inside the enclosure against ingress of solid foreign objects (falling dirt); to provide a degree of protection with respect to harmful effects on the equipment due to the ingress of water (rain, sleet, snow); and that will be undamaged by the external formation of ice on the enclosure.

Type 3S Enclosures constructed for either indoor or outdoor use to provide a degree of protection to personnel against access to hazardous parts; to provide a degree of protection of the equipment inside the enclosure against ingress of solid foreign objects (falling dirt and windblown dust); to provide a degree of protection with respect to harmful effects on the equipment due to the ingress of water (rain, sleet, snow); and for which the external mechanism(s) remain operable when ice laden.

Type 3X Enclosures constructed for either indoor or outdoor use to provide a degree of protection to personnel against access to hazardous parts; to provide a degree of protection of the equipment inside the enclosure against ingress of solid foreign objects (falling dirt and windblown dust); to provide a degree of protection with respect to harmful effects on the equipment due to the ingress of water (rain, sleet, snow); that provides an additional level of protection against corrosion and that will be undamaged by the external formation of ice on the enclosure.

Type 3RX Enclosures constructed for either indoor or outdoor use to provide a degree of protection to personnel against access to hazardous parts; to provide a degree of protection of the equipment inside the enclosure against ingress of solid foreign objects (falling dirt); to provide a degree of protection with respect to harmful effects on the equipment due to the

ingress of water (rain, sleet, snow); that will be undamaged by the external formation of ice on the enclosure that provides an additional level of protection against corrosion; and that will be undamaged by the external formation of ice on the enclosure.

Type 3SX Enclosures constructed for either indoor or outdoor use to provide a degree of protection to personnel against access to hazardous parts; to provide a degree of protection of the equipment inside the enclosure against ingress of solid foreign objects (falling dirt and windblown dust); to provide a degree of protection with respect to harmful effects on the equipment due to the ingress of water (rain, sleet, snow); that provides an additional level of protection against corrosion; and for which the external mechanism(s) remain operable when ice laden.

Type 4 Enclosures constructed for either indoor or outdoor use to provide a degree of protection to personnel against access to hazardous parts; to provide a degree of protection of the equipment inside the enclosure against ingress of solid foreign objects (falling dirt and windblown dust); to provide a degree of protection with respect to harmful effects on the equipment due to the ingress of water (rain, sleet, snow, splashing water, and hose directed water); and that will be undamaged by the external formation of ice on the enclosure.

Type 4X Enclosures constructed for either indoor or outdoor use to provide a degree of protection to personnel against access to hazardous parts; to provide a degree of protection of the equipment inside the enclosure against ingress of solid foreign objects (windblown dust); to provide a degree of protection with respect to harmful effects on the equipment due to the ingress of water (rain, sleet, snow, splashing water, and hose directed water); that provides an additional level of protection against corrosion; and that will be undamaged by the external formation of ice on the enclosure.

Type 5 Enclosures constructed for indoor use to provide a degree of protection to personnel against access to hazardous parts; to provide a degree of protection of the equipment inside the enclosure against ingress of solid foreign objects (falling dirt and settling airborne dust, lint, fibers, and flyings); and to provide a degree of protection with respect to harmful effects on the equipment due to the ingress of water (dripping and light splashing).

Type 6 Enclosures constructed for either indoor or outdoor use to provide a degree of protection to personnel against access to hazardous parts; to provide a degree of protection of the equipment inside the enclosure against ingress of solid foreign objects (falling dirt); to provide a degree of protection with respect to harmful effects on the equipment due to the ingress of water (hose directed water and the entry of water during occasional temporary submersion at a limited depth); and that will be undamaged by the external formation of ice on the enclosure.

Type 6P Enclosures constructed for either indoor or outdoor use to provide a degree of protection to personnel against access to hazardous parts; to provide a degree of protection of

the equipment inside the enclosure against ingress of solid foreign objects (falling dirt); to provide a degree of protection with respect to harmful effects on the equipment due to the ingress of water (hose directed water and the entry of water during prolonged submersion at a limited depth); that provides an additional level of protection against corrosion and that will be undamaged by the external formation of ice on the enclosure.

Type 12 Enclosures constructed (without knockouts) for indoor use to provide a degree of protection to personnel against access to hazardous parts; to provide a degree of protection of the equipment inside the enclosure against ingress of solid foreign objects (falling dirt and circulating dust, lint, fibers, and flyings); and to provide a degree of protection with respect to harmful effects on the equipment due to the ingress of water (dripping and light splashing).

Type 12K Enclosures constructed (with knockouts) for indoor use to provide a degree of protection to personnel against access to hazardous parts; to provide a degree of protection of the equipment inside the enclosure against ingress of solid foreign objects (falling dirt and circulating dust, lint, fibers, and flyings); and to provide a degree of protection with respect to harmful effects on the equipment due to the ingress of water (dripping and light splashing).

Type 13 Enclosures constructed for indoor use to provide a degree of protection to personnel against access to hazardous parts; to provide a degree of protection of the equipment inside the enclosure against ingress of solid foreign objects (falling dirt and circulating dust, lint, fibers, and flyings); to provide a degree of protection with respect to harmful effects on the equipment due to the ingress of water (dripping and light splashing); and to provide a degree of protection against the spraying, splashing, and seepage of oil and non-corrosive coolants.

Table 8

[From NEMA 250-2003]

Comparison of Specific Applications of Enclosures

for Indoor Nonhazardous Locations

Provides a Degree of Protection Against the Following Conditions	Type of Enclosure									
	1 *	2 *	4	4X	5	6	6P	12	12K	13
Access to hazardous parts	X	X	X	X	X	X	X	X	X	X
Ingress of solid foreign objects (falling dirt)	X	X	X	X	X	X	X	X	X	X
Ingress of water (Dripping and light splashing)	...	X	X	X	X	X	X	X	X	X
Ingress of solid foreign objects (Circulating dust, lint, fibers, and flyings **)	...	...	X	X	...	X	X	X	X	X
Ingress of solid foreign objects (Settling airborne dust, lint, fibers, and flyings **)	...	...	X	X	X	X	X	X	X	X
Ingress of water (Hosedown and splashing water)	...	...	X	X	...	X	X	...	...	...
Oil and coolant seepage	...	...	...	..	...	...	...	X	X	X
Oil or coolant spraying and splashing	...	...	...	...	...	...	...	...	...	X
Corrosive agents	...	...	...	X	...	...	X	...	...	...
Ingress of water (Occasional temporary submersion)	...	...	...	...	...	X	X	...	...	...
Ingress of water (Occasional prolonged submersion)	...	...	...	...	...	...	X	...	...	...

* These enclosures may be ventilated.

** These fibers and flyings are nonhazardous materials and are not considered Class III type ignitable fibers or combustible flyings. For Class III type ignitable fibers or combustible flyings see the National Electrical Code, Article 500.

Table 9

[From NEMA 250-2003]

Comparison of Specific Applications of Enclosures for Outdoor Nonhazardous Locations

Provides a Degree of Protection Against the Following Conditions	Type of Enclosure									
	3	3X	3R*	3RX*	3S	3SX	4	4X	6	6P
Access to hazardous parts	X	X	X	X	X	X	X	X	X	X
Ingress of water (Rain, snow, and sleet **)	X	X	X	X	X	X	X	X	X	X
Sleet ***	...	...	...	...	X	X	...	...	...	...
Ingress of solid foreign objects (Windblown dust, lint, fibers, and flyings)	X	X	...	...	X	X	X	X	X	X
Ingress of water (Hosedown)	...	...	...	...	...	...	X	X	X	X
Corrosive agents	...	X	...	X	...	X	...	X	...	X
Ingress of water (Occasional temporary submersion)	...	...	...	...	...	...	...	...	X	X
Ingress of water (Occasional prolonged submersion)	...	...	...	...	...	...	...	...	...	X

* These enclosures may be ventilated.
** External operating mechanisms are not required to be operable when the enclosure is ice covered.
*** External operating mechanisms are operable when the enclosure is ice covered.

In **Hazardous Locations,** when completely and properly installed and maintained, Type 7 and 10 enclosures are designed to contain an internal explosion without causing an external hazard. Type 8 enclosures are designed to prevent combustion through the use of oil-immersed equipment. Type 9 enclosures are designed to prevent the ignition of combustible dust.

Type 7 Enclosures constructed for indoor use in hazardous (classified) locations classified as Class I, Division 1, Groups A, B, C, or D as defined in NFPA 70.

Type 8 Enclosures constructed for either indoor or outdoor use in hazardous (classified) locations classified as Class I, Division 1, Groups A, B, C, and D as defined in NFPA 70.

Type 9 Enclosures constructed for indoor use in hazardous (classified) locations classified as Class II, Division 1, Groups E, F, or G as defined in NFPA 70.

Type 10 Enclosures constructed to meet the requirements of the Mine Safety and Health Administration, 30 CFR, Part 18.

Table 10

[From NEMA 250-2003]

Comparison of Specific Applications of Enclosures for Indoor Hazardous Locations

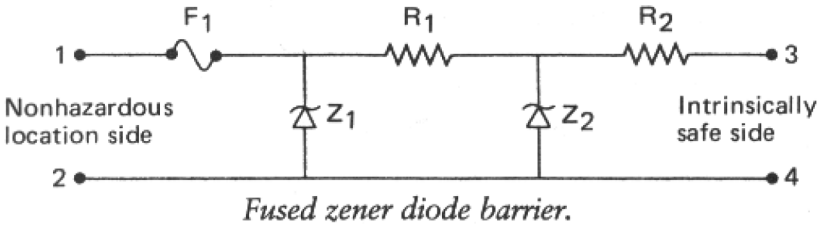
(If the installation is outdoors and/or additional protection is required by Table 1 and Table 2, a combination-type enclosure is required.)

Provides a Degree of Protection Against Atmospheres Typically Containing (See NFPA 497M for Complete Listing)	Class	Enclosure Types 7 and 8, Class I Groups **				Enclosure Type 9, Class II Groups			
		A	B	C	D	E	F	G	10
Acetylene	I	X	...	...	...	...	...	...	...
Hydrogen, manufactured gas	I	...	X	...	...	...	...	...	...
Diethyl ether, ethylene, cyclopropane	I	...	...	X	...	...	...	...	...
Gasoline, hexane, butane, naphtha, propane, acetone, toluene, isoprene	I	...	...	...	X	...	...	...	...
Metal dust	II	...	...	...	...	X	...	...	...
Carbon black, coal dust, coke dust	II	...	...	...	...	...	X	...	...
Flour, starch, grain dust	II	...	...	...	...	...	...	X	...
Fibers, flyings *	III	...	...	...	...	...	...	X	...
Methane with or without coal dust	MSHA	...	...	...	...	...	...	...	X

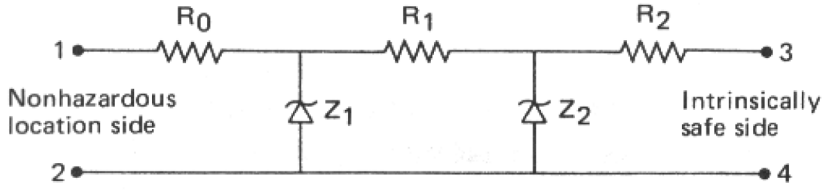
* For Class III type ignitable fibers or combustible flyings see the National Electrical Code, Article 500.

** Due to the characteristics of the gas, vapor, or dust, a product suitable for one Class or Group may not be suitable for another Class or Group unless marked on the product.

Intrinsically Safe Systems – Zener diode barrier



Fused zener diode barrier.

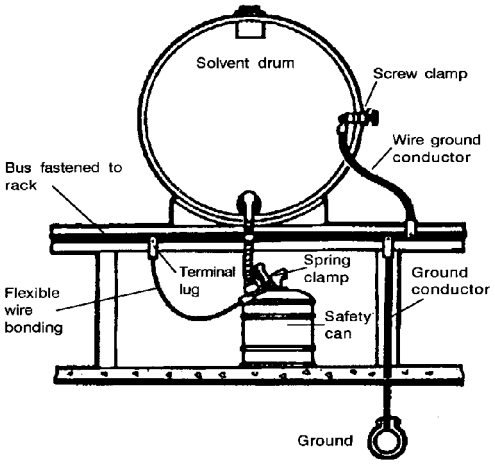


Unfused zener diode barrier.

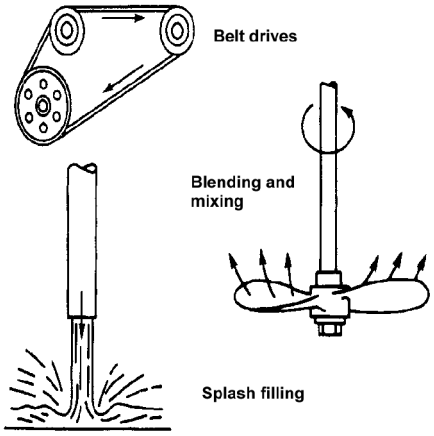
NFPA 77 Static Electricity

The buildup of static electricity in flowing applications is a major concern. It is important that proper grounding be implemented to protect personnel from shock and possible explosions due to sparks.

NFPA 77 covers proper grounding techniques for loading stations, where these hazards may occur.



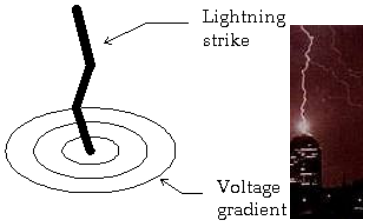
Proper Protection Grounding



Static Electric Generators

NFPA 78 Lightning Protection

The lightning strike can generate up to 300,000 volts and shoot through a concrete wall 2 feet thick. A direct lightning strike can cause an enormous amount of physical damage. Lightning strikes that hit equipment and storage or process vessels containing flammable materials can cause devastating accidents at refineries, bulk plants, processing sites, and other facilities.



However, the indirect effects from a nearby strike can also cause damage by inducing voltage surges onto mains and data cables. Lightning-induced voltage surges are often described as a "secondary effect" of lightning, and there are three recognized means by which these surges are induced in mains or data/telecommunications cables:

a) Resistive coupling
b) Inductive coupling
c) Capacitive coupling

NFPA 78 covers proper grounding techniques for lightning protection. Lightning surge arrestors and lightning protection equipment should be used to protect the process control systems and ensure it continues to function correctly.

NFPA 79 Industrial Machinery

The wire sizing and color codes for wires and buttons are covered in industrial machinery NFPA 79.

Conductor sizing:

Conductors shall not be smaller than:

(a) Power circuits . . . . . #14 awg

(b) Lighting and Control circuits on machine and in raceways . . . . #16 awg

Exception: in jacket multiconductor cable assembly . . . . . #18 awg

(c) Control circuits . . . . . #18 awg

(d) Electronic, control conductors in raceways . . . . . #24 awg

Conductor colors:

Black - Line, load and control circuits at line voltage

Red - AC control circuits less than line voltage

Blue - DC control circuits

Yellow - Interlock control circuits powered from external power supply

Green - Equipment ground conductor where insulated or covered

Pushbutton functions for color:

Red - Emergency stop, Stop, Off

Yellow - Return, Emergency return, Intervention – suppress abnormal conditions

Green - Start, On

Black - No specified function assigned

White - Any function not covered above

Clear - Any function not covered above

Blue - Any function not covered above

Grey - Any function not covered above

NFPA 496 Purged and Pressurized Systems

This standard shall apply to all purged and pressurized enclosures. The standard's intent is to provide information on the methods for purging and pressurizing enclosures to prevent ignition of flammable atmospheres.

Purging for Class I hazardous locations (NEC/NFPA):

Type X Purging - Reduces the classification from Division 1 to nonhazardous

Type Y Purging - Reduces the classification from Division 1 to Division 2

Type Z Purging - Reduces the classification from Division 2 to nonhazardous

Note: At least ***four volumes*** of purge gas must pass through the enclosure, while maintaining a minimum pressure of ***0.1 inches of water***, before operation of the equipment inside. A ***minimum*** of 0.1 inches of water pressure must be maintained in the enclosure when operating.

A warning label shall be mounted on the enclosure. On Type Y and Type Z purge failure, an alarm or pressure switch can be used to remove power from the enclosure. With Type X purge, this power must be removed with an explosion proof switch.

The Fisher Control Valve Handbook

Guide to Using the Control Valve Handbook

One of the required books, *Fisher Control Valve Handbook*, is necessary to work many of the examples in this book and the CSE examination. The information and tables in the *Fisher Control Valve Handbook* will be constantly referenced. The book may be downloaded in PDF format from the Fisher Controls public website at the following address: http://www.documentation.emersonprocess.com/groups/public/documents/book/cvh99.pdf

A hard copy is recommended for the test and can be acquired for free, or for less than $20. See **Required Books** for more information on how to obtain a hard copy of the book.

I suggest tabbing the FCVH for quick reference during the CSE examination.

Import Sections and Pages in the FCVH

Important Sections to Review

Chapter 3 – Valve and Actuator Types 41
Chapter 5 – Control Valve Selection (and sizing) 41

Important Pages to Tab

Pressure-Temperature Ratings for Standard Class. 76
Valve Trim Material Temperature Limits. 93
Ambient Temperature Corrosion information 95
Fluid Compatibility 103
Representative Sizing Coefficients for Single-Ported Globe Valve Bodies 125
Representative Sizing Coefficients for Rotary-Shaft Valve Bodies. 126
Physical Constants of Various Fluids 203
Properties of Water 211
Properties of Saturated Steam 212
Flow of Water Through Schedule 40 pipe 228
Flow of Air Through Schedule 40 pipe 232
Flow Correction Formulas for Steam, Vapor, Temperature and Pressure 235
Pipe Data – Carbon and Alloy Steel – Stainless Steel. 238

Appendix

Table A1. Specific Gravity for Some Common Fluids

Product	Temperature		Specific Gravity *SG* [1]
	°F	°C	
Acetaldehyde CH_3CHO	61 68	16.1 20	0.79 0.76
Acetic acid 5% - vinegar	59	15	1.006
Acetic acid - 10%	59	15	1.014
Acetic acid - 50%	59	15	1.061
Acetic acid - 80%	59	15	1.075
Acetic acid - concentrated	59	15	1.055
Acetic acid anhydride $(CH_3COO)_2O$	59	15	1.087
Acetone CH_3COCH_3	68	20	0.792
Alcohol - allyl	68	20	0.855
Alcohol - butyl-n	68 158	20 70	0.81 0.78
Alcohol - ethyl (grain) C_2H_5OH	68 104	20 40	0.789 0.772
Alcohol - methyl (wood) CH_3OH	68	20	0.79
Alcohol - propyl	68 32	20 0	0.804 0.817
Aluminum sulfate 36% solution	60	15.6	1.055
Ammonia	0	-17.8	0.662
Aniline	68 32	20 0	1.022 1.035
Automotive crankcase oils SAE-5W/10W/20W/30W/40W/50W	60	15.6	0.88-0.94
Automotive gear oils SAE-75W/80W/85W/90W/140W/150W	60	15.6	0.88-0.94
Beer	60	15.6	1.01
Benzene (benzol) C_6H_6	32 60	0 15.6	0.899 0.885
Bone oil	60	15.6	0.918
Boric acid H_3BO_3	46.4 59	8 15	1.014 1.025
Bromine	32	0	2.9
Butane-n	60	15.6	0.584
Butyric acid	68	20	0.959
Calcium chloride 5%	65	18.3	1.040
Calcium chloride 25%	60	15.6	1.23
Carbolic acid (phenol)	65	18.3	1.08
Carbon tetrachloride CCl_4	68	20	1.594

Product (Continued)	°F	°C	S.G.
Carbon disulfide CS_2	32 68	0 20	1.293 1.263
Castor Oil	68 104	20 40	0.96 0.95
China wood oil	60	15.6	0.943
Chloroform	68 140	20 60	1.489 1.413
Coconut oil	60	15.6	0.925
Cod liver oil	59	15	0.920-0.925
Corn oil	60	15.6	0.924
Cotton seed oil	60	15.6	0.88-0.93
Creosote	60	15.6	1.04-1.10
Crude oil 48° API	60 130	15.6 54.4	0.79 0.76
Crude oil 40° API	60 130	15.6 54.4	0.825 0.805
Crude oil 35.6° API	60 130	15.6 54.4	0.847 0.824
Crude oil 32.6° API	60 130	15.6 54.4	0.832 0.84
Crude oil Salt creek	60 130	15.6 54.4	0.843 0.82
Decane-n	68	20	0.73
Diethylene glycol	60	15.6	1.12
Diethyl ether	68	20	0.714
Diphenylamine			1.16
Diesel Fuel Oil 2D/3D/4D/5D	60	15.6	0.81 - 0.96
Dowtherm	77	25	1.056
Ethyl acetate $CH_3COOC_2H_3$	59 68	15 20	0.907 0.90
Ethyl bromide C_2H_3Br	59	15	1.45
Ethylene bromide	68	20	2.18
Ethylene chloride	68	20	1.246
Ethylene glycol	60	15.6	1.125
Formic acid - 10%	68	20	1.025
Formic acid - 50%	68	20	1.121
Formic acid - 80%	68	20	1.186
Formic acid - concentrated	68	20	1.221
Freon - 11	70	21.1	1.49
Freon - 12	70	21.1	1.33
Freon - 21	70	21.1	1.37
Furfurol	68	20	1.159
Fuel oils 1/2/3/5A/5B/6	60	15.6	0.82-0.95
Gas oils	60	15.6	0.89

Product (Continued)	°F	°C	S.G.
Gasoline a	60	15.6	0.74
Gasoline b	60	15.6	0.72
Gasoline c	60	15.6	0.68
Glycerin 100%	68	20	1.26
Glycerin 50% water	68	20	1.13
Glucose	60	15.6	1.35-1.44
Heptane-n	60	15.6	0.688
Hexane-n	60	15.6	0.664
Ink printers	60	15.6	1.0-1.4
Kerosene	60	15.6	0.78-0.82
Jet fuel	60	15.6	0.82
Lard	60	15.6	0.96
Lard oil	60	15.6	0.91-0.93
Linseed oil	60	15.6	0.92-0.94
Mercury	60	15.6	13.6
Methyl acetate	68	20	0.93
Methyl iodide	68	20	2.28
Milk	60	15.6	1.02-1.05
Molasses A first	60	15.6	1.40-1.46
Molasses B second	60	15.6	1.43-1.48
Molasses C blackstrap	60	15.6	1.46-1.49
Naphthalene	68	20	1.145
Neatsfoot oil	60	15.6	0.917
Nitrobentzene	68 59	20 15	1.203 1.205
Nonane-n	60 68	15.6 20	0.722 0.718
Octane-n	60	15.6	0.707
Olive oil	60	15.6	0.91 - 0.92
Palm oil	60	15.6	0.924
Peanut oil	60	15.6	0.92
Pentane-n	32 60	0 15.6	0.650 0.631
Potassium hydrate			1.24
Sodium chloride			1.19
Sodium hydrate			1.27
Tuluol			0.87
Turpentine			0.87
Water. fresh			**1**
Water. sea 36 °F			1.02
Xylene			0.87

[1)] Based on water at 60°F and SG = 1

Table A2. Specific Gravity and Gas Constants for Some Common Gases

The specific gravity of some common gases can be found in the table below:

Gas	Specific Gravity[1)] *- SG -*	Molecular Weight *- M -*	Ratio of specific heat *- k -*
Acetylene (ethyne) - C_2H_2	0.907	26.038	1.234
Air[1)]	**1.000**	**28.967**	**1.399**
Ammonia - NH_3	0.588	17.032	1.304
Argon - Ar	1.379	39.944	1.668
Arsine	2.69		
Benzene	2.559	78.114	1.113
Blast Furnace gas	1.02		
Butadiene	1.869		
n-Butane - C_4H_{10}	2.007	58.124	1.093
I-Butene - C_4H_8	1.937	56.108	1.111
Carbon dioxide - CO_2	1.519	44.011	1.288
Carbon monoxide - CO	0.967	28.011	1.399
Carbureted Water Gas	0.63		
Chlorine - Cl_2	2.486	70.910	
Coke Oven Gas	0.44		
Cyclobutane	1.938		
Cyclohextane	2.905	84.161	1.07
Cyclopentane	2.422	70.135	1.08
Cyclopropane	1.451		
DoDecane – $C_{12}H_{26}$	5.88	170.340	1.031
Digestive Gas (Sewage or Biogas)	0.8		
Ethane - C_2H_6	1.038	30.070	1.188
Ethylene (Ethene) - C_2H_4	0.9685	28.054	1.236
Fluorine	1.31	38.000	
Freon, F-12		120.925	1.136
Helium - He	0.138	4.003	1.667
n-Heptane – C_7H_{16}	3.459	100.205	1.053
n-Hexane – C_6H_{14}	2.9753	86.178	1.062
Hydrogen	0.069	2.016	1.405
Hydrogen chloride - HCl	1.268	36.470	
Hydrogen sulfide - H_2S	1.177	34.082	
Isobutane - C_4H_{10}	2.007	58.124	1.094
Isopentane – C_5H_{12}	2.4911	72.151	1.074
Krypton	2.89		
Methane - CH_4	0.554	16.043	1.304
Methyl Chloride	1.74	50.490	
Natural Gas (typical)	0.60 - 0.70 (0.65)	(18.829)	(1.32)
Neon	0.696	20.183	1.667

Gas (Continued)	Specific Gravity[1)] - *SG* -	Molecular Weight - *M* -	Ratio of specific heat - *k* -
Nitric oxide - NO	1.0359	30.008	1.386
Nitrogen - N_2	0.967	28.016	1.40
Nitrous oxide - N_2O	1.530	44.020	
Nonane	4.428	128.258	1.04
Octane	3.944	114.232	1.046
Oxygen – O_2	1.105	32.000	1.396
Ozone	1.660		
n-Pentane – C_5H_{12}	2.4908	72.151	1.074
Phosgene	1.39		
Propane – C_3H_8	1.522	44.097	1.128
Propene (Propylene) – C_3H_6	1.4527	42.081	1.187
Sasol	0.42		
Silane	1.11		
Sulfur Dioxide - SO_2	2.2117	64.066	1.264
Toluene-Methylbenzene	3.176		
Water Gas (bituminous)	0.71		
Water Vapor	0.622	18.016	1.329
Xenon	4.533	131.300	1.667

[1)] NTP - Normal Temperature and Pressure - is defined as air at 20°C (293.15 K, 68°F) and 1 atm (101.325 kN/m2, 101.325 kPa, 14.7 psia, 0 psig, 30 in Hg, 760 torr)

Since specific gravity is the ratio between the density (mass per unit volume) of the actual gas and the density of air, specific gravity has no dimension.

Table A3. The kinematic viscosity for some common fluids

Fluid	Temperature		Kinematic Viscosity	
	(^oF)	(^oC)	CentiStokes (cSt)	Seconds Saybolt Universal (SSU)
Acetaldehyde CH_3CHO	61 68	16.1 20	0.305 0.295	36
Acetic acid - vinegar - 10% CH_3COOH	59	15	1.35	31.7
Acetic acid - 50%	59	15	2.27	33
Acetic acid - 80%	59	15	2.85	35
Acetic acid - concentrated glacial	59	15	1.34	31.7
Acetic acid anhydride $(CH_3COO)_2O$	59	15	0.88	
Acetone CH_3COCH_3	68	20	0.41	
Alcohol - allyl	68 104	20 40	1.60 0.90 cp	31.8

Fluids (Continued)	(^{o}F)	(^{o}C)	CentiStokes (cSt)	Seconds Saybolt Universal (SSU)
Alcohol - butyl-n	68	20	3.64	38
Alcohol – ethyl (grain) C_2H_5OH	68 100	20 37.8	1.52 1.2	31.7 31.5
Alcohol - methyl (wood) CH_3OH	59 32	15 0	0.74 1.04	
Alcohol - propyl	68 122	20 50	2.8 1.4	35 31.7
Aluminum sulfate - 36% solution	68	20	1.41	31.7
Ammonia	0	-17.8	0.30	
Aniline	68 50	20 10	4.37 6.4	40 46.4
Asphalt RC-0, MC-0, SC-0	77 100	25 37.8	159-324 60-108	737-1.5M 280-500
Automatic crankcase oil SAE 10W	0	-17.8	1295-max	6M-max
Automatic crankcase oil SAE 10W	0	-17.8	1295-2590	6M-12M
Automatic crankcase oil SAE 20W	0	-17.8	2590-10350	12M-48M
Automatic crankcase oil SAE 20	210	98.9	5.7-9.6	45-58
Automatic crankcase oil SAE 30	210	98.9	9.6-12.9	58-70
Automatic crankcase oil SAE 40	210	98.9	12.9-16.8	70-85
Automatic crankcase oil SAE 50	210	98.9	16.8-22.7	85-110
Automotive gear oil SAE 75W	210	98.9	4.2 min	40 min
Automotive gear oil SAE 80W	210	98.9	7.0 min	49 min
Automotive gear oil SAE 85W	210	98.9	11.0 min	63 min
Automotive gear oil SAE 90W	210	98.9	14-25	74-120
Automotive gear oil SAE 140	210	98.9	25-43	120-200
Automotive gear oil SAE150	210	98.9	43 - min	200 min
Beer	68	20	1.8	32
Benzene (Benzol) C_6H_6	32 68	0 20	1.0 0.74	31
Bone oil	130 212	54.4 100	47.5 11.6	220 65

Fluids (Continued)	(^{o}F)	(^{o}C)	CentiStokes (cSt)	Seconds Saybolt Universal (SSU)
Bromine	68	20	0.34	
Butane-n	-50 30	-1.1	0.52 0.35	
Butyric acid n	68 32	20 0	1.61 2.3 cp	31.6
Calcium chloride 5%	65	18.3	1.156	
Calcium chloride 25%	60	15.6	4.0	39
Carbolic acid (phenol)	65 194	18.3 90	11.83 1.26 cp	65
Carbon tetrachloride CCl_4	68 100	20 37.8	0.612 0.53	
Carbon disulfide CS_2	32 68	0 20	0.33 0.298	
Castor oil	100 130	37.8 54.4	259-325 98-130	1200-1500 450-600
China wood oil	69 100	20.6 37.8	308.5 125.5	1425 580
Chloroform	68 140	20 60	0.38 0.35	
Coconut oil	100 130	37.8 54.4	29.8-31.6 14.7-15.7	140-148 76-80
Cod oil (fish oil)	100 130	37.8 54.4	32.1 19.4	150 95
Corn oil	130 212	54.4 100	28.7 8.6	135 54
Corn starch solution 22 Baume	70 100	21.1 37.8	32.1 27.5	150 130
Corn starch solution 24 Baume	70 100	21.1 37.8	129.8 95.2	600 440
Corn starch solution 25 Baume	70 100	21.1 37.8	303 173.2	1400 800
Cotton seed oil	100 130	37.8 54.4	37.9 20.6	176 100
Crude oil 48° API	60 130	15.6 54.4	3.8 1.6	39 31.8
Crude oil 40° API	60 130	15.6 54.4	9.7 3.5	55.7 38
Crude oil 35.6^{o} API	60 130	15.6 54.4	17.8 4.9	88.4 42.3
Crude oil 32.6^{o} API	60 130	15.6 54.4	23.2 7.1	110 46.8
Decane-n	0 100	17.8 37.8	2.36 1.001	34 31
Diethyl glycol	70	21.1	32	149.7
Diethyl ether	68	20	0.32	

Fluids (Continued)	(^{o}F)	(^{o}C)	CentiStokes (cSt)	Seconds Saybolt Universal (SSU)
Diesel fuel 20	100 130	37.8 54.4	2-6 1.-3.97	32.6-45.5 -39
Diesel fuel 30	100 130	37.8 54.4	6-11.75 3.97-6.78	45.5-65 39-48
Diesel fuel 40	100 130	37.8 54.4	29.8 max 13.1 max	140 max 70 max
Diesel fuel 60	122 160	50 71.1	86.6 max 35.2 max	400 max 165 max
Ethyl acetate $CH_3COOC_2H_3$	59 68	15 20	0.4 0.49	
Ethyl bromide C_2H_5Br	68	20	0.27	
Ethylene bromide	68	20	0.787	
Ethylene chloride	68	20	0.668	
Ethylene glycol	70	21.1	17.8	88.4
Formic acid 10%	68	20	1.04	31
Formic acid 50%	68	20	1.2	31.5
Formic acid 80%	68	20	1.4	31.7
Formic acid concentrated	68 77	20 25	1.48 1.57cp	31.7
Freon -11	70	21.1	0.21	
Freon -12	70	21.1	0.27	
Freon -21	70	21.1	1.45	
Furfurol	68 77	20 25	1.45 1.49cp	31.7
Fuel oil 1	70 100	21.1 37.8	2.39-4.28 -2.69	34-40 32-35
Fuel oil 2	70 100	21.1 37.8	3.0-7.4 2.11-4.28	36-50 33-40
Fuel oil 3	70 100	21.1 37.8	2.69-5.84 2.06-3.97	35-45 32.8-39
Fuel oil 5A	70 100	21.1 37.8	7.4-26.4 4.91-13.7	50-125 42-72
Fuel oil 5B	70 100	21.1 37.8	26.4- 13.6-67.1	125- 72-310
Fuel oil 6	122 160	50 71.1	97.4-660 37.5-172	450-3M 175-780
Gas oils	70 100	21.1 37.8	13.9 7.4	73 50
Gasoline a	60 100	15.6 37.8	0.88 0.71	
Gasoline b	60 100	15.6 37.8	0.64	
Gasoline c	60 100	15.6 37.8	0.46 0.40	

Fluids (Continued)	(^{o}F)	(^{o}C)	CentiStokes (cSt)	Seconds Saybolt Universal (SSU)
Glycerin 100%	68.6 100	20.3 37.8	648 176	2950 813
Glycerin 50% water	68 140	20 60	5.29 1.85 cp	43
Glucose	100 150	37.8 65.6	7.7M-22M 880-2420	35M-100M 4M-11M
Heptanes-n	0 100	-17.8 37.8	0.928 0.511	
Hexane-n	0 100	-17.8 37.8	0.683 0.401	
Honey	100	37.8	73.6	349
Ink, printers	100 130	37.8 54.4	550-2200 238-660	2500-10M 1100-3M
Insulating oil	70 100	21.1 37.8	24.1 max 11.75 max	115 max 65 max
Kerosene	68	20	2.71	35
Jet Fuel	-30.	-34.4	7.9	52
Lard	100 130	37.8 54.4	62.1 34.3	287 160
Kinematic Viscosity: 1 cSt (centiStokes) = 10^{-6} m^2/s				
Lard oil	100 130	37.8 54.4	41-47.5 23.4-27.1	190-220 112-128
Linseed oil	100 130	37.8 54.4	30.5 18.94	143 93
Mercury	70 100	21.1 37.8	0.118 0.11	
Methyl acetate	68 104	20 40	0.44 0.32 cp	
Methyl iodide	68 104	20 40	0.213 0.42 cp	
Menhaden oil	100 130	37.8 54.4	29.8 18.2	140 90
Milk	68	20	1.13	31.5
Molasses A, first	100 130	37.8 54.4	281-5070 151-1760	1300-23500 700-8160
B, second	100 130	37.8 54.4	1410-13.2M 660-3.3M	6535-61180 3058-15294
C, blackstrap	100 130	37.8 54.4	2630-55M 1320-16.5M	12190-255M 6120-76.5M
Naphthalene	176 212	80 100	0.9 0.78 cp	
Neatstool oil	100 130	37.8 54.4	49.7 27.5	230 130
Nitrobenzene	68	20	1.67	31.8

Fluids (Continued)	(°F)	(°C)	CentiStokes (cSt)	Seconds Saybolt Universal (SSU)
Nonane-n	0 100	-17.8 37.8	1.728 0.807	32
Octane-n	0 100	-17.8 37.8	1.266 0.645	31.7
Olive oil	100 130	37.8 54.4	43.2 24.1	200
Palms oil	100 130	37.8 54.4	47.8 26.4	
Peanut oil	100 130	37.8 54.4	42 23.4	200
Pentane-n	0 80	17.8 26.7	0.508 0.342	
Petrolatum	130 160	54.4 71.1	20.5 15	100 77
Petroleum ether	60	15.6	31(est)	1.1
Propionic acid	32 68	0 20	1.52 cp 1.13	31.5
Propylene glycol	70	21.1	52	241
Quenching oil (typical)			100-120	20.5-25
Rapeseed oil	100 130	37.8 54.4	54.1 31	250 145
Rosin oil	100 130	37.8 54.4	324.7 129.9	1500 600
Rosin (wood)	100 200	37.8 93.3	216-11M 108-4400	1M-50M 500-20M
Sesame seed oil	100 130	37.8 54.4	39.6 23	184 110
Sodium chloride 5%	68	20	1.097	31.1
Sodium chloride 25%	60	15.6	2.4	34
Sodium hydroxide (caustic soda) 20%	65	18.3	4.0	39.4
Sodium hydroxide (caustic soda) 30%	65	18.3	10.0	58.1
Sodium hydroxide (caustic soda) 40%	65	18.3		
Soya bean oil	100 130	37.8 5.4	35.4 19.64	165 96
Sperm oil	100 130	37.5 54.4	21-23 15.2	110 78
Sulphuric acid 100%	68 140	20 60	14.56 7.2 cp	76
Sulphuric acid 95%	68	20	14.5	75
Sulphuric acid 60%	68	20	4.4	41
Sulphuric acid 20%				3M-8M 650-1400

Fluids (Continued)	(°F)	(°C)	CentiStokes (cSt)	Seconds Saybolt Universal (SSU)
Tar, coke oven	70 100	21.1 37.8	600-1760 141-308	15M-300M 2M-20M
Kinematic Viscosity: 1 cSt (centiStokes) = 10^{-6} m^2/s				
Tar, gas house	70 100	21.1 37.8	3300-66M 440-4400	2500 500
Tar, pine	100 132	37.8 55.6	559 108.2	200-300 55-60
Toluene	68 140	20 60	0.68 0.38 cp	185.7
Triethylene glycol	70	21.1	40	400-440 185-205
Turpentine	100 130	37.8 54.4	86.5-95.2 39.9-44.3	1425 650
Varnish, spar	68 100	20 37.8	313 143	
Water, distilled	**68**	**20**	**1.0038**	**31**
Water, fresh	**60** **130**	**15.6** **54.4**	**1.13** **0.55**	**31.5**
Water, sea			**1.15**	**31.5**
Whale oil	100 130	37.8 54.4	35-39.6 19.9-23.4	163-184 97-112
Xylene-o	68 104	20 40	0.93 0.623 cp	

[1] Where centistokes are greater than 50:

CentiPoises (cp)	=	CentiStokes (cSt) x Density
SSU[1]	=	Centistokes (cSt) * 4.55
Degree Engler[1] * 7.45	=	Centistokes (cSt)
Seconds Redwood[1] - 4.05	=	Centistokes (cSt)

Table A4. The absolute viscosity for some common liquids

Temperature °F	Viscosity Centipoise	Temperature °F	Viscosity Centipoise	Temperature °F	Viscosity Centipois
Acetic Acid		Ethylene Oxide		Isohexane	
59	1.31	-57	0.577	32	0.376
64	1.30	-37	0.488	68	0.306
77	1.155	-5.8	0.394	104	0.254
86	1.04	32	0.320	Isopentane	
106	1.00	Fluorbenzene		32	0.273
212	0.43	68	0.598	68	0.223
Acetic Anhydride		140	0.389	Kerosene	
32	1.24	212	0.275	68	2.69
59	0.971	Fuel Oil, #2		100	2.0
64	0.90	70	3.0 - 7.4	Methyl Alcohol	
86	0.783	100	2.11 - 4.28	-48	1.98
212	0.490	Fuel Oil, #6		32	0.82
Acetone		122	97.4 - 660	59	0.623
14	0.450	160	37.5 - 172	68	0.597
32	0.399	Gasoline		77	0.546
59	0.337	60	0.46 - 0.88	86	0.510
77	0.316	100	0.40 - 0.71	Methyl Chloride	
Ammonia		Glycerin		0	0.25
-92	0.475	32	12,110	20	0.23
-58	0.317	43	6,260	40	0.21
-40	0.276	59	2,330	60	0.19
-28	0.255	68	1,490	100	0.16
Benzene		77	954	Naphthalene	
32	0.912	86	629	176	0.967
50	0.758	Heptane		212	0.776
68	0.652	32	0.524	Nitric Acid	
86	0.564	63	0.461	32	2.275
104	0.503	68	0.409	Nitrobenzene	
122	0.542	77	0.386	37	2.91
Carbon Tetrachloride		104	0.341	42	2.71
32	1.329	Hexane		50	2.48
59	1.038	32	0.401	68	2.03
68	0.969	68	0.326	Nitromethane	
86	0.843	77	0.386	32	0.853
104	0.739	104	0.341	77	0.620
140	0.585	Hydrochloric Acid, 31.5%		n-Octane	
Chlorine Liquid		0	3.4	32	0.706
-40	0.505	20	2.9	68	0.240
-20	0.462	40	2.5	104	0.433
20	0.400	60	2.0	Pentane	
60	0.350	80	1.8	32	0.289
100	0.313	100	1.6	68	0.524
Ethylbenzene		140	1.2	Phenol	
63	0.691	Iodine Liquid		65	12.7
Ethylene Glycol		241	2.27	122	3.49
68	19.9	Isoheptane		158	2.03
104	9.13	32	0.481	194	1.26
140	4.95	68	0.384		
176	3.02	104	0.315		

Table A4. The absolute viscosity for some common liquids (continued)

Temperature °F	Viscosity Centipoise	Temperature °F	Viscosity Centipoise	Temperature °F	Viscosity Centipoise
Phosphorous Liquid		Sodium Liquid		Sulfuric Acid	
71	2.34	0	2.4	32	48.4
88	2.01	26	1.3	59	32.8
110	1.73	40	1.2	68	25.4
123	1.60	60	1.1	86	15.7
140	1.45	100	1.0	104	11.5
158	1.32	140	0.85	122	8.82
Sodium Hydroxide		Sulfur (gas free)		140	7.22
70	100	253	10.9	Turpentine	
100	40	276	8.7	60	2.11
120	25			100	2.0
140	15	301	7.1		
160	9.5	314	7.2	Water	
200	3.7	317	7.6	60	1.13
220	2.4	319	14.5	130	0.55
250	1.4	Sulfur Dioxide			
		-28	0.5508		
		13	0.4285		
		32	0.3936		

Table A5. The absolute viscosity for some common gases

Temperature °F	Viscosity Centipoise	Temperature °F	Viscosity Centipoise	Temperature °F	Viscosity Centipoise
Acetylene		Carbon Monoxide		Nitrogen	
32	0.00935	32	0.0166	-6.7	0.0156
Air		59	0.0172	51.6	0.0171
32	0.0171	260.8	0.0218	81	0.0178
104	0.0190	Chlorine		261	0.0219
444	0.0264	122	0.0147	440	0.0256
633	0.0305	212	0.0168	Oxygen	
674	0.0312	302	0.0187	32	0.0189
768	0.0341	392	0.0208	67	0.0202
Ammonia		Ethane		262	0.0257
32	0.0092	32	0.0085	440	0.0302
68	0.0098	63	0.0090	n-Pentane	
212	0.0128	Ethylene		77	0.0068
302	0.0146	32	0.0091	212	0.0084
482	0.0181	68	0.0101	Propane	
Argon		122	0.0110	64.2	0.0079
32	0.0210	212	0.0126	213	0.0101
68	0.0222	Helium		Propylene	
212	0.0269	32	0.0186	62	0.0083
392	0.0322	68	0.0194	122	0.0093
Benzene		Hydrogen		Sulfur Dioxide	
0	0.0065	-172	0.0057	32	0.0116
40	0.0070	-143.5	0.0062	64.4	0.0124
70	0.0075	-25	0.0077	68.9	0.0125
100	0.0080	32	0.0084	213	0.0161
200	0.0091	69	0.0088		
Butene		264	0.0108		
0	0.0075	Hydrogen Chloride			
40	0.0080	54.4	0.0139		
70	0.0085	61.8	0.0141		
100	0.0090	212	0.0182		
200	0.0104	Hydrogen Sulfide			
Butylene		32	0.0117		
66	0.0074	62.6	0.0124		
212	0.0095	212	0.0159		
Carbon Dioxide		Methane			
-144	0.0090	32	0.0102		
-76	0.0106	68	0.0109		
32	0.0139	212	0.0133		
68	0.0148				
86	0.0153				

Table A6. Thermocouple Table (Type J)

Thermoelectric Voltage in Millivolts

°F	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1	0	°F
-340					-8.095	-8.085	-8.074	-8.063	-8.052	-8.041	-8.030	-340
-330	-8.030	-8.019	-8.008	-7.996	-7.985	-7.973	-7.962	-7.950	-7.938	-7.927	-7.915	-330
-320	-7.915	-7.903	-7.890	-7.878	-7.866	-7.854	-7.841	-7.829	-7.816	-7.804	-7.791	-320
-310	-7.791	-7.778	-7.765	-7.752	-7.739	-7.726	-7.713	-7.699	-7.686	-7.672	-7.659	-310
-300	-7.659	-7.645	-7.632	-7.618	-7.604	-7.590	-7.576	-7.562	-7.548	-7.534	-7.519	-300
-290	-7.519	-7.505	-7.491	-7.476	-7.462	-7.447	-7.432	-7.417	-7.403	-7.388	-7.373	-290
-280	-7.373	-7.357	-7.342	-7.327	-7.312	-7.296	-7.281	-7.265	-7.250	-7.234	-7.219	-280
-270	-7.219	-7.203	-7.187	-7.171	-7.155	-7.139	-7.123	-7.107	-7.090	-7.074	-7.058	-270
-260	-7.058	-7.041	-7.025	-7.008	-6.991	-6.975	-6.958	-6.941	-6.924	-6.907	-6.890	-260
-250	-6.890	-6.873	-6.856	-6.839	-6.821	-6.804	-6.787	-6.769	-6.752	-6.734	-6.716	-250
-240	-6.716	-6.699	-6.681	-6.663	-6.645	-6.627	-6.609	-6.591	-6.573	-6.555	-6.536	-240
-230	-6.536	-6.518	-6.500	-6.481	-6.463	-6.444	-6.426	-6.407	-6.388	-6.370	-6.351	-230
-220	-6.351	-6.332	-6.313	-6.294	-6.275	-6.256	-6.236	-6.217	-6.198	-6.179	-6.159	-220
-210	-6.159	-6.140	-6.120	-6.101	-6.081	-6.061	-6.042	-6.022	-6.002	-5.982	-5.962	-210
-200	-5.962	-5.942	-5.922	-5.902	-5.882	-5.862	-5.842	-5.821	-5.801	-5.781	-5.760	-200
-190	-5.760	-5.740	-5.719	-5.699	-5.678	-5.657	-5.637	-5.616	-5.595	-5.574	-5.553	-190
-180	-5.553	-5.532	-5.511	-5.490	-5.469	-5.448	-5.426	-5.405	-5.384	-5.363	-5.341	-180
-170	-5.341	-5.320	-5.298	-5.277	-5.255	-5.233	-5.212	-5.190	-5.168	-5.146	-5.125	-170
-160	-5.125	-5.103	-5.081	-5.059	-5.037	-5.015	-4.992	-4.970	-4.948	-4.926	-4.903	-160
-150	-4.903	-4.881	-4.859	-4.836	-4.814	-4.791	-4.769	-4.746	-4.724	-4.701	-4.678	-150
-140	-4.678	-4.655	-4.633	-4.610	-4.587	-4.564	-4.541	-4.518	-4.495	-4.472	-4.449	-140
-130	-4.449	-4.425	-4.402	-4.379	-4.356	-4.332	-4.309	-4.286	-4.262	-4.239	-4.215	-130
-120	-4.215	-4.192	-4.168	-4.144	-4.121	-4.097	-4.073	-4.050	-4.026	-4.002	-3.978	-120
-110	-3.978	-3.954	-3.930	-3.906	-3.882	-3.858	-3.834	-3.810	-3.786	-3.761	-3.737	-110
-100	-3.737	-3.713	-3.688	-3.664	-3.640	-3.615	-3.591	-3.566	-3.542	-3.517	-3.493	-100
-90	-3.493	-3.468	-3.443	-3.419	-3.394	-3.369	-3.344	-3.320	-3.295	-3.270	-3.245	-90
-80	-3.245	-3.220	-3.195	-3.170	-3.145	-3.120	-3.095	-3.070	-3.044	-3.019	-2.994	-80
-70	-2.994	-2.969	-2.943	-2.918	-2.893	-2.867	-2.842	-2.817	-2.791	-2.766	-2.740	-70
-60	-2.740	-2.714	-2.689	-2.663	-2.638	-2.612	-2.586	-2.560	-2.535	-2.509	-2.483	-60
-50	-2.483	-2.457	-2.431	-2.405	-2.379	-2.353	-2.327	-2.301	-2.275	-2.249	-2.223	-50
-40	-2.223	-2.197	-2.171	-2.145	-2.118	-2.092	-2.066	-2.040	-2.013	-1.987	-1.961	-40
-30	-1.961	-1.934	-1.908	-1.881	-1.855	-1.828	-1.802	-1.775	-1.749	-1.722	-1.695	-30
-20	-1.695	-1.669	-1.642	-1.615	-1.589	-1.562	-1.535	-1.508	-1.482	-1.455	-1.428	-20
-10	-1.428	-1.401	-1.374	-1.347	-1.320	-1.293	-1.266	-1.239	-1.212	-1.185	-1.158	-10
0	-1.158	-1.131	-1.104	-1.076	-1.049	-1.022	-0.995	-0.967	-0.940	-0.913	-0.886	0
0	-0.886	-0.858	-0.831	-0.803	-0.776	-0.749	-0.721	-0.694	-0.666	-0.639	-0.611	0
10	-0.611	-0.583	-0.556	-0.528	-0.501	-0.473	-0.445	-0.418	-0.390	-0.362	-0.334	10
20	-0.334	-0.307	-0.279	-0.251	-0.223	-0.195	-0.168	-0.140	-0.112	-0.084	-0.056	20
30	-0.056	-0.028	0.000	0.028	0.056	0.084	0.112	0.140	0.168	0.196	0.225	30
40	0.225	0.253	0.281	0.309	0.337	0.365	0.394	0.422	0.450	0.478	0.507	40
50	0.507	0.535	0.563	0.592	0.620	0.649	0.677	0.705	0.734	0.762	0.791	50
60	0.791	0.819	0.848	0.876	0.905	0.933	0.962	0.991	1.019	1.048	1.076	60
70	1.076	1.105	1.134	1.162	1.191	1.220	1.249	1.277	1.306	1.335	1.364	70
80	1.364	1.392	1.421	1.450	1.479	1.508	1.537	1.566	1.594	1.623	1.652	80
90	1.652	1.681	1.710	1.739	1.768	1.797	1.826	1.855	1.884	1.913	1.942	90
100	1.942	1.972	2.001	2.030	2.059	2.088	2.117	2.146	2.175	2.205	2.234	100
110	2.234	2.263	2.292	2.322	2.351	2.380	2.409	2.439	2.468	2.497	2.527	110
120	2.527	2.556	2.585	2.615	2.644	2.673	2.703	2.732	2.762	2.791	2.821	120
130	2.821	2.850	2.880	2.909	2.938	2.968	2.997	3.027	3.057	3.086	3.116	130
140	3.116	3.145	3.175	3.204	3.234	3.264	3.293	3.323	3.353	3.382	3.412	140
150	3.412	3.442	3.471	3.501	3.531	3.560	3.590	3.620	3.650	3.679	3.709	150
160	3.709	3.739	3.769	3.798	3.828	3.858	3.888	3.918	3.948	3.977	4.007	160
170	4.007	4.037	4.067	4.097	4.127	4.157	4.187	4.217	4.246	4.276	4.306	170
180	4.306	4.336	4.366	4.396	4.426	4.456	4.486	4.516	4.546	4.576	4.606	180
190	4.606	4.636	4.666	4.696	4.726	4.757	4.787	4.817	4.847	4.877	4.907	190
200	4.907	4.937	4.967	4.997	5.028	5.058	5.088	5.118	5.148	5.178	5.209	200
210	5.209	5.239	5.269	5.299	5.329	5.360	5.390	5.420	5.450	5.480	5.511	210
220	5.511	5.541	5.571	5.602	5.632	5.662	5.692	5.723	5.753	5.783	5.814	220
230	5.814	5.844	5.874	5.905	5.935	5.965	5.996	6.026	6.056	6.087	6.117	230
240	6.117	6.147	6.178	6.208	6.239	6.269	6.299	6.330	6.360	6.391	6.421	240
250	6.421	6.452	6.482	6.512	6.543	6.573	6.604	6.634	6.665	6.695	6.726	250
260	6.726	6.756	6.787	6.817	6.848	6.878	6.909	6.939	6.970	7.000	7.031	260
270	7.031	7.061	7.092	7.122	7.153	7.184	7.214	7.245	7.275	7.306	7.336	270
280	7.336	7.367	7.398	7.428	7.459	7.489	7.520	7.550	7.581	7.612	7.642	280
290	7.642	7.673	7.704	7.734	7.765	7.795	7.826	7.857	7.887	7.918	7.949	290
°F	0	1	2	3	4	5	6	7	8	9	10	°F

°F	0	1	2	3	4	5	6	7	8	9	10	°F
300	7.949	7.979	8.010	8.041	8.071	8.102	8.133	8.163	8.194	8.225	8.255	300
310	8.255	8.286	8.317	8.347	8.378	8.409	8.439	8.470	8.501	8.532	8.562	310
320	8.562	8.593	8.624	8.654	8.685	8.716	8.747	8.777	8.808	8.839	8.869	320
330	8.869	8.900	8.931	8.962	8.992	9.023	9.054	9.085	9.115	9.146	9.177	330
340	9.177	9.208	9.238	9.269	9.300	9.331	9.362	9.392	9.423	9.454	9.485	340
350	9.485	9.515	9.546	9.577	9.608	9.639	9.669	9.700	9.731	9.762	9.793	350
360	9.793	9.823	9.854	9.885	9.916	9.947	9.977	10.008	10.039	10.070	10.101	360
370	10.101	10.131	10.162	10.193	10.224	10.255	10.285	10.316	10.347	10.378	10.409	370
380	10.409	10.440	10.470	10.501	10.532	10.563	10.594	10.625	10.655	10.686	10.717	380
390	10.717	10.748	10.779	10.810	10.840	10.871	10.902	10.933	10.964	10.995	11.025	390
400	11.025	11.056	11.087	11.118	11.149	11.180	11.211	11.241	11.272	11.303	11.334	400
410	11.334	11.365	11.396	11.426	11.457	11.488	11.519	11.550	11.581	11.612	11.642	410
420	11.642	11.673	11.704	11.735	11.766	11.797	11.828	11.858	11.889	11.920	11.951	420
430	11.951	11.982	12.013	12.044	12.074	12.105	12.136	12.167	12.198	12.229	12.260	430
440	12.260	12.290	12.321	12.352	12.383	12.414	12.445	12.476	12.506	12.537	12.568	440
450	12.568	12.599	12.630	12.661	12.691	12.722	12.753	12.784	12.815	12.846	12.877	450
460	12.877	12.907	12.938	12.969	13.000	13.031	13.062	13.093	13.123	13.154	13.185	460
470	13.185	13.216	13.247	13.278	13.308	13.339	13.370	13.401	13.432	13.463	13.494	470
480	13.494	13.524	13.555	13.586	13.617	13.648	13.679	13.709	13.740	13.771	13.802	480
490	13.802	13.833	13.864	13.894	13.925	13.956	13.987	14.018	14.049	14.079	14.110	490
500	14.110	14.141	14.172	14.203	14.233	14.264	14.295	14.326	14.357	14.388	14.418	500
510	14.418	14.449	14.480	14.511	14.542	14.573	14.603	14.634	14.665	14.696	14.727	510
520	14.727	14.757	14.788	14.819	14.850	14.881	14.911	14.942	14.973	15.004	15.035	520
530	15.035	15.065	15.096	15.127	15.158	15.189	15.219	15.250	15.281	15.312	15.343	530
540	15.343	15.373	15.404	15.435	15.466	15.496	15.527	15.558	15.589	15.620	15.650	540
550	15.650	15.681	15.712	15.743	15.773	15.804	15.835	15.866	15.897	15.927	15.958	550
560	15.958	15.989	16.020	16.050	16.081	16.112	16.143	16.173	16.204	16.235	16.266	560
570	16.266	16.296	16.327	16.358	16.389	16.419	16.450	16.481	16.512	16.542	16.573	570
580	16.573	16.604	16.635	16.665	16.696	16.727	16.758	16.788	16.819	16.850	16.881	580
590	16.881	16.911	16.942	16.973	17.003	17.034	17.065	17.096	17.126	17.157	17.188	590
600	17.188	17.219	17.249	17.280	17.311	17.341	17.372	17.403	17.434	17.464	17.495	600
610	17.495	17.526	17.556	17.587	17.618	17.649	17.679	17.710	17.741	17.771	17.802	610
620	17.802	17.833	17.863	17.894	17.925	17.955	17.986	18.017	18.048	18.078	18.109	620
630	18.109	18.140	18.170	18.201	18.232	18.262	18.293	18.324	18.354	18.385	18.416	630
640	18.416	18.446	18.477	18.508	18.538	18.569	18.600	18.630	18.661	18.692	18.722	640
650	18.722	18.753	18.784	18.814	18.845	18.876	18.906	18.937	18.968	18.998	19.029	650
660	19.029	19.060	19.090	19.121	19.152	19.182	19.213	19.244	19.274	19.305	19.336	660
670	19.336	19.366	19.397	19.428	19.458	19.489	19.520	19.550	19.581	19.612	19.642	670
680	19.642	19.673	19.704	19.734	19.765	19.795	19.826	19.857	19.887	19.918	19.949	680
690	19.949	19.979	20.010	20.041	20.071	20.102	20.132	20.163	20.194	20.224	20.255	690
700	20.255	20.286	20.316	20.347	20.378	20.408	20.439	20.469	20.500	20.531	20.561	700
710	20.561	20.592	20.623	20.653	20.684	20.715	20.745	20.776	20.806	20.837	20.868	710
720	20.868	20.898	20.929	20.960	20.990	21.021	21.052	21.082	21.113	21.143	21.174	720
730	21.174	21.205	21.235	21.266	21.297	21.327	21.358	21.389	21.419	21.450	21.480	730
740	21.480	21.511	21.542	21.572	21.603	21.634	21.664	21.695	21.726	21.756	21.787	740
750	21.787	21.817	21.848	21.879	21.909	21.940	21.971	22.001	22.032	22.063	22.093	750
760	22.093	22.124	22.154	22.185	22.216	22.246	22.277	22.308	22.338	22.369	22.400	760
770	22.400	22.430	22.461	22.492	22.522	22.553	22.584	22.614	22.645	22.676	22.706	770
780	22.706	22.737	22.768	22.798	22.829	22.860	22.890	22.921	22.952	22.982	23.013	780
790	23.013	23.044	23.074	23.105	23.136	23.166	23.197	23.228	23.258	23.289	23.320	790
800	23.320	23.350	23.381	23.412	23.442	23.473	23.504	23.535	23.565	23.596	23.627	800
810	23.627	23.657	23.688	23.719	23.749	23.780	23.811	23.842	23.872	23.903	23.934	810
820	23.934	23.964	23.995	24.026	24.057	24.087	24.118	24.149	24.180	24.210	24.241	820
830	24.241	24.272	24.303	24.333	24.364	24.395	24.426	24.456	24.487	24.518	24.549	830
840	24.549	24.579	24.610	24.641	24.672	24.702	24.733	24.764	24.795	24.826	24.856	840
850	24.856	24.887	24.918	24.949	24.979	25.010	25.041	25.072	25.103	25.134	25.164	850
860	25.164	25.195	25.226	25.257	25.288	25.318	25.349	25.380	25.411	25.442	25.473	860
870	25.473	25.504	25.534	25.565	25.596	25.627	25.658	25.689	25.720	25.750	25.781	870
880	25.781	25.812	25.843	25.874	25.905	25.936	25.967	25.998	26.028	26.059	26.090	880
890	26.090	26.121	26.152	26.183	26.214	26.245	26.276	26.307	26.338	26.369	26.400	890
900	26.400	26.431	26.462	26.493	26.524	26.555	26.586	26.617	26.648	26.679	26.710	900
910	26.710	26.741	26.772	26.803	26.834	26.865	26.896	26.927	26.958	26.989	27.020	910
920	27.020	27.051	27.082	27.113	27.144	27.175	27.206	27.237	27.268	27.299	27.330	920
930	27.330	27.362	27.393	27.424	27.455	27.486	27.517	27.548	27.579	27.610	27.642	930
940	27.642	27.673	27.704	27.735	27.766	27.797	27.829	27.860	27.891	27.922	27.953	940
°F	0	1	2	3	4	5	6	7	8	9	10	°F

Table A6. Thermocouple Table (Type J) Continued

Thermoelectric Voltage in Millivolts

°F	0	1	2	3	4	5	6	7	8	9	10	°F
950	27.953	27.985	28.016	28.047	28.078	28.109	28.141	28.172	28.203	28.234	28.266	950
960	28.266	28.297	28.328	28.359	28.391	28.422	28.453	28.485	28.516	28.547	28.579	960
970	28.579	28.610	28.641	28.672	28.704	28.735	28.767	28.798	28.829	28.861	28.892	970
980	28.892	28.923	28.955	28.986	29.018	29.049	29.080	29.112	29.143	29.175	29.206	980
990	29.206	29.238	29.269	29.301	29.332	29.363	29.395	29.426	29.458	29.489	29.521	990
1000	29.521	29.552	29.584	29.616	29.647	29.679	29.710	29.742	29.773	29.805	29.836	1000
1010	29.836	29.868	29.900	29.931	29.963	29.995	30.026	30.058	30.089	30.121	30.153	1010
1020	30.153	30.184	30.216	30.248	30.279	30.311	30.343	30.375	30.406	30.438	30.470	1020
1030	30.470	30.502	30.533	30.565	30.597	30.629	30.660	30.692	30.724	30.756	30.788	1030
1040	30.788	30.819	30.851	30.883	30.915	30.947	30.979	31.011	31.043	31.074	31.106	1040
1050	31.106	31.138	31.170	31.202	31.234	31.266	31.298	31.330	31.362	31.394	31.426	1050
1060	31.426	31.458	31.490	31.522	31.554	31.586	31.618	31.650	31.682	31.714	31.746	1060
1070	31.746	31.778	31.811	31.843	31.875	31.907	31.939	31.971	32.003	32.035	32.068	1070
1080	32.068	32.100	32.132	32.164	32.196	32.229	32.261	32.293	32.325	32.358	32.390	1080
1090	32.390	32.422	32.455	32.487	32.519	32.551	32.584	32.616	32.648	32.681	32.713	1090
1100	32.713	32.746	32.778	32.810	32.843	32.875	32.908	32.940	32.973	33.005	33.037	1100
1110	33.037	33.070	33.102	33.135	33.167	33.200	33.232	33.265	33.298	33.330	33.363	1110
1120	33.363	33.395	33.428	33.460	33.493	33.526	33.558	33.591	33.624	33.656	33.689	1120
1130	33.689	33.722	33.754	33.787	33.820	33.853	33.885	33.918	33.951	33.984	34.016	1130
1140	34.016	34.049	34.082	34.115	34.148	34.180	34.213	34.246	34.279	34.312	34.345	1140
1150	34.345	34.378	34.411	34.444	34.476	34.509	34.542	34.575	34.608	34.641	34.674	1150
1160	34.674	34.707	34.740	34.773	34.806	34.840	34.873	34.906	34.939	34.972	35.005	1160
1170	35.005	35.038	35.071	35.104	35.138	35.171	35.204	35.237	35.270	35.304	35.337	1170
1180	35.337	35.370	35.403	35.437	35.470	35.503	35.536	35.570	35.603	35.636	35.670	1180
1190	35.670	35.703	35.736	35.770	35.803	35.837	35.870	35.903	35.937	35.970	36.004	1190
1200	36.004	36.037	36.071	36.104	36.138	36.171	36.205	36.238	36.272	36.305	36.339	1200
1210	36.339	36.373	36.406	36.440	36.473	36.507	36.541	36.574	36.608	36.642	36.675	1210
1220	36.675	36.709	36.743	36.777	36.810	36.844	36.878	36.912	36.945	36.979	37.013	1220
1230	37.013	37.047	37.081	37.114	37.148	37.182	37.216	37.250	37.284	37.318	37.352	1230
1240	37.352	37.386	37.420	37.454	37.488	37.522	37.556	37.590	37.624	37.658	37.692	1240
1250	37.692	37.726	37.760	37.794	37.828	37.862	37.896	37.930	37.964	37.999	38.033	1250
1260	38.033	38.067	38.101	38.135	38.169	38.204	38.238	38.272	38.306	38.341	38.375	1260
1270	38.375	38.409	38.444	38.478	38.512	38.546	38.581	38.615	38.650	38.684	38.718	1270
1280	38.718	38.753	38.787	38.822	38.856	38.890	38.925	38.959	38.994	39.028	39.063	1280
1290	39.063	39.097	39.132	39.166	39.201	39.235	39.270	39.305	39.339	39.374	39.408	1290
1300	39.408	39.443	39.478	39.512	39.547	39.582	39.616	39.651	39.686	39.720	39.755	1300
1310	39.755	39.790	39.825	39.859	39.894	39.929	39.964	39.998	40.033	40.068	40.103	1310
1320	40.103	40.138	40.173	40.207	40.242	40.277	40.312	40.347	40.382	40.417	40.452	1320
1330	40.452	40.487	40.522	40.556	40.591	40.626	40.661	40.696	40.731	40.766	40.801	1330
1340	40.801	40.836	40.872	40.907	40.942	40.977	41.012	41.047	41.082	41.117	41.152	1340
1350	41.152	41.187	41.222	41.258	41.293	41.328	41.363	41.398	41.433	41.469	41.504	1350
1360	41.504	41.539	41.574	41.610	41.645	41.680	41.715	41.751	41.786	41.821	41.856	1360
1370	41.856	41.892	41.927	41.962	41.998	42.033	42.068	42.104	42.139	42.174	42.210	1370
1380	42.210	42.245	42.281	42.316	42.351	42.387	42.422	42.458	42.493	42.528	42.564	1380
1390	42.564	42.599	42.635	42.670	42.706	42.741	42.777	42.812	42.848	42.883	42.919	1390
1400	42.919	42.954	42.990	43.025	43.061	43.096	43.132	43.167	43.203	43.239	43.274	1400
1410	43.274	43.310	43.346	43.381	43.417	43.452	43.488	43.524	43.559	43.595	43.631	1410
1420	43.631	43.667	43.702	43.738	43.774	43.809	43.845	43.881	43.917	43.953	43.988	1420
1430	43.988	44.024	44.060	44.096	44.131	44.167	44.203	44.239	44.275	44.310	44.346	1430
1440	44.346	44.382	44.418	44.454	44.490	44.525	44.561	44.597	44.633	44.669	44.705	1440
1450	44.705	44.741	44.777	44.812	44.848	44.884	44.920	44.956	44.992	45.028	45.064	1450
1460	45.064	45.099	45.135	45.171	45.207	45.243	45.279	45.315	45.351	45.387	45.423	1460
1470	45.423	45.458	45.494	45.530	45.566	45.602	45.638	45.674	45.710	45.746	45.782	1470
1480	45.782	45.818	45.853	45.889	45.925	45.961	45.997	46.033	46.069	46.105	46.141	1480
1490	46.141	46.177	46.212	46.248	46.284	46.320	46.356	46.392	46.428	46.464	46.500	1490
1500	46.500	46.535	46.571	46.607	46.643	46.679	46.715	46.751	46.786	46.822	46.858	1500
1510	46.858	46.894	46.930	46.966	47.001	47.037	47.073	47.109	47.145	47.181	47.216	1510
1520	47.216	47.252	47.288	47.324	47.359	47.395	47.431	47.467	47.503	47.538	47.574	1520
1530	47.574	47.610	47.646	47.681	47.717	47.753	47.788	47.824	47.860	47.896	47.931	1530
1540	47.931	47.967	48.003	48.038	48.074	48.110	48.145	48.181	48.217	48.252	48.288	1540
1550	48.288	48.324	48.359	48.395	48.430	48.466	48.502	48.537	48.573	48.608	48.644	1550
1560	48.644	48.679	48.715	48.750	48.786	48.822	48.857	48.893	48.928	48.964	48.999	1560
1570	48.999	49.034	49.070	49.105	49.141	49.176	49.212	49.247	49.283	49.318	49.353	1570
1580	49.353	49.389	49.424	49.460	49.495	49.530	49.566	49.601	49.636	49.672	49.707	1580
1590	49.707	49.742	49.778	49.813	49.848	49.883	49.919	49.954	49.989	50.024	50.060	1590
°F	0	1	2	3	4	5	6	7	8	9	10	°F

°F	0	1	2	3	4	5	6	7	8	9	10	°F
1600	50.060	50.095	50.130	50.165	50.200	50.235	50.271	50.306	50.341	50.376	50.411	1600
1610	50.411	50.446	50.481	50.517	50.552	50.587	50.622	50.657	50.692	50.727	50.762	1610
1620	50.762	50.797	50.832	50.867	50.902	50.937	50.972	51.007	51.042	51.077	51.112	1620
1630	51.112	51.147	51.181	51.216	51.251	51.286	51.321	51.356	51.391	51.425	51.460	1630
1640	51.460	51.495	51.530	51.565	51.599	51.634	51.669	51.704	51.738	51.773	51.808	1640
1650	51.808	51.843	51.877	51.912	51.947	51.981	52.016	52.051	52.085	52.120	52.154	1650
1660	52.154	52.189	52.224	52.258	52.293	52.327	52.362	52.396	52.431	52.465	52.500	1660
1670	52.500	52.534	52.569	52.603	52.638	52.672	52.707	52.741	52.776	52.810	52.844	1670
1680	52.844	52.879	52.913	52.947	52.982	53.016	53.050	53.085	53.119	53.153	53.188	1680
1690	53.188	53.222	53.256	53.290	53.325	53.359	53.393	53.427	53.462	53.496	53.530	1690
1700	53.530	53.564	53.598	53.632	53.667	53.701	53.735	53.769	53.803	53.837	53.871	1700
1710	53.871	53.905	53.939	53.973	54.007	54.041	54.075	54.109	54.143	54.177	54.211	1710
1720	54.211	54.245	54.279	54.313	54.347	54.381	54.415	54.449	54.483	54.516	54.550	1720
1730	54.550	54.584	54.618	54.652	54.686	54.719	54.753	54.787	54.821	54.855	54.888	1730
1740	54.888	54.922	54.956	54.990	55.023	55.057	55.091	55.124	55.158	55.192	55.225	1740
1750	55.225	55.259	55.293	55.326	55.360	55.393	55.427	55.461	55.494	55.528	55.561	1750
1760	55.561	55.595	55.628	55.662	55.695	55.729	55.762	55.796	55.829	55.863	55.896	1760
1770	55.896	55.930	55.963	55.997	56.030	56.063	56.097	56.130	56.164	56.197	56.230	1770
1780	56.230	56.264	56.297	56.330	56.364	56.397	56.430	56.464	56.497	56.530	56.564	1780
1790	56.564	56.597	56.630	56.663	56.697	56.730	56.763	56.796	56.829	56.863	56.896	1790
1800	56.896	56.929	56.962	56.995	57.028	57.062	57.095	57.128	57.161	57.194	57.227	1800
1810	57.227	57.260	57.293	57.326	57.360	57.393	57.426	57.459	57.492	57.525	57.558	1810
1820	57.558	57.591	57.624	57.657	57.690	57.723	57.756	57.789	57.822	57.855	57.888	1820
1830	57.888	57.920	57.953	57.986	58.019	58.052	58.085	58.118	58.151	58.184	58.217	1830
1840	58.217	58.249	58.282	58.315	58.348	58.381	58.414	58.446	58.479	58.512	58.545	1840
1850	58.545	58.578	58.610	58.643	58.676	58.709	58.741	58.774	58.807	58.840	58.872	1850
1860	58.872	58.905	58.938	58.971	59.003	59.036	59.069	59.101	59.134	59.167	59.199	1860
1870	59.199	59.232	59.265	59.297	59.330	59.363	59.395	59.428	59.460	59.493	59.526	1870
1880	59.526	59.558	59.591	59.623	59.656	59.689	59.721	59.754	59.786	59.819	59.851	1880
1890	59.851	59.884	59.916	59.949	59.982	60.014	60.047	60.079	60.112	60.144	60.177	1890
1900	60.177	60.209	60.242	60.274	60.307	60.339	60.371	60.404	60.436	60.469	60.501	1900
1910	60.501	60.534	60.566	60.599	60.631	60.663	60.696	60.728	60.761	60.793	60.826	1910
1920	60.826	60.858	60.890	60.923	60.955	60.987	61.020	61.052	61.085	61.117	61.149	1920
1930	61.149	61.182	61.214	61.246	61.279	61.311	61.343	61.376	61.408	61.440	61.473	1930
1940	61.473	61.505	61.537	61.570	61.602	61.634	61.667	61.699	61.731	61.763	61.796	1940
1950	61.796	61.828	61.860	61.893	61.925	61.957	61.989	62.022	62.054	62.086	62.118	1950
1960	62.118	62.151	62.183	62.215	62.247	62.280	62.312	62.344	62.376	62.409	62.441	1960
1970	62.441	62.473	62.505	62.537	62.570	62.602	62.634	62.666	62.699	62.731	62.763	1970
1980	62.763	62.795	62.827	62.860	62.892	62.924	62.956	62.988	63.020	63.053	63.085	1980
1990	63.085	63.117	63.149	63.181	63.214	63.246	63.278	63.310	63.342	63.374	63.406	1990
2000	63.406	63.439	63.471	63.503	63.535	63.567	63.599	63.632	63.664	63.696	63.728	2000
2010	63.728	63.760	63.792	63.824	63.856	63.889	63.921	63.953	63.985	64.017	64.049	2010
2020	64.049	64.081	64.113	64.146	64.178	64.210	64.242	64.274	64.306	64.338	64.370	2020
2030	64.370	64.402	64.435	64.467	64.499	64.531	64.563	64.595	64.627	64.659	64.691	2030
2040	64.691	64.723	64.756	64.788	64.820	64.852	64.884	64.916	64.948	64.980	65.012	2040
2050	65.012	65.044	65.076	65.109	65.141	65.173	65.205	65.237	65.269	65.301	65.333	2050
2060	65.333	65.365	65.397	65.429	65.461	65.493	65.525	65.557	65.590	65.622	65.654	2060
2070	65.654	65.686	65.718	65.750	65.782	65.814	65.846	65.878	65.910	65.942	65.974	2070
2080	65.974	66.006	66.038	66.070	66.102	66.134	66.166	66.199	66.231	66.263	66.295	2080
2090	66.295	66.327	66.359	66.391	66.423	66.455	66.487	66.519	66.551	66.583	66.615	2090
2100	66.615	66.647	66.679	66.711	66.743	66.775	66.807	66.839	66.871	66.903	66.935	2100
2110	66.935	66.967	66.999	67.031	67.063	67.095	67.127	67.159	67.191	67.223	67.255	2110
2120	67.255	67.287	67.319	67.351	67.383	67.415	67.447	67.479	67.511	67.543	67.575	2120
2130	67.575	67.607	67.639	67.671	67.703	67.735	67.767	67.799	67.831	67.863	67.895	2130
2140	67.895	67.927	67.959	67.991	68.023	68.055	68.087	68.119	68.150	68.182	68.214	2140
2150	68.214	68.246	68.278	68.310	68.342	68.374	68.406	68.438	68.470	68.502	68.534	2150
2160	68.534	68.566	68.597	68.629	68.661	68.693	68.725	68.757	68.789	68.821	68.853	2160
2170	68.853	68.884	68.916	68.948	68.980	69.012	69.044	69.076	69.108	69.139	69.171	2170
2180	69.171	69.203	69.235	69.267	69.299	69.330	69.362	69.394	69.426	69.458	69.490	2180
2190	69.490	69.521	69.553									2190
°F	0	1	2	3	4	5	6	7	8	9	10	°F

Table A7. Thermocouple Table (Type K)

Thermoelectric Voltage in Millivolts

°F	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1	0	°F
-450							-6.458	-6.457	-6.457	-6.456	-6.456	-450
-440	-6.456	-6.455	-6.454	-6.454	-6.453	-6.452	-6.451	-6.450	-6.449	-6.448	-6.446	-440
-430	-6.446	-6.445	-6.444	-6.443	-6.441	-6.440	-6.438	-6.436	-6.435	-6.433	-6.431	-430
-420	-6.431	-6.429	-6.427	-6.425	-6.423	-6.421	-6.419	-6.416	-6.414	-6.411	-6.409	-420
-410	-6.409	-6.406	-6.404	-6.401	-6.398	-6.395	-6.392	-6.389	-6.386	-6.383	-6.380	-410
-400	-6.380	-6.377	-6.373	-6.370	-6.366	-6.363	-6.359	-6.355	-6.352	-6.348	-6.344	-400
-390	-6.344	-6.340	-6.336	-6.332	-6.328	-6.323	-6.319	-6.315	-6.310	-6.306	-6.301	-390
-380	-6.301	-6.296	-6.292	-6.287	-6.282	-6.277	-6.272	-6.267	-6.262	-6.257	-6.251	-380
-370	-6.251	-6.246	-6.241	-6.235	-6.230	-6.224	-6.218	-6.213	-6.207	-6.201	-6.195	-370
-360	-6.195	-6.189	-6.183	-6.177	-6.171	-6.165	-6.158	-6.152	-6.146	-6.139	-6.133	-360
-350	-6.133	-6.126	-6.119	-6.113	-6.106	-6.099	-6.092	-6.085	-6.078	-6.071	-6.064	-350
-340	-6.064	-6.057	-6.049	-6.042	-6.035	-6.027	-6.020	-6.012	-6.004	-5.997	-5.989	-340
-330	-5.989	-5.981	-5.973	-5.965	-5.957	-5.949	-5.941	-5.933	-5.925	-5.917	-5.908	-330
-320	-5.908	-5.900	-5.891	-5.883	-5.874	-5.866	-5.857	-5.848	-5.840	-5.831	-5.822	-320
-310	-5.822	-5.813	-5.804	-5.795	-5.786	-5.776	-5.767	-5.758	-5.749	-5.739	-5.730	-310
-300	-5.730	-5.720	-5.711	-5.701	-5.691	-5.682	-5.672	-5.662	-5.652	-5.642	-5.632	-300
-290	-5.632	-5.622	-5.612	-5.602	-5.592	-5.581	-5.571	-5.561	-5.550	-5.540	-5.529	-290
-280	-5.529	-5.519	-5.508	-5.497	-5.487	-5.476	-5.465	-5.454	-5.443	-5.432	-5.421	-280
-270	-5.421	-5.410	-5.399	-5.388	-5.377	-5.365	-5.354	-5.343	-5.331	-5.320	-5.308	-270
-260	-5.308	-5.296	-5.285	-5.273	-5.261	-5.250	-5.238	-5.226	-5.214	-5.202	-5.190	-260
-250	-5.190	-5.178	-5.166	-5.153	-5.141	-5.129	-5.117	-5.104	-5.092	-5.079	-5.067	-250
-240	-5.067	-5.054	-5.042	-5.029	-5.016	-5.003	-4.991	-4.978	-4.965	-4.952	-4.939	-240
-230	-4.939	-4.926	-4.913	-4.900	-4.886	-4.873	-4.860	-4.847	-4.833	-4.820	-4.806	-230
-220	-4.806	-4.793	-4.779	-4.766	-4.752	-4.738	-4.724	-4.711	-4.697	-4.683	-4.669	-220
-210	-4.669	-4.655	-4.641	-4.627	-4.613	-4.599	-4.584	-4.570	-4.556	-4.542	-4.527	-210
-200	-4.527	-4.513	-4.498	-4.484	-4.469	-4.455	-4.440	-4.425	-4.411	-4.396	-4.381	-200
-190	-4.381	-4.366	-4.351	-4.336	-4.321	-4.306	-4.291	-4.276	-4.261	-4.246	-4.231	-190
-180	-4.231	-4.215	-4.200	-4.185	-4.169	-4.154	-4.138	-4.123	-4.107	-4.091	-4.076	-180
-170	-4.076	-4.060	-4.044	-4.029	-4.013	-3.997	-3.981	-3.965	-3.949	-3.933	-3.917	-170
-160	-3.917	-3.901	-3.885	-3.869	-3.852	-3.836	-3.820	-3.803	-3.787	-3.771	-3.754	-160
-150	-3.754	-3.738	-3.721	-3.705	-3.688	-3.671	-3.655	-3.638	-3.621	-3.604	-3.587	-150
-140	-3.587	-3.571	-3.554	-3.537	-3.520	-3.503	-3.486	-3.468	-3.451	-3.434	-3.417	-140
-130	-3.417	-3.400	-3.382	-3.365	-3.348	-3.330	-3.313	-3.295	-3.278	-3.260	-3.243	-130
-120	-3.243	-3.225	-3.207	-3.190	-3.172	-3.154	-3.136	-3.119	-3.101	-3.083	-3.065	-120
-110	-3.065	-3.047	-3.029	-3.011	-2.993	-2.975	-2.957	-2.938	-2.920	-2.902	-2.884	-110
-100	-2.884	-2.865	-2.847	-2.829	-2.810	-2.792	-2.773	-2.755	-2.736	-2.718	-2.699	-100
-90	-2.699	-2.680	-2.662	-2.643	-2.624	-2.605	-2.587	-2.568	-2.549	-2.530	-2.511	-90
-80	-2.511	-2.492	-2.473	-2.454	-2.435	-2.416	-2.397	-2.378	-2.359	-2.339	-2.320	-80
-70	-2.320	-2.301	-2.282	-2.262	-2.243	-2.223	-2.204	-2.185	-2.165	-2.146	-2.126	-70
-60	-2.126	-2.106	-2.087	-2.067	-2.048	-2.028	-2.008	-1.988	-1.969	-1.949	-1.929	-60
-50	-1.929	-1.909	-1.889	-1.869	-1.850	-1.830	-1.810	-1.790	-1.770	-1.749	-1.729	-50
-40	-1.729	-1.709	-1.689	-1.669	-1.649	-1.628	-1.608	-1.588	-1.568	-1.547	-1.527	-40
-30	-1.527	-1.507	-1.486	-1.466	-1.445	-1.425	-1.404	-1.384	-1.363	-1.343	-1.322	-30
-20	-1.322	-1.301	-1.281	-1.260	-1.239	-1.218	-1.198	-1.177	-1.156	-1.135	-1.114	-20
-10	-1.114	-1.094	-1.073	-1.052	-1.031	-1.010	-0.989	-0.968	-0.947	-0.926	-0.905	-10
0	-0.905	-0.883	-0.862	-0.841	-0.820	-0.799	-0.778	-0.756	-0.735	-0.714	-0.692	0
0	-0.692	-0.671	-0.650	-0.628	-0.607	-0.586	-0.564	-0.543	-0.521	-0.500	-0.478	0
10	-0.478	-0.457	-0.435	-0.413	-0.392	-0.370	-0.349	-0.327	-0.305	-0.284	-0.262	10
20	-0.262	-0.240	-0.218	-0.197	-0.175	-0.153	-0.131	-0.109	-0.088	-0.066	-0.044	20
30	-0.044	-0.022	0.000	0.022	0.044	0.066	0.088	0.110	0.132	0.154	0.176	30
40	0.176	0.198	0.220	0.242	0.264	0.286	0.308	0.330	0.353	0.375	0.397	40
50	0.397	0.419	0.441	0.463	0.486	0.508	0.530	0.552	0.575	0.597	0.619	50
60	0.619	0.642	0.664	0.686	0.709	0.731	0.753	0.776	0.798	0.821	0.843	60
70	0.843	0.865	0.888	0.910	0.933	0.955	0.978	1.000	1.023	1.045	1.068	70
80	1.068	1.090	1.113	1.136	1.158	1.181	1.203	1.226	1.249	1.271	1.294	80
90	1.294	1.316	1.339	1.362	1.384	1.407	1.430	1.453	1.475	1.498	1.521	90
°F	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**°F**

°F	0	1	2	3	4	5	6	7	8	9	10	°F
100	1.521	1.543	1.566	1.589	1.612	1.635	1.657	1.680	1.703	1.726	1.749	100
110	1.749	1.771	1.794	1.817	1.840	1.863	1.886	1.909	1.931	1.954	1.977	110
120	1.977	2.000	2.023	2.046	2.069	2.092	2.115	2.138	2.161	2.184	2.207	120
130	2.207	2.230	2.253	2.276	2.298	2.321	2.344	2.367	2.390	2.413	2.436	130
140	2.436	2.459	2.483	2.506	2.529	2.552	2.575	2.598	2.621	2.644	2.667	140
150	2.667	2.690	2.713	2.736	2.759	2.782	2.805	2.828	2.851	2.874	2.897	150
160	2.897	2.920	2.944	2.967	2.990	3.013	3.036	3.059	3.082	3.105	3.128	160
170	3.128	3.151	3.174	3.197	3.220	3.244	3.267	3.290	3.313	3.336	3.359	170
180	3.359	3.382	3.405	3.428	3.451	3.474	3.497	3.520	3.544	3.567	3.590	180
190	3.590	3.613	3.636	3.659	3.682	3.705	3.728	3.751	3.774	3.797	3.820	190
200	3.820	3.843	3.866	3.889	3.912	3.935	3.958	3.981	4.004	4.027	4.050	200
210	4.050	4.073	4.096	4.119	4.142	4.165	4.188	4.211	4.234	4.257	4.280	210
220	4.280	4.303	4.326	4.349	4.372	4.395	4.417	4.440	4.463	4.486	4.509	220
230	4.509	4.532	4.555	4.578	4.601	4.623	4.646	4.669	4.692	4.715	4.738	230
240	4.738	4.760	4.783	4.806	4.829	4.852	4.874	4.897	4.920	4.943	4.965	240
250	4.965	4.988	5.011	5.034	5.056	5.079	5.102	5.124	5.147	5.170	5.192	250
260	5.192	5.215	5.238	5.260	5.283	5.306	5.328	5.351	5.374	5.396	5.419	260
270	5.419	5.441	5.464	5.487	5.509	5.532	5.554	5.577	5.599	5.622	5.644	270
280	5.644	5.667	5.690	5.712	5.735	5.757	5.779	5.802	5.824	5.847	5.869	280
290	5.869	5.892	5.914	5.937	5.959	5.982	6.004	6.026	6.049	6.071	6.094	290
300	6.094	6.116	6.138	6.161	6.183	6.205	6.228	6.250	6.272	6.295	6.317	300
310	6.317	6.339	6.362	6.384	6.406	6.429	6.451	6.473	6.496	6.518	6.540	310
320	6.540	6.562	6.585	6.607	6.629	6.652	6.674	6.696	6.718	6.741	6.763	320
330	6.763	6.785	6.807	6.829	6.852	6.874	6.896	6.918	6.941	6.963	6.985	330
340	6.985	7.007	7.029	7.052	7.074	7.096	7.118	7.140	7.163	7.185	7.207	340
350	7.207	7.229	7.251	7.273	7.296	7.318	7.340	7.362	7.384	7.407	7.429	350
360	7.429	7.451	7.473	7.495	7.517	7.540	7.562	7.584	7.606	7.628	7.650	360
370	7.650	7.673	7.695	7.717	7.739	7.761	7.783	7.806	7.828	7.850	7.872	370
380	7.872	7.894	7.917	7.939	7.961	7.983	8.005	8.027	8.050	8.072	8.094	380
390	8.094	8.116	8.138	8.161	8.183	8.205	8.227	8.250	8.272	8.294	8.316	390
400	8.316	8.338	8.361	8.383	8.405	8.427	8.450	8.472	8.494	8.516	8.539	400
410	8.539	8.561	8.583	8.605	8.628	8.650	8.672	8.694	8.717	8.739	8.761	410
420	8.761	8.784	8.806	8.828	8.851	8.873	8.895	8.918	8.940	8.962	8.985	420
430	8.985	9.007	9.029	9.052	9.074	9.096	9.119	9.141	9.163	9.186	9.208	430
440	9.208	9.231	9.253	9.275	9.298	9.320	9.343	9.365	9.388	9.410	9.432	440
450	9.432	9.455	9.477	9.500	9.522	9.545	9.567	9.590	9.612	9.635	9.657	450
460	9.657	9.680	9.702	9.725	9.747	9.770	9.792	9.815	9.837	9.860	9.882	460
470	9.882	9.905	9.927	9.950	9.973	9.995	10.018	10.040	10.063	10.086	10.108	470
480	10.108	10.131	10.153	10.176	10.199	10.221	10.244	10.267	10.289	10.312	10.334	480
490	10.334	10.357	10.380	10.402	10.425	10.448	10.471	10.493	10.516	10.539	10.561	490
500	10.561	10.584	10.607	10.629	10.652	10.675	10.698	10.720	10.743	10.766	10.789	500
510	10.789	10.811	10.834	10.857	10.880	10.903	10.925	10.948	10.971	10.994	11.017	510
520	11.017	11.039	11.062	11.085	11.108	11.131	11.154	11.176	11.199	11.222	11.245	520
530	11.245	11.268	11.291	11.313	11.336	11.359	11.382	11.405	11.428	11.451	11.474	530
540	11.474	11.497	11.519	11.542	11.565	11.588	11.611	11.634	11.657	11.680	11.703	540
550	11.703	11.726	11.749	11.772	11.795	11.818	11.841	11.864	11.887	11.910	11.933	550
560	11.933	11.956	11.978	12.001	12.024	12.047	12.070	12.093	12.116	12.140	12.163	560
570	12.163	12.186	12.209	12.232	12.255	12.278	12.301	12.324	12.347	12.370	12.393	570
580	12.393	12.416	12.439	12.462	12.485	12.508	12.531	12.554	12.577	12.600	12.624	580
590	12.624	12.647	12.670	12.693	12.716	12.739	12.762	12.785	12.808	12.831	12.855	590
600	12.855	12.878	12.901	12.924	12.947	12.970	12.993	13.016	13.040	13.063	13.086	600
610	13.086	13.109	13.132	13.155	13.179	13.202	13.225	13.248	13.271	13.294	13.318	610
620	13.318	13.341	13.364	13.387	13.410	13.433	13.457	13.480	13.503	13.526	13.549	620
630	13.549	13.573	13.596	13.619	13.642	13.665	13.689	13.712	13.735	13.758	13.782	630
640	13.782	13.805	13.828	13.851	13.874	13.898	13.921	13.944	13.967	13.991	14.014	640
650	14.014	14.037	14.060	14.084	14.107	14.130	14.154	14.177	14.200	14.223	14.247	650
660	14.247	14.270	14.293	14.316	14.340	14.363	14.386	14.410	14.433	14.456	14.479	660
670	14.479	14.503	14.526	14.549	14.573	14.596	14.619	14.643	14.666	14.689	14.713	670
680	14.713	14.736	14.759	14.783	14.806	14.829	14.853	14.876	14.899	14.923	14.946	680
690	14.946	14.969	14.993	15.016	15.039	15.063	15.086	15.109	15.133	15.156	15.179	690
°F	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**°F**

Table A7. Thermocouple Table (Type K) Continued

Thermoelectric Voltage in Millivolts

°F	0	1	2	3	4	5	6	7	8	9	10	°F
700	15.179	15.203	15.226	15.250	15.273	15.296	15.320	15.343	15.366	15.390	15.413	700
710	15.413	15.437	15.460	15.483	15.507	15.530	15.554	15.577	15.600	15.624	15.647	710
720	15.647	15.671	15.694	15.717	15.741	15.764	15.788	15.811	15.834	15.858	15.881	720
730	15.881	15.905	15.928	15.952	15.975	15.998	16.022	16.045	16.069	16.092	16.116	730
740	16.116	16.139	16.163	16.186	16.209	16.233	16.256	16.280	16.303	16.327	16.350	740
750	16.350	16.374	16.397	16.421	16.444	16.468	16.491	16.514	16.538	16.561	16.585	750
760	16.585	16.608	16.632	16.655	16.679	16.702	16.726	16.749	16.773	16.796	16.820	760
770	16.820	16.843	16.867	16.890	16.914	16.937	16.961	16.984	17.008	17.031	17.055	770
780	17.055	17.078	17.102	17.125	17.149	17.173	17.196	17.220	17.243	17.267	17.290	780
790	17.290	17.314	17.337	17.361	17.384	17.408	17.431	17.455	17.478	17.502	17.526	790
800	17.526	17.549	17.573	17.596	17.620	17.643	17.667	17.690	17.714	17.738	17.761	800
810	17.761	17.785	17.808	17.832	17.855	17.879	17.902	17.926	17.950	17.973	17.997	810
820	17.997	18.020	18.044	18.068	18.091	18.115	18.138	18.162	18.185	18.209	18.233	820
830	18.233	18.256	18.280	18.303	18.327	18.351	18.374	18.398	18.421	18.445	18.469	830
840	18.469	18.492	18.516	18.539	18.563	18.587	18.610	18.634	18.657	18.681	18.705	840
850	18.705	18.728	18.752	18.776	18.799	18.823	18.846	18.870	18.894	18.917	18.941	850
860	18.941	18.965	18.988	19.012	19.035	19.059	19.083	19.106	19.130	19.154	19.177	860
870	19.177	19.201	19.224	19.248	19.272	19.295	19.319	19.343	19.366	19.390	19.414	870
880	19.414	19.437	19.461	19.485	19.508	19.532	19.556	19.579	19.603	19.626	19.650	880
890	19.650	19.674	19.697	19.721	19.745	19.768	19.792	19.816	19.839	19.863	19.887	890
900	19.887	19.910	19.934	19.958	19.981	20.005	20.029	20.052	20.076	20.100	20.123	900
910	20.123	20.147	20.171	20.194	20.218	20.242	20.265	20.289	20.313	20.336	20.360	910
920	20.360	20.384	20.407	20.431	20.455	20.479	20.502	20.526	20.550	20.573	20.597	920
930	20.597	20.621	20.644	20.668	20.692	20.715	20.739	20.763	20.786	20.810	20.834	930
940	20.834	20.857	20.881	20.905	20.929	20.952	20.976	21.000	21.023	21.047	21.071	940
950	21.071	21.094	21.118	21.142	21.165	21.189	21.213	21.236	21.260	21.284	21.308	950
960	21.308	21.331	21.355	21.379	21.402	21.426	21.450	21.473	21.497	21.521	21.544	960
970	21.544	21.568	21.592	21.616	21.639	21.663	21.687	21.710	21.734	21.758	21.781	970
980	21.781	21.805	21.829	21.852	21.876	21.900	21.924	21.947	21.971	21.995	22.018	980
990	22.018	22.042	22.066	22.089	22.113	22.137	22.160	22.184	22.208	22.232	22.255	990
1000	22.255	22.279	22.303	22.326	22.350	22.374	22.397	22.421	22.445	22.468	22.492	1000
1010	22.492	22.516	22.540	22.563	22.587	22.611	22.634	22.658	22.682	22.705	22.729	1010
1020	22.729	22.753	22.776	22.800	22.824	22.847	22.871	22.895	22.919	22.942	22.966	1020
1030	22.966	22.990	23.013	23.037	23.061	23.084	23.108	23.132	23.155	23.179	23.203	1030
1040	23.203	23.226	23.250	23.274	23.297	23.321	23.345	23.368	23.392	23.416	23.439	1040
1050	23.439	23.463	23.487	23.510	23.534	23.558	23.581	23.605	23.629	23.652	23.676	1050
1060	23.676	23.700	23.723	23.747	23.771	23.794	23.818	23.842	23.865	23.889	23.913	1060
1070	23.913	23.936	23.960	23.984	24.007	24.031	24.055	24.078	24.102	24.126	24.149	1070
1080	24.149	24.173	24.197	24.220	24.244	24.267	24.291	24.315	24.338	24.362	24.386	1080
1090	24.386	24.409	24.433	24.457	24.480	24.504	24.527	24.551	24.575	24.598	24.622	1090
1100	24.622	24.646	24.669	24.693	24.717	24.740	24.764	24.787	24.811	24.835	24.858	1100
1110	24.858	24.882	24.905	24.929	24.953	24.976	25.000	25.024	25.047	25.071	25.094	1110
1120	25.094	25.118	25.142	25.165	25.189	25.212	25.236	25.260	25.283	25.307	25.330	1120
1130	25.330	25.354	25.377	25.401	25.425	25.448	25.472	25.495	25.519	25.543	25.566	1130
1140	25.566	25.590	25.613	25.637	25.660	25.684	25.708	25.731	25.755	25.778	25.802	1140
1150	25.802	25.825	25.849	25.873	25.896	25.920	25.943	25.967	25.990	26.014	26.037	1150
1160	26.037	26.061	26.084	26.108	26.132	26.155	26.179	26.202	26.226	26.249	26.273	1160
1170	26.273	26.296	26.320	26.343	26.367	26.390	26.414	26.437	26.461	26.484	26.508	1170
1180	26.508	26.532	26.555	26.579	26.602	26.626	26.649	26.673	26.696	26.720	26.743	1180
1190	26.743	26.767	26.790	26.814	26.837	26.861	26.884	26.907	26.931	26.954	26.978	1190
1200	26.978	27.001	27.025	27.048	27.072	27.095	27.119	27.142	27.166	27.189	27.213	1200
1210	27.213	27.236	27.259	27.283	27.306	27.330	27.353	27.377	27.400	27.424	27.447	1210
1220	27.447	27.471	27.494	27.517	27.541	27.564	27.588	27.611	27.635	27.658	27.681	1220
1230	27.681	27.705	27.728	27.752	27.775	27.798	27.822	27.845	27.869	27.892	27.915	1230
1240	27.915	27.939	27.962	27.986	28.009	28.032	28.056	28.079	28.103	28.126	28.149	1240
1250	28.149	28.173	28.196	28.219	28.243	28.266	28.289	28.313	28.336	28.360	28.383	1250
1260	28.383	28.406	28.430	28.453	28.476	28.500	28.523	28.546	28.570	28.593	28.616	1260
1270	28.616	28.640	28.663	28.686	28.710	28.733	28.756	28.780	28.803	28.826	28.849	1270
1280	28.849	28.873	28.896	28.919	28.943	28.966	28.989	29.013	29.036	29.059	29.082	1280
1290	29.082	29.106	29.129	29.152	29.176	29.199	29.222	29.245	29.269	29.292	29.315	1290
°F	0	1	2	3	4	5	6	7	8	9	10	°F

°F	0	1	2	3	4	5	6	7	8	9	10	°F
1300	29.315	29.338	29.362	29.385	29.408	29.431	29.455	29.478	29.501	29.524	29.548	1300
1310	29.548	29.571	29.594	29.617	29.640	29.664	29.687	29.710	29.733	29.757	29.780	1310
1320	29.780	29.803	29.826	29.849	29.873	29.896	29.919	29.942	29.965	29.989	30.012	1320
1330	30.012	30.035	30.058	30.081	30.104	30.128	30.151	30.174	30.197	30.220	30.243	1330
1340	30.243	30.267	30.290	30.313	30.336	30.359	30.382	30.405	30.429	30.452	30.475	1340
1350	30.475	30.498	30.521	30.544	30.567	30.590	30.613	30.637	30.660	30.683	30.706	1350
1360	30.706	30.729	30.752	30.775	30.798	30.821	30.844	30.868	30.891	30.914	30.937	1360
1370	30.937	30.960	30.983	31.006	31.029	31.052	31.075	31.098	31.121	31.144	31.167	1370
1380	31.167	31.190	31.213	31.236	31.260	31.283	31.306	31.329	31.352	31.375	31.398	1380
1390	31.398	31.421	31.444	31.467	31.490	31.513	31.536	31.559	31.582	31.605	31.628	1390
1400	31.628	31.651	31.674	31.697	31.720	31.743	31.766	31.789	31.812	31.834	31.857	1400
1410	31.857	31.880	31.903	31.926	31.949	31.972	31.995	32.018	32.041	32.064	32.087	1410
1420	32.087	32.110	32.133	32.156	32.179	32.202	32.224	32.247	32.270	32.293	32.316	1420
1430	32.316	32.339	32.362	32.385	32.408	32.431	32.453	32.476	32.499	32.522	32.545	1430
1440	32.545	32.568	32.591	32.614	32.636	32.659	32.682	32.705	32.728	32.751	32.774	1440
1450	32.774	32.796	32.819	32.842	32.865	32.888	32.911	32.933	32.956	32.979	33.002	1450
1460	33.002	33.025	33.047	33.070	33.093	33.116	33.139	33.161	33.184	33.207	33.230	1460
1470	33.230	33.253	33.275	33.298	33.321	33.344	33.366	33.389	33.412	33.435	33.458	1470
1480	33.458	33.480	33.503	33.526	33.548	33.571	33.594	33.617	33.639	33.662	33.685	1480
1490	33.685	33.708	33.730	33.753	33.776	33.798	33.821	33.844	33.867	33.889	33.912	1490
1500	33.912	33.935	33.957	33.980	34.003	34.025	34.048	34.071	34.093	34.116	34.139	1500
1510	34.139	34.161	34.184	34.207	34.229	34.252	34.275	34.297	34.320	34.343	34.365	1510
1520	34.365	34.388	34.410	34.433	34.456	34.478	34.501	34.524	34.546	34.569	34.591	1520
1530	34.591	34.614	34.637	34.659	34.682	34.704	34.727	34.750	34.772	34.795	34.817	1530
1540	34.817	34.840	34.862	34.885	34.908	34.930	34.953	34.975	34.998	35.020	35.043	1540
1550	35.043	35.065	35.088	35.110	35.133	35.156	35.178	35.201	35.223	35.246	35.268	1550
1560	35.268	35.291	35.313	35.336	35.358	35.381	35.403	35.426	35.448	35.471	35.493	1560
1570	35.493	35.516	35.538	35.560	35.583	35.605	35.628	35.650	35.673	35.695	35.718	1570
1580	35.718	35.740	35.763	35.785	35.807	35.830	35.852	35.875	35.897	35.920	35.942	1580
1590	35.942	35.964	35.987	36.009	36.032	36.054	36.076	36.099	36.121	36.144	36.166	1590
1600	36.166	36.188	36.211	36.233	36.256	36.278	36.300	36.323	36.345	36.367	36.390	1600
1610	36.390	36.412	36.434	36.457	36.479	36.501	36.524	36.546	36.568	36.591	36.613	1610
1620	36.613	36.635	36.658	36.680	36.702	36.725	36.747	36.769	36.792	36.814	36.836	1620
1630	36.836	36.859	36.881	36.903	36.925	36.948	36.970	36.992	37.014	37.037	37.059	1630
1640	37.059	37.081	37.104	37.126	37.148	37.170	37.193	37.215	37.237	37.259	37.281	1640
1650	37.281	37.304	37.326	37.348	37.370	37.393	37.415	37.437	37.459	37.481	37.504	1650
1660	37.504	37.526	37.548	37.570	37.592	37.615	37.637	37.659	37.681	37.703	37.725	1660
1670	37.725	37.748	37.770	37.792	37.814	37.836	37.858	37.881	37.903	37.925	37.947	1670
1680	37.947	37.969	37.991	38.013	38.036	38.058	38.080	38.102	38.124	38.146	38.168	1680
1690	38.168	38.190	38.212	38.235	38.257	38.279	38.301	38.323	38.345	38.367	38.389	1690
1700	38.389	38.411	38.433	38.455	38.477	38.499	38.522	38.544	38.566	38.588	38.610	1700
1710	38.610	38.632	38.654	38.676	38.698	38.720	38.742	38.764	38.786	38.808	38.830	1710
1720	38.830	38.852	38.874	38.896	38.918	38.940	38.962	38.984	39.006	39.028	39.050	1720
1730	39.050	39.072	39.094	39.116	39.138	39.160	39.182	39.204	39.226	39.248	39.270	1730
1740	39.270	39.292	39.314	39.335	39.357	39.379	39.401	39.423	39.445	39.467	39.489	1740
1750	39.489	39.511	39.533	39.555	39.577	39.599	39.620	39.642	39.664	39.686	39.708	1750
1760	39.708	39.730	39.752	39.774	39.796	39.817	39.839	39.861	39.883	39.905	39.927	1760
1770	39.927	39.949	39.970	39.992	40.014	40.036	40.058	40.080	40.101	40.123	40.145	1770
1780	40.145	40.167	40.189	40.211	40.232	40.254	40.276	40.298	40.320	40.341	40.363	1780
1790	40.363	40.385	40.407	40.429	40.450	40.472	40.494	40.516	40.537	40.559	40.581	1790
1800	40.581	40.603	40.624	40.646	40.668	40.690	40.711	40.733	40.755	40.777	40.798	1800
1810	40.798	40.820	40.842	40.864	40.885	40.907	40.929	40.950	40.972	40.994	41.015	1810
1820	41.015	41.037	41.059	41.081	41.102	41.124	41.146	41.167	41.189	41.211	41.232	1820
1830	41.232	41.254	41.276	41.297	41.319	41.341	41.362	41.384	41.405	41.427	41.449	1830
1840	41.449	41.470	41.492	41.514	41.535	41.557	41.578	41.600	41.622	41.643	41.665	1840
1850	41.665	41.686	41.708	41.730	41.751	41.773	41.794	41.816	41.838	41.859	41.881	1850
1860	41.881	41.902	41.924	41.945	41.967	41.988	42.010	42.032	42.053	42.075	42.096	1860
1870	42.096	42.118	42.139	42.161	42.182	42.204	42.225	42.247	42.268	42.290	42.311	1870
1880	42.311	42.333	42.354	42.376	42.397	42.419	42.440	42.462	42.483	42.505	42.526	1880
1890	42.526	42.548	42.569	42.591	42.612	42.633	42.655	42.676	42.698	42.719	42.741	1890
°F	0	1	2	3	4	5	6	7	8	9	10	°F

Table A7. Thermocouple Table (Type K) Continued

Thermoelectric Voltage in Millivolts

°F	0	1	2	3	4	5	6	7	8	9	10	°F
1900	42.741	42.762	42.783	42.805	42.826	42.848	42.869	42.891	42.912	42.933	42.955	1900
1910	42.955	42.976	42.998	43.019	43.040	43.062	43.083	43.104	43.126	43.147	43.169	1910
1920	43.169	43.190	43.211	43.233	43.254	43.275	43.297	43.318	43.339	43.361	43.382	1920
1930	43.382	43.403	43.425	43.446	43.467	43.489	43.510	43.531	43.552	43.574	43.595	1930
1940	43.595	43.616	43.638	43.659	43.680	43.701	43.723	43.744	43.765	43.787	43.808	1940
1950	43.808	43.829	43.850	43.872	43.893	43.914	43.935	43.957	43.978	43.999	44.020	1950
1960	44.020	44.041	44.063	44.084	44.105	44.126	44.147	44.169	44.190	44.211	44.232	1960
1970	44.232	44.253	44.275	44.296	44.317	44.338	44.359	44.380	44.402	44.423	44.444	1970
1980	44.444	44.465	44.486	44.507	44.528	44.550	44.571	44.592	44.613	44.634	44.655	1980
1990	44.655	44.676	44.697	44.719	44.740	44.761	44.782	44.803	44.824	44.845	44.866	1990
2000	44.866	44.887	44.908	44.929	44.950	44.971	44.992	45.014	45.035	45.056	45.077	2000
2010	45.077	45.098	45.119	45.140	45.161	45.182	45.203	45.224	45.245	45.266	45.287	2010
2020	45.287	45.308	45.329	45.350	45.371	45.392	45.413	45.434	45.455	45.476	45.497	2020
2030	45.497	45.518	45.539	45.560	45.580	45.601	45.622	45.643	45.664	45.685	45.706	2030
2040	45.706	45.727	45.748	45.769	45.790	45.811	45.832	45.852	45.873	45.894	45.915	2040
2050	45.915	45.936	45.957	45.978	45.999	46.019	46.040	46.061	46.082	46.103	46.124	2050
2060	46.124	46.145	46.165	46.186	46.207	46.228	46.249	46.269	46.290	46.311	46.332	2060
2070	46.332	46.353	46.373	46.394	46.415	46.436	46.457	46.477	46.498	46.519	46.540	2070
2080	46.540	46.560	46.581	46.602	46.623	46.643	46.664	46.685	46.706	46.726	46.747	2080
2090	46.747	46.768	46.789	46.809	46.830	46.851	46.871	46.892	46.913	46.933	46.954	2090
2100	46.954	46.975	46.995	47.016	47.037	47.057	47.078	47.099	47.119	47.140	47.161	2100
2110	47.161	47.181	47.202	47.223	47.243	47.264	47.284	47.305	47.326	47.346	47.367	2110
2120	47.367	47.387	47.408	47.429	47.449	47.470	47.490	47.511	47.531	47.552	47.573	2120
2130	47.573	47.593	47.614	47.634	47.655	47.675	47.696	47.716	47.737	47.757	47.778	2130
2140	47.778	47.798	47.819	47.839	47.860	47.880	47.901	47.921	47.942	47.962	47.983	2140
2150	47.983	48.003	48.024	48.044	48.065	48.085	48.105	48.126	48.146	48.167	48.187	2150
2160	48.187	48.208	48.228	48.248	48.269	48.289	48.310	48.330	48.350	48.371	48.391	2160
2170	48.391	48.411	48.432	48.452	48.473	48.493	48.513	48.534	48.554	48.574	48.595	2170
2180	48.595	48.615	48.635	48.656	48.676	48.696	48.717	48.737	48.757	48.777	48.798	2180
2190	48.798	48.818	48.838	48.859	48.879	48.899	48.919	48.940	48.960	48.980	49.000	2190
2200	49.000	49.021	49.041	49.061	49.081	49.101	49.122	49.142	49.162	49.182	49.202	2200
2210	49.202	49.223	49.243	49.263	49.283	49.303	49.323	49.344	49.364	49.384	49.404	2210
2220	49.404	49.424	49.444	49.465	49.485	49.505	49.525	49.545	49.565	49.585	49.605	2220
2230	49.605	49.625	49.645	49.666	49.686	49.706	49.726	49.746	49.766	49.786	49.806	2230
2240	49.806	49.826	49.846	49.866	49.886	49.906	49.926	49.946	49.966	49.986	50.006	2240
°F	0	1	2	3	4	5	6	7	8	9	10	°F

°F	0	1	2	3	4	5	6	7	8	9	10	°F
2250	50.006	50.026	50.046	50.066	50.086	50.106	50.126	50.146	50.166	50.186	50.206	2250
2260	50.206	50.226	50.246	50.266	50.286	50.306	50.326	50.346	50.366	50.385	50.405	2260
2270	50.405	50.425	50.445	50.465	50.485	50.505	50.525	50.545	50.564	50.584	50.604	2270
2280	50.604	50.624	50.644	50.664	50.684	50.703	50.723	50.743	50.763	50.783	50.802	2280
2290	50.802	50.822	50.842	50.862	50.882	50.901	50.921	50.941	50.961	50.981	51.000	2290
2300	51.000	51.020	51.040	51.060	51.079	51.099	51.119	51.139	51.158	51.178	51.198	2300
2310	51.198	51.217	51.237	51.257	51.276	51.296	51.316	51.336	51.355	51.375	51.395	2310
2320	51.395	51.414	51.434	51.453	51.473	51.493	51.512	51.532	51.552	51.571	51.591	2320
2330	51.591	51.611	51.630	51.650	51.669	51.689	51.708	51.728	51.748	51.767	51.787	2330
2340	51.787	51.806	51.826	51.845	51.865	51.885	51.904	51.924	51.943	51.963	51.982	2340
2350	51.982	52.002	52.021	52.041	52.060	52.080	52.099	52.119	52.138	52.158	52.177	2350
2360	52.177	52.197	52.216	52.235	52.255	52.274	52.294	52.313	52.333	52.352	52.371	2360
2370	52.371	52.391	52.410	52.430	52.449	52.468	52.488	52.507	52.527	52.546	52.565	2370
2380	52.565	52.585	52.604	52.623	52.643	52.662	52.681	52.701	52.720	52.739	52.759	2380
2390	52.759	52.778	52.797	52.817	52.836	52.855	52.875	52.894	52.913	52.932	52.952	2390
2400	52.952	52.971	52.990	53.010	53.029	53.048	53.067	53.087	53.106	53.125	53.144	2400
2410	53.144	53.163	53.183	53.202	53.221	53.240	53.260	53.279	53.298	53.317	53.336	2410
2420	53.336	53.355	53.375	53.394	53.413	53.432	53.451	53.470	53.490	53.509	53.528	2420
2430	53.528	53.547	53.566	53.585	53.604	53.623	53.643	53.662	53.681	53.700	53.719	2430
2440	53.719	53.738	53.757	53.776	53.795	53.814	53.833	53.852	53.871	53.890	53.910	2440
2450	53.910	53.929	53.948	53.967	53.986	54.005	54.024	54.043	54.062	54.081	54.100	2450
2460	54.100	54.119	54.138	54.157	54.176	54.195	54.214	54.233	54.252	54.271	54.289	2460
2470	54.289	54.308	54.327	54.346	54.365	54.384	54.403	54.422	54.441	54.460	54.479	2470
2480	54.479	54.498	54.517	54.536	54.554	54.573	54.592	54.611	54.630	54.649	54.668	2480
2490	54.668	54.687	54.705	54.724	54.743	54.762	54.781	54.800	54.819	54.837	54.856	2490
2500	54.856	54.875	54.894									2500

Table A8. Thermocouple Table (Type E)

Thermoelectric Voltage in Millivolts

°F	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1	0	-10	°F
-450							-9.835	-9.834	-9.833	-9.832	-9.830	-450	
-440	-9.830	-9.829	-9.827	-9.825	-9.823	-9.821	-9.819	-9.817	-9.814	-9.812	-9.809	-440	
-430	-9.809	-9.806	-9.803	-9.800	-9.797	-9.793	-9.790	-9.786	-9.782	-9.779	-9.775	-430	
-420	-9.775	-9.771	-9.766	-9.762	-9.758	-9.753	-9.749	-9.744	-9.739	-9.734	-9.729	-420	
-410	-9.729	-9.724	-9.718	-9.713	-9.707	-9.702	-9.696	-9.690	-9.684	-9.678	-9.672	-410	
-400	-9.672	-9.666	-9.659	-9.653	-9.646	-9.639	-9.632	-9.625	-9.618	-9.611	-9.604	-400	
-390	-9.604	-9.597	-9.589	-9.581	-9.574	-9.566	-9.558	-9.550	-9.542	-9.534	-9.525	-390	
-380	-9.525	-9.517	-9.508	-9.500	-9.491	-9.482	-9.473	-9.464	-9.455	-9.446	-9.436	-380	
-370	-9.436	-9.427	-9.417	-9.408	-9.398	-9.388	-9.378	-9.368	-9.358	-9.348	-9.338	-370	
-360	-9.338	-9.327	-9.317	-9.306	-9.295	-9.285	-9.274	-9.263	-9.252	-9.241	-9.229	-360	
-350	-9.229	-9.218	-9.207	-9.195	-9.184	-9.172	-9.160	-9.148	-9.136	-9.124	-9.112	-350	
-340	-9.112	-9.100	-9.088	-9.075	-9.063	-9.050	-9.038	-9.025	-9.012	-8.999	-8.986	-340	
-330	-8.986	-8.973	-8.960	-8.947	-8.934	-8.920	-8.907	-8.893	-8.880	-8.866	-8.852	-330	
-320	-8.852	-8.839	-8.825	-8.811	-8.797	-8.782	-8.768	-8.754	-8.739	-8.725	-8.710	-320	
-310	-8.710	-8.696	-8.681	-8.666	-8.652	-8.637	-8.622	-8.607	-8.591	-8.576	-8.561	-310	
-300	-8.561	-8.546	-8.530	-8.515	-8.499	-8.483	-8.468	-8.452	-8.436	-8.420	-8.404	-300	
-290	-8.404	-8.388	-8.372	-8.356	-8.339	-8.323	-8.307	-8.290	-8.273	-8.257	-8.240	-290	
-280	-8.240	-8.223	-8.206	-8.189	-8.173	-8.155	-8.138	-8.121	-8.104	-8.087	-8.069	-280	
-270	-8.069	-8.052	-8.034	-8.017	-7.999	-7.981	-7.963	-7.945	-7.928	-7.910	-7.891	-270	
-260	-7.891	-7.873	-7.855	-7.837	-7.819	-7.800	-7.782	-7.763	-7.745	-7.726	-7.707	-260	
-250	-7.707	-7.688	-7.670	-7.651	-7.632	-7.613	-7.593	-7.574	-7.555	-7.536	-7.516	-250	
-240	-7.516	-7.497	-7.478	-7.458	-7.438	-7.419	-7.399	-7.379	-7.359	-7.339	-7.319	-240	
-230	-7.319	-7.299	-7.279	-7.259	-7.239	-7.219	-7.198	-7.178	-7.157	-7.137	-7.116	-230	
-220	-7.116	-7.096	-7.075	-7.054	-7.033	-7.013	-6.992	-6.971	-6.950	-6.928	-6.907	-220	
-210	-6.907	-6.886	-6.865	-6.843	-6.822	-6.801	-6.779	-6.757	-6.736	-6.714	-6.692	-210	
-200	-6.692	-6.671	-6.649	-6.627	-6.605	-6.583	-6.561	-6.539	-6.516	-6.494	-6.472	-200	
-190	-6.472	-6.449	-6.427	-6.405	-6.382	-6.359	-6.337	-6.314	-6.291	-6.269	-6.246	-190	
-180	-6.246	-6.223	-6.200	-6.177	-6.154	-6.130	-6.107	-6.084	-6.061	-6.037	-6.014	-180	
-170	-6.014	-5.991	-5.967	-5.943	-5.920	-5.896	-5.872	-5.849	-5.825	-5.801	-5.777	-170	
-160	-5.777	-5.753	-5.729	-5.705	-5.681	-5.656	-5.632	-5.608	-5.584	-5.559	-5.535	-160	
-150	-5.535	-5.510	-5.486	-5.461	-5.436	-5.412	-5.387	-5.362	-5.337	-5.312	-5.287	-150	
-140	-5.287	-5.262	-5.237	-5.212	-5.187	-5.162	-5.136	-5.111	-5.086	-5.060	-5.035	-140	
-130	-5.035	-5.009	-4.984	-4.958	-4.932	-4.907	-4.881	-4.855	-4.829	-4.803	-4.777	-130	
-120	-4.777	-4.751	-4.725	-4.699	-4.673	-4.647	-4.621	-4.594	-4.568	-4.542	-4.515	-120	
-110	-4.515	-4.489	-4.462	-4.436	-4.409	-4.382	-4.355	-4.329	-4.302	-4.275	-4.248	-110	
-100	-4.248	-4.221	-4.194	-4.167	-4.140	-4.113	-4.086	-4.058	-4.031	-4.004	-3.976	-100	
-90	-3.976	-3.949	-3.922	-3.894	-3.867	-3.839	-3.811	-3.784	-3.756	-3.728	-3.700	-90	
-80	-3.700	-3.672	-3.645	-3.617	-3.589	-3.561	-3.532	-3.504	-3.476	-3.448	-3.420	-80	
-70	-3.420	-3.391	-3.363	-3.335	-3.306	-3.278	-3.249	-3.221	-3.192	-3.163	-3.135	-70	
-60	-3.135	-3.106	-3.077	-3.048	-3.020	-2.991	-2.962	-2.933	-2.904	-2.875	-2.846	-60	
-50	-2.846	-2.816	-2.787	-2.758	-2.729	-2.699	-2.670	-2.641	-2.611	-2.582	-2.552	-50	
-40	-2.552	-2.523	-2.493	-2.463	-2.434	-2.404	-2.374	-2.344	-2.315	-2.285	-2.255	-40	
-30	-2.255	-2.225	-2.195	-2.165	-2.135	-2.105	-2.074	-2.044	-2.014	-1.984	-1.953	-30	
-20	-1.953	-1.923	-1.893	-1.862	-1.832	-1.801	-1.771	-1.740	-1.709	-1.679	-1.648	-20	
-10	-1.648	-1.617	-1.587	-1.556	-1.525	-1.494	-1.463	-1.432	-1.401	-1.370	-1.339	-10	
0	-1.339	-1.308	-1.277	-1.245	-1.214	-1.183	-1.152	-1.120	-1.089	-1.057	-1.026	0	
0	-1.026	-0.994	-0.963	-0.931	-0.900	-0.868	-0.836	-0.805	-0.773	-0.741	-0.709	0	
10	-0.709	-0.677	-0.645	-0.614	-0.582	-0.550	-0.517	-0.485	-0.453	-0.421	-0.389	10	
20	-0.389	-0.357	-0.324	-0.292	-0.260	-0.227	-0.195	-0.163	-0.130	-0.098	-0.065	20	
30	-0.065	-0.033	0.000	0.033	0.065	0.098	0.131	0.163	0.196	0.229	0.262	30	
40	0.262	0.294	0.327	0.360	0.393	0.426	0.459	0.492	0.525	0.558	0.591	40	
50	0.591	0.624	0.657	0.691	0.724	0.757	0.790	0.824	0.857	0.890	0.924	50	
60	0.924	0.957	0.990	1.024	1.057	1.091	1.124	1.158	1.192	1.225	1.259	60	
70	1.259	1.292	1.326	1.360	1.394	1.427	1.461	1.495	1.529	1.563	1.597	70	
80	1.597	1.631	1.665	1.699	1.733	1.767	1.801	1.835	1.869	1.904	1.938	80	
90	1.938	1.972	2.006	2.041	2.075	2.109	2.144	2.178	2.212	2.247	2.281	90	
°F	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**-10**	**°F**

°F	0	1	2	3	4	5	6	7	8	9	10	°F
100	2.281	2.316	2.351	2.385	2.420	2.454	2.489	2.524	2.558	2.593	2.628	100
110	2.628	2.663	2.698	2.733	2.767	2.802	2.837	2.872	2.907	2.942	2.977	110
120	2.977	3.012	3.048	3.083	3.118	3.153	3.188	3.224	3.259	3.294	3.330	120
130	3.330	3.365	3.400	3.436	3.471	3.507	3.542	3.578	3.613	3.649	3.685	130
140	3.685	3.720	3.756	3.792	3.827	3.863	3.899	3.935	3.970	4.006	4.042	140
150	4.042	4.078	4.114	4.150	4.186	4.222	4.258	4.294	4.330	4.366	4.403	150
160	4.403	4.439	4.475	4.511	4.547	4.584	4.620	4.656	4.693	4.729	4.766	160
170	4.766	4.802	4.839	4.875	4.912	4.948	4.985	5.021	5.058	5.095	5.131	170
180	5.131	5.168	5.205	5.242	5.278	5.315	5.352	5.389	5.426	5.463	5.500	180
190	5.500	5.537	5.574	5.611	5.648	5.685	5.722	5.759	5.796	5.833	5.871	190
200	5.871	5.908	5.945	5.982	6.020	6.057	6.094	6.132	6.169	6.207	6.244	200
210	6.244	6.281	6.319	6.356	6.394	6.432	6.469	6.507	6.544	6.582	6.620	210
220	6.620	6.658	6.695	6.733	6.771	6.809	6.847	6.884	6.922	6.960	6.998	220
230	6.998	7.036	7.074	7.112	7.150	7.188	7.226	7.264	7.302	7.341	7.379	230
240	7.379	7.417	7.455	7.493	7.532	7.570	7.608	7.647	7.685	7.723	7.762	240
250	7.762	7.800	7.839	7.877	7.916	7.954	7.993	8.031	8.070	8.108	8.147	250
260	8.147	8.186	8.224	8.263	8.302	8.340	8.379	8.418	8.457	8.496	8.535	260
270	8.535	8.573	8.612	8.651	8.690	8.729	8.768	8.807	8.846	8.885	8.924	270
280	8.924	8.963	9.002	9.041	9.081	9.120	9.159	9.198	9.237	9.277	9.316	280
290	9.316	9.355	9.395	9.434	9.473	9.513	9.552	9.591	9.631	9.670	9.710	290
300	9.710	9.749	9.789	9.828	9.868	9.907	9.947	9.987	10.026	10.066	10.106	300
310	10.106	10.145	10.185	10.225	10.265	10.304	10.344	10.384	10.424	10.464	10.503	310
320	10.503	10.543	10.583	10.623	10.663	10.703	10.743	10.783	10.823	10.863	10.903	320
330	10.903	10.943	10.983	11.024	11.064	11.104	11.144	11.184	11.224	11.265	11.305	330
340	11.305	11.345	11.385	11.426	11.466	11.506	11.547	11.587	11.627	11.668	11.708	340
350	11.708	11.749	11.789	11.830	11.870	11.911	11.951	11.992	12.032	12.073	12.113	350
360	12.113	12.154	12.195	12.235	12.276	12.317	12.357	12.398	12.439	12.480	12.520	360
370	12.520	12.561	12.602	12.643	12.684	12.724	12.765	12.806	12.847	12.888	12.929	370
380	12.929	12.970	13.011	13.052	13.093	13.134	13.175	13.216	13.257	13.298	13.339	380
390	13.339	13.380	13.421	13.462	13.504	13.545	13.586	13.627	13.668	13.710	13.751	390
400	13.751	13.792	13.833	13.875	13.916	13.957	13.999	14.040	14.081	14.123	14.164	400
410	14.164	14.205	14.247	14.288	14.330	14.371	14.413	14.454	14.496	14.537	14.579	410
420	14.579	14.620	14.662	14.704	14.745	14.787	14.828	14.870	14.912	14.953	14.995	420
430	14.995	15.037	15.078	15.120	15.162	15.204	15.245	15.287	15.329	15.371	15.413	430
440	15.413	15.454	15.496	15.538	15.580	15.622	15.664	15.706	15.748	15.790	15.831	440
450	15.831	15.873	15.915	15.957	15.999	16.041	16.083	16.125	16.168	16.210	16.252	450
460	16.252	16.294	16.336	16.378	16.420	16.462	16.504	16.547	16.589	16.631	16.673	460
470	16.673	16.715	16.758	16.800	16.842	16.884	16.927	16.969	17.011	17.054	17.096	470
480	17.096	17.138	17.181	17.223	17.265	17.308	17.350	17.392	17.435	17.477	17.520	480
490	17.520	17.562	17.605	17.647	17.690	17.732	17.775	17.817	17.860	17.902	17.945	490
500	17.945	17.987	18.030	18.073	18.115	18.158	18.200	18.243	18.286	18.328	18.371	500
510	18.371	18.414	18.456	18.499	18.542	18.585	18.627	18.670	18.713	18.756	18.798	510
520	18.798	18.841	18.884	18.927	18.969	19.012	19.055	19.098	19.141	19.184	19.227	520
530	19.227	19.269	19.312	19.355	19.398	19.441	19.484	19.527	19.570	19.613	19.656	530
540	19.656	19.699	19.742	19.785	19.828	19.871	19.914	19.957	20.000	20.043	20.086	540
550	20.086	20.129	20.172	20.216	20.259	20.302	20.345	20.388	20.431	20.474	20.517	550
560	20.517	20.561	20.604	20.647	20.690	20.733	20.777	20.820	20.863	20.906	20.950	560
570	20.950	20.993	21.036	21.080	21.123	21.166	21.209	21.253	21.296	21.339	21.383	570
580	21.383	21.426	21.470	21.513	21.556	21.600	21.643	21.686	21.730	21.773	21.817	580
590	21.817	21.860	21.904	21.947	21.991	22.034	22.078	22.121	22.165	22.208	22.252	590
600	22.252	22.295	22.339	22.382	22.426	22.469	22.513	22.556	22.600	22.644	22.687	600
610	22.687	22.731	22.774	22.818	22.862	22.905	22.949	22.993	23.036	23.080	23.124	610
620	23.124	23.167	23.211	23.255	23.298	23.342	23.386	23.429	23.473	23.517	23.561	620
630	23.561	23.604	23.648	23.692	23.736	23.780	23.823	23.867	23.911	23.955	23.999	630
640	23.999	24.042	24.086	24.130	24.174	24.218	24.262	24.305	24.349	24.393	24.437	640
650	24.437	24.481	24.525	24.569	24.613	24.657	24.701	24.745	24.789	24.832	24.876	650
660	24.876	24.920	24.964	25.008	25.052	25.096	25.140	25.184	25.228	25.272	25.316	660
670	25.316	25.360	25.404	25.448	25.493	25.537	25.581	25.625	25.669	25.713	25.757	670
680	25.757	25.801	25.845	25.889	25.933	25.977	26.022	26.066	26.110	26.154	26.198	680
690	26.198	26.242	26.286	26.331	26.375	26.419	26.463	26.507	26.552	26.596	26.640	690
°F	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**°F**

Table A8. Thermocouple Table (Type E) Continued

Thermoelectric Voltage in Millivolts

°F	0	1	2	3	4	5	6	7	8	9	10	°F
700	26.640	26.684	26.728	26.773	26.817	26.861	26.905	26.950	26.994	27.038	27.082	700
710	27.082	27.127	27.171	27.215	27.259	27.304	27.348	27.392	27.437	27.481	27.525	710
720	27.525	27.570	27.614	27.658	27.703	27.747	27.791	27.836	27.880	27.924	27.969	720
730	27.969	28.013	28.057	28.102	28.146	28.191	28.235	28.279	28.324	28.368	28.413	730
740	28.413	28.457	28.501	28.546	28.590	28.635	28.679	28.724	28.768	28.813	28.857	740
750	28.857	28.901	28.946	28.990	29.035	29.079	29.124	29.168	29.213	29.257	29.302	750
760	29.302	29.346	29.391	29.435	29.480	29.525	29.569	29.614	29.658	29.703	29.747	760
770	29.747	29.792	29.836	29.881	29.925	29.970	30.015	30.059	30.104	30.148	30.193	770
780	30.193	30.238	30.282	30.327	30.371	30.416	30.461	30.505	30.550	30.595	30.639	780
790	30.639	30.684	30.728	30.773	30.818	30.862	30.907	30.952	30.996	31.041	31.086	790
800	31.086	31.130	31.175	31.220	31.264	31.309	31.354	31.398	31.443	31.488	31.533	800
810	31.533	31.577	31.622	31.667	31.711	31.756	31.801	31.846	31.890	31.935	31.980	810
820	31.980	32.025	32.069	32.114	32.159	32.204	32.248	32.293	32.338	32.383	32.427	820
830	32.427	32.472	32.517	32.562	32.606	32.651	32.696	32.741	32.786	32.830	32.875	830
840	32.875	32.920	32.965	33.010	33.054	33.099	33.144	33.189	33.234	33.278	33.323	840
850	33.323	33.368	33.413	33.458	33.503	33.547	33.592	33.637	33.682	33.727	33.772	850
860	33.772	33.816	33.861	33.906	33.951	33.996	34.041	34.086	34.130	34.175	34.220	860
870	34.220	34.265	34.310	34.355	34.400	34.445	34.489	34.534	34.579	34.624	34.669	870
880	34.669	34.714	34.759	34.804	34.849	34.893	34.938	34.983	35.028	35.073	35.118	880
890	35.118	35.163	35.208	35.253	35.298	35.343	35.387	35.432	35.477	35.522	35.567	890
900	35.567	35.612	35.657	35.702	35.747	35.792	35.837	35.882	35.927	35.972	36.016	900
910	36.016	36.061	36.106	36.151	36.196	36.241	36.286	36.331	36.376	36.421	36.466	910
920	36.466	36.511	36.556	36.601	36.646	36.691	36.736	36.781	36.826	36.870	36.915	920
930	36.915	36.960	37.005	37.050	37.095	37.140	37.185	37.230	37.275	37.320	37.365	930
940	37.365	37.410	37.455	37.500	37.545	37.590	37.635	37.680	37.725	37.770	37.815	940
950	37.815	37.860	37.905	37.950	37.995	38.040	38.085	38.130	38.175	38.220	38.265	950
960	38.265	38.309	38.354	38.399	38.444	38.489	38.534	38.579	38.624	38.669	38.714	960
970	38.714	38.759	38.804	38.849	38.894	38.939	38.984	39.029	39.074	39.119	39.164	970
980	39.164	39.209	39.254	39.299	39.344	39.389	39.434	39.479	39.524	39.569	39.614	980
990	39.614	39.659	39.704	39.749	39.794	39.839	39.884	39.929	39.974	40.019	40.064	990
1000	40.064	40.109	40.154	40.199	40.243	40.288	40.333	40.378	40.423	40.468	40.513	1000
1010	40.513	40.558	40.603	40.648	40.693	40.738	40.783	40.828	40.873	40.918	40.963	1010
1020	40.963	41.008	41.053	41.098	41.143	41.188	41.233	41.278	41.323	41.368	41.412	1020
1030	41.412	41.457	41.502	41.547	41.592	41.637	41.682	41.727	41.772	41.817	41.862	1030
1040	41.862	41.907	41.952	41.997	42.042	42.087	42.132	42.176	42.221	42.266	42.311	1040
1050	42.311	42.356	42.401	42.446	42.491	42.536	42.581	42.626	42.671	42.715	42.760	1050
1060	42.760	42.805	42.850	42.895	42.940	42.985	43.030	43.075	43.120	43.165	43.209	1060
1070	43.209	43.254	43.299	43.344	43.389	43.434	43.479	43.524	43.569	43.613	43.658	1070
1080	43.658	43.703	43.748	43.793	43.838	43.883	43.928	43.972	44.017	44.062	44.107	1080
1090	44.107	44.152	44.197	44.242	44.286	44.331	44.376	44.421	44.466	44.511	44.555	1090
1100	44.555	44.600	44.645	44.690	44.735	44.780	44.824	44.869	44.914	44.959	45.004	1100
1110	45.004	45.049	45.093	45.138	45.183	45.228	45.273	45.317	45.362	45.407	45.452	1110
1120	45.452	45.497	45.541	45.586	45.631	45.676	45.720	45.765	45.810	45.855	45.900	1120
1130	45.900	45.944	45.989	46.034	46.079	46.123	46.168	46.213	46.258	46.302	46.347	1130
1140	46.347	46.392	46.437	46.481	46.526	46.571	46.616	46.660	46.705	46.750	46.794	1140
1150	46.794	46.839	46.884	46.929	46.973	47.018	47.063	47.107	47.152	47.197	47.241	1150
1160	47.241	47.286	47.331	47.375	47.420	47.465	47.509	47.554	47.599	47.643	47.688	1160
1170	47.688	47.733	47.777	47.822	47.867	47.911	47.956	48.001	48.045	48.090	48.135	1170
1180	48.135	48.179	48.224	48.268	48.313	48.358	48.402	48.447	48.492	48.536	48.581	1180
1190	48.581	48.625	48.670	48.715	48.759	48.804	48.848	48.893	48.937	48.982	49.027	1190
1200	49.027	49.071	49.116	49.160	49.205	49.249	49.294	49.338	49.383	49.428	49.472	1200
1210	49.472	49.517	49.561	49.606	49.650	49.695	49.739	49.784	49.828	49.873	49.917	1210
1220	49.917	49.962	50.006	50.051	50.095	50.140	50.184	50.229	50.273	50.318	50.362	1220
1230	50.362	50.407	50.451	50.495	50.540	50.584	50.629	50.673	50.718	50.762	50.807	1230
1240	50.807	50.851	50.895	50.940	50.984	51.029	51.073	51.118	51.162	51.206	51.251	1240
1250	51.251	51.295	51.340	51.384	51.428	51.473	51.517	51.561	51.606	51.650	51.695	1250
1260	51.695	51.739	51.783	51.828	51.872	51.916	51.961	52.005	52.049	52.094	52.138	1260
1270	52.138	52.182	52.227	52.271	52.315	52.360	52.404	52.448	52.493	52.537	52.581	1270
1280	52.581	52.625	52.670	52.714	52.758	52.803	52.847	52.891	52.935	52.980	53.024	1280
1290	53.024	53.068	53.112	53.157	53.201	53.245	53.289	53.334	53.378	53.422	53.466	1290
°F	0	1	2	3	4	5	6	7	8	9	10	°F

°F	0	1	2	3	4	5	6	7	8	9	10	°F
1300	53.466	53.510	53.555	53.599	53.643	53.687	53.732	53.776	53.820	53.864	53.908	1300
1310	53.908	53.952	53.997	54.041	54.085	54.129	54.173	54.218	54.262	54.306	54.350	1310
1320	54.350	54.394	54.438	54.482	54.527	54.571	54.615	54.659	54.703	54.747	54.791	1320
1330	54.791	54.835	54.879	54.924	54.968	55.012	55.056	55.100	55.144	55.188	55.232	1330
1340	55.232	55.276	55.320	55.364	55.408	55.453	55.497	55.541	55.585	55.629	55.673	1340
1350	55.673	55.717	55.761	55.805	55.849	55.893	55.937	55.981	56.025	56.069	56.113	1350
1360	56.113	56.157	56.201	56.245	56.289	56.333	56.377	56.421	56.465	56.509	56.553	1360
1370	56.553	56.597	56.641	56.685	56.729	56.773	56.816	56.860	56.904	56.948	56.992	1370
1380	56.992	57.036	57.080	57.124	57.168	57.212	57.256	57.300	57.344	57.387	57.431	1380
1390	57.431	57.475	57.519	57.563	57.607	57.651	57.695	57.738	57.782	57.826	57.870	1390
1400	57.870	57.914	57.958	58.002	58.045	58.089	58.133	58.177	58.221	58.265	58.308	1400
1410	58.308	58.352	58.396	58.440	58.484	58.527	58.571	58.615	58.659	58.702	58.746	1410
1420	58.746	58.790	58.834	58.878	58.921	58.965	59.009	59.053	59.096	59.140	59.184	1420
1430	59.184	59.228	59.271	59.315	59.359	59.402	59.446	59.490	59.534	59.577	59.621	1430
1440	59.621	59.665	59.708	59.752	59.796	59.839	59.883	59.927	59.970	60.014	60.058	1440
1450	60.058	60.101	60.145	60.189	60.232	60.276	60.320	60.363	60.407	60.451	60.494	1450
1460	60.494	60.538	60.581	60.625	60.669	60.712	60.756	60.799	60.843	60.887	60.930	1460
1470	60.930	60.974	61.017	61.061	61.105	61.148	61.192	61.235	61.279	61.322	61.366	1470
1480	61.366	61.409	61.453	61.496	61.540	61.583	61.627	61.671	61.714	61.758	61.801	1480
1490	61.801	61.845	61.888	61.932	61.975	62.018	62.062	62.105	62.149	62.192	62.236	1490
1500	62.236	62.279	62.323	62.366	62.410	62.453	62.496	62.540	62.583	62.627	62.670	1500
1510	62.670	62.714	62.757	62.800	62.844	62.887	62.931	62.974	63.017	63.061	63.104	1510
1520	63.104	63.148	63.191	63.234	63.278	63.321	63.364	63.408	63.451	63.494	63.538	1520
1530	63.538	63.581	63.624	63.668	63.711	63.754	63.798	63.841	63.884	63.927	63.971	1530
1540	63.971	64.014	64.057	64.101	64.144	64.187	64.230	64.274	64.317	64.360	64.403	1540
1550	64.403	64.447	64.490	64.533	64.576	64.619	64.663	64.706	64.749	64.792	64.835	1550
1560	64.835	64.879	64.922	64.965	65.008	65.051	65.094	65.138	65.181	65.224	65.267	1560
1570	65.267	65.310	65.353	65.396	65.440	65.483	65.526	65.569	65.612	65.655	65.698	1570
1580	65.698	65.741	65.784	65.827	65.871	65.914	65.957	66.000	66.043	66.086	66.129	1580
1590	66.129	66.172	66.215	66.258	66.301	66.344	66.387	66.430	66.473	66.516	66.559	1590
1600	66.559	66.602	66.645	66.688	66.731	66.774	66.817	66.860	66.903	66.946	66.989	1600
1610	66.989	67.031	67.074	67.117	67.160	67.203	67.246	67.289	67.332	67.375	67.418	1610
1620	67.418	67.460	67.503	67.546	67.589	67.632	67.675	67.718	67.760	67.803	67.846	1620
1630	67.846	67.889	67.932	67.974	68.017	68.060	68.103	68.146	68.188	68.231	68.274	1630
1640	68.274	68.317	68.359	68.402	68.445	68.488	68.530	68.573	68.616	68.659	68.701	1640
1650	68.701	68.744	68.787	68.829	68.872	68.915	68.957	69.000	69.043	69.085	69.128	1650
1660	69.128	69.171	69.213	69.256	69.298	69.341	69.384	69.426	69.469	69.511	69.554	1660
1670	69.554	69.597	69.639	69.682	69.724	69.767	69.809	69.852	69.894	69.937	69.979	1670
1680	69.979	70.022	70.064	70.107	70.149	70.192	70.234	70.277	70.319	70.362	70.404	1680
1690	70.404	70.447	70.489	70.531	70.574	70.616	70.659	70.701	70.744	70.786	70.828	1690
1700	70.828	70.871	70.913	70.955	70.998	71.040	71.082	71.125	71.167	71.209	71.252	1700
1710	71.252	71.294	71.336	71.379	71.421	71.463	71.506	71.548	71.590	71.632	71.675	1710
1720	71.675	71.717	71.759	71.801	71.844	71.886	71.928	71.970	72.012	72.055	72.097	1720
1730	72.097	72.139	72.181	72.223	72.266	72.308	72.350	72.392	72.434	72.476	72.518	1730
1740	72.518	72.561	72.603	72.645	72.687	72.729	72.771	72.813	72.855	72.897	72.939	1740
1750	72.939	72.981	73.023	73.066	73.108	73.150	73.192	73.234	73.276	73.318	73.360	1750
1760	73.360	73.402	73.444	73.486	73.528	73.570	73.612	73.654	73.696	73.738	73.780	1760
1770	73.780	73.821	73.863	73.905	73.947	73.989	74.031	74.073	74.115	74.157	74.199	1770
1780	74.199	74.241	74.283	74.324	74.366	74.408	74.450	74.492	74.534	74.576	74.618	1780
1790	74.618	74.659	74.701	74.743	74.785	74.827	74.869	74.910	74.952	74.994	75.036	1790
1800	75.036	75.078	75.120	75.161	75.203	75.245	75.287	75.329	75.370	75.412	75.454	1800
1810	75.454	75.496	75.538	75.579	75.621	75.663	75.705	75.746	75.788	75.830	75.872	1810
1820	75.872	75.913	75.955	75.997	76.039	76.081	76.122	76.164	76.206	76.248	76.289	1820
1830	76.289	76.331	76.373									1830
°F	0	1	2	3	4	5	6	7	8	9	10	°F

Table A9. Thermocouple Table (Type T)

Thermoelectric Voltage in Millivolts

°F	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1	0	°F
-450							-6.258	-6.257	-6.256	-6.255	-6.254	-450
-440	-6.254	-6.253	-6.252	-6.251	-6.250	-6.248	-6.247	-6.245	-6.243	-6.242	-6.240	-440
-430	-6.240	-6.238	-6.236	-6.234	-6.232	-6.230	-6.227	-6.225	-6.222	-6.220	-6.217	-430
-420	-6.217	-6.215	-6.212	-6.209	-6.206	-6.203	-6.200	-6.197	-6.194	-6.191	-6.187	-420
-410	-6.187	-6.184	-6.180	-6.177	-6.173	-6.170	-6.166	-6.162	-6.158	-6.154	-6.150	-410
-400	-6.150	-6.146	-6.141	-6.137	-6.133	-6.128	-6.124	-6.119	-6.115	-6.110	-6.105	-400
-390	-6.105	-6.100	-6.095	-6.090	-6.085	-6.080	-6.075	-6.069	-6.064	-6.059	-6.053	-390
-380	-6.053	-6.047	-6.042	-6.036	-6.030	-6.025	-6.019	-6.013	-6.007	-6.001	-5.994	-380
-370	-5.994	-5.988	-5.982	-5.976	-5.969	-5.963	-5.956	-5.950	-5.943	-5.937	-5.930	-370
-360	-5.930	-5.923	-5.916	-5.909	-5.902	-5.896	-5.888	-5.881	-5.874	-5.867	-5.860	-360
-350	-5.860	-5.853	-5.845	-5.838	-5.830	-5.823	-5.815	-5.808	-5.800	-5.792	-5.785	-350
-340	-5.785	-5.777	-5.769	-5.761	-5.753	-5.745	-5.737	-5.729	-5.721	-5.713	-5.705	-340
-330	-5.705	-5.697	-5.688	-5.680	-5.672	-5.663	-5.655	-5.646	-5.638	-5.629	-5.620	-330
-320	-5.620	-5.612	-5.603	-5.594	-5.585	-5.577	-5.568	-5.559	-5.550	-5.541	-5.532	-320
-310	-5.532	-5.523	-5.513	-5.504	-5.495	-5.486	-5.476	-5.467	-5.458	-5.448	-5.439	-310
-300	-5.439	-5.429	-5.420	-5.410	-5.400	-5.391	-5.381	-5.371	-5.361	-5.351	-5.341	-300
-290	-5.341	-5.332	-5.322	-5.312	-5.301	-5.291	-5.281	-5.271	-5.261	-5.250	-5.240	-290
-280	-5.240	-5.230	-5.219	-5.209	-5.198	-5.188	-5.177	-5.167	-5.156	-5.145	-5.135	-280
-270	-5.135	-5.124	-5.113	-5.102	-5.091	-5.081	-5.070	-5.059	-5.048	-5.036	-5.025	-270
-260	-5.025	-5.014	-5.003	-4.992	-4.980	-4.969	-4.958	-4.946	-4.935	-4.923	-4.912	-260
-250	-4.912	-4.900	-4.889	-4.877	-4.865	-4.854	-4.842	-4.830	-4.818	-4.806	-4.794	-250
-240	-4.794	-4.783	-4.771	-4.759	-4.746	-4.734	-4.722	-4.710	-4.698	-4.685	-4.673	-240
-230	-4.673	-4.661	-4.648	-4.636	-4.624	-4.611	-4.599	-4.586	-4.573	-4.561	-4.548	-230
-220	-4.548	-4.535	-4.523	-4.510	-4.497	-4.484	-4.471	-4.458	-4.445	-4.432	-4.419	-220
-210	-4.419	-4.406	-4.393	-4.380	-4.366	-4.353	-4.340	-4.326	-4.313	-4.300	-4.286	-210
-200	-4.286	-4.273	-4.259	-4.246	-4.232	-4.218	-4.205	-4.191	-4.177	-4.163	-4.149	-200
-190	-4.149	-4.136	-4.122	-4.108	-4.094	-4.080	-4.066	-4.052	-4.037	-4.023	-4.009	-190
-180	-4.009	-3.995	-3.980	-3.966	-3.952	-3.937	-3.923	-3.908	-3.894	-3.879	-3.865	-180
-170	-3.865	-3.850	-3.836	-3.821	-3.806	-3.791	-3.777	-3.762	-3.747	-3.732	-3.717	-170
-160	-3.717	-3.702	-3.687	-3.672	-3.657	-3.642	-3.626	-3.611	-3.596	-3.581	-3.565	-160
-150	-3.565	-3.550	-3.535	-3.519	-3.504	-3.488	-3.473	-3.457	-3.441	-3.426	-3.410	-150
-140	-3.410	-3.394	-3.379	-3.363	-3.347	-3.331	-3.315	-3.299	-3.283	-3.267	-3.251	-140
-130	-3.251	-3.235	-3.219	-3.203	-3.187	-3.171	-3.154	-3.138	-3.122	-3.105	-3.089	-130
-120	-3.089	-3.072	-3.056	-3.040	-3.023	-3.006	-2.990	-2.973	-2.956	-2.940	-2.923	-120
-110	-2.923	-2.906	-2.889	-2.873	-2.856	-2.839	-2.822	-2.805	-2.788	-2.771	-2.754	-110
-100	-2.754	-2.737	-2.719	-2.702	-2.685	-2.668	-2.651	-2.633	-2.616	-2.598	-2.581	-100
-90	-2.581	-2.564	-2.546	-2.529	-2.511	-2.493	-2.476	-2.458	-2.440	-2.423	-2.405	-90
-80	-2.405	-2.387	-2.369	-2.351	-2.334	-2.316	-2.298	-2.280	-2.262	-2.244	-2.225	-80
-70	-2.225	-2.207	-2.189	-2.171	-2.153	-2.134	-2.116	-2.098	-2.079	-2.061	-2.043	-70
-60	-2.043	-2.024	-2.006	-1.987	-1.969	-1.950	-1.931	-1.913	-1.894	-1.875	-1.857	-60
-50	-1.857	-1.838	-1.819	-1.800	-1.781	-1.762	-1.743	-1.724	-1.705	-1.686	-1.667	-50
-40	-1.667	-1.648	-1.629	-1.610	-1.591	-1.572	-1.552	-1.533	-1.514	-1.494	-1.475	-40
-30	-1.475	-1.456	-1.436	-1.417	-1.397	-1.378	-1.358	-1.338	-1.319	-1.299	-1.279	-30
-20	-1.279	-1.260	-1.240	-1.220	-1.200	-1.181	-1.161	-1.141	-1.121	-1.101	-1.081	-20
-10	-1.081	-1.061	-1.041	-1.021	-1.001	-0.980	-0.960	-0.940	-0.920	-0.900	-0.879	-10
0	-0.879	-0.859	-0.839	-0.818	-0.798	-0.777	-0.757	-0.736	-0.716	-0.695	-0.675	0
0	-0.675	-0.654	-0.633	-0.613	-0.592	-0.571	-0.550	-0.530	-0.509	-0.488	-0.467	0
10	-0.467	-0.446	-0.425	-0.404	-0.383	-0.362	-0.341	-0.320	-0.299	-0.278	-0.256	10
20	-0.256	-0.235	-0.214	-0.193	-0.171	-0.150	-0.129	-0.107	-0.086	-0.064	-0.043	20
30	-0.043	-0.022	0.000	0.022	0.043	0.065	0.086	0.108	0.130	0.151	0.173	30
40	0.173	0.195	0.216	0.238	0.260	0.282	0.303	0.325	0.347	0.369	0.391	40
50	0.391	0.413	0.435	0.457	0.479	0.501	0.523	0.545	0.567	0.589	0.611	50
60	0.611	0.634	0.656	0.678	0.700	0.723	0.745	0.767	0.790	0.812	0.834	60
70	0.834	0.857	0.879	0.902	0.924	0.947	0.969	0.992	1.015	1.037	1.060	70
80	1.060	1.083	1.105	1.128	1.151	1.174	1.196	1.219	1.242	1.265	1.288	80
90	1.288	1.311	1.334	1.357	1.380	1.403	1.426	1.449	1.472	1.496	1.519	90
100	1.519	1.542	1.565	1.588	1.612	1.635	1.658	1.682	1.705	1.729	1.752	100
110	1.752	1.776	1.799	1.823	1.846	1.870	1.893	1.917	1.941	1.964	1.988	110
120	1.988	2.012	2.036	2.060	2.083	2.107	2.131	2.155	2.179	2.203	2.227	120
130	2.227	2.251	2.275	2.299	2.323	2.347	2.371	2.395	2.420	2.444	2.468	130
140	2.468	2.492	2.517	2.541	2.565	2.590	2.614	2.639	2.663	2.687	2.712	140

°F	0	1	2	3	4	5	6	7	8	9	10	°F
150	2.712	2.737	2.761	2.786	2.810	2.835	2.860	2.884	2.909	2.934	2.958	150
160	2.958	2.983	3.008	3.033	3.058	3.082	3.107	3.132	3.157	3.182	3.207	160
170	3.207	3.232	3.257	3.282	3.307	3.333	3.358	3.383	3.408	3.433	3.459	170
180	3.459	3.484	3.509	3.534	3.560	3.585	3.610	3.636	3.661	3.687	3.712	180
190	3.712	3.738	3.763	3.789	3.814	3.840	3.866	3.891	3.917	3.943	3.968	190
200	3.968	3.994	4.020	4.046	4.071	4.097	4.123	4.149	4.175	4.201	4.227	200
210	4.227	4.253	4.279	4.305	4.331	4.357	4.383	4.409	4.435	4.461	4.487	210
220	4.487	4.513	4.540	4.566	4.592	4.618	4.645	4.671	4.697	4.724	4.750	220
230	4.750	4.776	4.803	4.829	4.856	4.882	4.909	4.935	4.962	4.988	5.015	230
240	5.015	5.042	5.068	5.095	5.122	5.148	5.175	5.202	5.228	5.255	5.282	240
250	5.282	5.309	5.336	5.363	5.389	5.416	5.443	5.470	5.497	5.524	5.551	250
260	5.551	5.578	5.605	5.632	5.660	5.687	5.714	5.741	5.768	5.795	5.823	260
270	5.823	5.850	5.877	5.904	5.932	5.959	5.986	6.014	6.041	6.068	6.096	270
280	6.096	6.123	6.151	6.178	6.206	6.233	6.261	6.288	6.316	6.343	6.371	280
290	6.371	6.399	6.426	6.454	6.482	6.510	6.537	6.565	6.593	6.621	6.648	290
300	6.648	6.676	6.704	6.732	6.760	6.788	6.816	6.844	6.872	6.900	6.928	300
310	6.928	6.956	6.984	7.012	7.040	7.068	7.096	7.124	7.152	7.181	7.209	310
320	7.209	7.237	7.265	7.294	7.322	7.350	7.378	7.407	7.435	7.463	7.492	320
330	7.492	7.520	7.549	7.577	7.606	7.634	7.663	7.691	7.720	7.748	7.777	330
340	7.777	7.805	7.834	7.863	7.891	7.920	7.949	7.977	8.006	8.035	8.064	340
350	8.064	8.092	8.121	8.150	8.179	8.208	8.237	8.266	8.294	8.323	8.352	350
360	8.352	8.381	8.410	8.439	8.468	8.497	8.526	8.555	8.585	8.614	8.643	360
370	8.643	8.672	8.701	8.730	8.759	8.789	8.818	8.847	8.876	8.906	8.935	370
380	8.935	8.964	8.994	9.023	9.052	9.082	9.111	9.141	9.170	9.200	9.229	380
390	9.229	9.259	9.288	9.318	9.347	9.377	9.406	9.436	9.466	9.495	9.525	390
400	9.525	9.555	9.584	9.614	9.644	9.673	9.703	9.733	9.763	9.793	9.822	400
410	9.822	9.852	9.882	9.912	9.942	9.972	10.002	10.032	10.062	10.092	10.122	410
420	10.122	10.152	10.182	10.212	10.242	10.272	10.302	10.332	10.362	10.392	10.423	420
430	10.423	10.453	10.483	10.513	10.543	10.574	10.604	10.634	10.664	10.695	10.725	430
440	10.725	10.755	10.786	10.816	10.847	10.877	10.907	10.938	10.968	10.999	11.029	440
450	11.029	11.060	11.090	11.121	11.151	11.182	11.213	11.243	11.274	11.304	11.335	450
460	11.335	11.366	11.396	11.427	11.458	11.489	11.519	11.550	11.581	11.612	11.643	460
470	11.643	11.673	11.704	11.735	11.766	11.797	11.828	11.859	11.890	11.920	11.951	470
480	11.951	11.982	12.013	12.044	12.075	12.106	12.138	12.169	12.200	12.231	12.262	480
490	12.262	12.293	12.324	12.355	12.386	12.418	12.449	12.480	12.511	12.543	12.574	490
500	12.574	12.605	12.636	12.668	12.699	12.730	12.762	12.793	12.824	12.856	12.887	500
510	12.887	12.919	12.950	12.982	13.013	13.045	13.076	13.108	13.139	13.171	13.202	510
520	13.202	13.234	13.265	13.297	13.328	13.360	13.392	13.423	13.455	13.487	13.518	520
530	13.518	13.550	13.582	13.614	13.645	13.677	13.709	13.741	13.772	13.804	13.836	530
540	13.836	13.868	13.900	13.932	13.964	13.995	14.027	14.059	14.091	14.123	14.155	540
550	14.155	14.187	14.219	14.251	14.283	14.315	14.347	14.379	14.411	14.444	14.476	550
560	14.476	14.508	14.540	14.572	14.604	14.636	14.669	14.701	14.733	14.765	14.797	560
570	14.797	14.830	14.862	14.894	14.926	14.959	14.991	15.023	15.056	15.088	15.121	570
580	15.121	15.153	15.185	15.218	15.250	15.283	15.315	15.347	15.380	15.412	15.445	580
590	15.445	15.477	15.510	15.543	15.575	15.608	15.640	15.673	15.705	15.738	15.771	590
600	15.771	15.803	15.836	15.869	15.901	15.934	15.967	15.999	16.032	16.065	16.098	600
610	16.098	16.130	16.163	16.196	16.229	16.262	16.295	16.327	16.360	16.393	16.426	610
620	16.426	16.459	16.492	16.525	16.558	16.591	16.624	16.657	16.690	16.723	16.756	620
630	16.756	16.789	16.822	16.855	16.888	16.921	16.954	16.987	17.020	17.053	17.086	630
640	17.086	17.120	17.153	17.186	17.219	17.252	17.286	17.319	17.352	17.385	17.418	640
650	17.418	17.452	17.485	17.518	17.552	17.585	17.618	17.652	17.685	17.718	17.752	650
660	17.752	17.785	17.819	17.852	17.886	17.919	17.952	17.986	18.019	18.053	18.086	660
670	18.086	18.120	18.153	18.187	18.221	18.254	18.288	18.321	18.355	18.389	18.422	670
680	18.422	18.456	18.490	18.523	18.557	18.591	18.624	18.658	18.692	18.725	18.759	680
690	18.759	18.793	18.827	18.861	18.894	18.928	18.962	18.996	19.030	19.064	19.097	690
700	19.097	19.131	19.165	19.199	19.233	19.267	19.301	19.335	19.369	19.403	19.437	700
710	19.437	19.471	19.505	19.539	19.573	19.607	19.641	19.675	19.709	19.743	19.777	710
720	19.777	19.811	19.845	19.879	19.913	19.947	19.982	20.016	20.050	20.084	20.118	720
730	20.118	20.152	20.187	20.221	20.255	20.289	20.323	20.358	20.392	20.426	20.460	730
740	20.460	20.495	20.529	20.563	20.597	20.632	20.666	20.700	20.735	20.769	20.803	740
750	20.803	20.838	20.872									750

Table A10. Platinum 100 Ohm RTD DIN Curve Table

Degrees Fahrenheit	Degrees Celsius	Ohms
0	-17.78	93.04
5	-15.00	94.12
10	-12.22	95.21
15	-9.44	96.31
20	-6.67	97.39
25	-3.89	98.48
30	-1.11	99.57
32	0.00	100.00
35	1.67	100.65
40	4.44	101.73
45	7.22	102.82
50	10.00	103.90
55	12.78	104.98
60	15.56	106.07
65	18.33	107.15
70	21.11	108.22
75	23.89	109.31
80	26.67	110.38
85	29.44	111.45
90	32.22	112.53
95	35.00	113.61
100	37.78	114.68
105	40.56	115.76
110	43.33	116.83
115	46.11	117.90
120	48.89	118.97
125	51.67	120.04
130	54.44	121.11
135	57.22	122.17
140	60.00	123.24
145	62.78	124.31
150	65.56	125.37
155	68.33	126.44
160	71.11	127.50
165	73.89	128.56
170	76.67	129.62
175	79.44	130.68
180	82.22	131.74
185	85.00	132.80
190	87.78	133.86
195	90.56	134.91
200	93.33	135.97
212	100.00	138.50
225	107.22	141.24
250	121.11	146.48
275	135.00	151.70
300	148.89	156.90

ISA SYMBOLS

ISA Identification Letters

	FIRST-LETTER (4)		SUCCEEDING-LETTERS (3)		
	MEASURED OR INITIATING VARIABLE	**MODIFIER**	**READOUT OR PASSIVE FUNCTION**	**OUTPUT FUNCTION**	**MODIFIER**
A	Analysis (5,19)		Alarm		
B	Burner, Combustion		User's Choice (1)	User's Choice (1)	User's Choice (1)
C	User's Choice (1)			Control (13)	
D	User's Choice (1)	Differential (4)			
E	Voltage		Sensor (Primary Element)		
F	Flow Rate	Ratio (Fraction) (4)			
G	User's Choice (1)		Glass, Viewing Device (9)		
H	Hand				High (7, 15, 16)
I	Current (Electrical)		Indicate (10)		
J	Power	Scan (7)			
K	Time, Time Schedule	Time Rate of Change (4, 21)		Control Station (22)	
L	Level		Light (11)		Low (7, 15, 16)
M	User's Choice (1)	Momentary (4)			Middle, Intermediate (7,15)
N	User's Choice (1)		User's Choice (1)	User's Choice (1)	User's Choice (1)
O	User's Choice (1)		Orifice, Restriction		
P	Pressure, Vacuum		Point (Test) Connection		
Q	Quantity	Integrate, Totalize (4)			
R	Radiation		Record (17)		
S	Speed, Frequency	Safety (8)		Switch (13)	
T	Temperature			Transmit (18)	
U	Multivariable (6)		Multifunction (12)	Multifunction (12)	Multifunction (12)
V	Vibration, Mechanical Analysis (19)			Valve, Damper, Louver (13)	
W	Weight, Force		Well		
X	Unclassified (2)	X Axis	Unclassified (2)	Unclassified (2)	Unclassified (2)
Y	Event, State or Presence (20)	Y Axis		Relay, Compute, Convert (13, 14, 18)	
Z	Position, Dimension	Z Axis		Driver, Actuator, Unclassified Final Control Element	

NOTE: Numbers in parentheses refer to specific explanatory notes in ANSI/ISA-5.1-1984(R1992) Section 5.1.

Typical Letter Combinations

First-Letters	Initiating or Measured Vaiable	Controllers				Readout Devices		Switches and Alarm Devices*			Transmitters			Solenoids, Relays, Computing Devices	Primary Element	Test Point	Well or Probe	Viewing Device, Glass	Safety Device	Final Element
		Recording	Indicating	Blind	Self-Actuated Control Valves	Recording	Indicating	High**	Low	Comb	Recording	Indicating	Blind							
A	Analysis	ARC	AIC	AC		AR	AI	ASH	ASL	ASHL	ART	AIT	AT	AY	AE	AP	AW			AV
B	Burner/Combustion	BRC	BIC	BC		BR	BI	BSH	BSL	BSHL	BRT	BIT	BT	BY	BE		BW	BG		BZ
C	User's Choice																			
D	User's Choice																			
E	Voltage	ERC	EIC	EC		ER	EI	ESH	ESL	ESHL	ERT	EIT	ET	EY	EE					EZ
F	Flow Rate	FRC	FIC	FC	FCV, FICV	FR	FI	FSH	FSL	FSHL	FRT	FIT	FT	FY	FE	FP		FG		FV
FQ	Flow Quantity	FQRC	FQIC			FQR	FQI	FQSH	FQSL			FQIT	FQT	FQY	FQE					FQV
FF	Flow Ratio	FFRC	FFIC	FFC		FFR	FFI	FFSH	FFSL						FE					FFV
G	User's Choice																			
H	Hand		HIC	HC						HS										HV
I	Current	IRC	IIC			IR	II	ISH	ISL	ISHL	IRT	IIT	IT	IY	IE					IZ
J	Power	JRC	JIC			JR	JI	JSH	JSL	JSHL	JRT	JIT	JT	JY	JE					JV
K	Time	KRC	KIC	KC	KCV	KR	KI	KSH	KSL	KSHL	KRT	KIT	KT	KY	KE					KV
L	Level	LRC	LIC	LC	LCV	LR	LI	LSH	LSL	LSHL	LRT	LIT	LT	LY	LE		LW	LG		LV
M	User's Choice																			
N	User's Choice																			
O	User's Choice																			
P	Pressure/ Vacuum	PRC	PIC	PC	PCV	PR	PI	PSH	PSL	PSHL	PRT	PIT	PT	PY	PE	PP			PSV, PSE	PV
PD	Pressure, Differential	PDRC	PDIC	PDC	PDCV	PDR	PDI	PDSH	PDSL		PDRT	PDIT	PDT	PDY	PE	PP				PDV
Q	Quantity	QRC	QIC			QR	QI	QSH	QSL	QSHL	QRT	QIT	QT	QY	QE					QZ
R	Radiation	RRC	RIC	RC		RR	RI	RSH	RSL	RSHL	RRT	RIT	RT	RY	RE		RW			RZ
S	Speed/Frequency	SRC	SIC	SC	SCV	SR	SI	SSH	SSL	SSHL	SRT	SIT	ST	SY	SE					SV
T	Temperature	TRC	TIC	TC	TCV	TR	TI	TSH	TSL	TSHL	TRT	TIT	TT	TY	TE	TP	TW		TSE	TV
TD	Temperature, Differential	TDRC	TDIC	TDC	TDCV	TDR	TDI	TDSH	TDSL		TDRT	TDIT	TDT	TDY	TE	TP	TW			TDV
U	Multivariable					UR	UI							UY						UV
V	Vibration/Machinery Analysis					VR	VI	VSH	VSL	VSHL	VRT	VIT	VT	VY	VE					VZ
W	Weight/Force	WRC	WIC	WC	WCV	WR	WI	WSH	WSL	WSHL	WRT	WIT	WT	WY	WE					WZ
WD	Weight/Force, Differential	WDRC	WDIC	WDC	WDCV	WDR	WDI	WDSH	WDSL		WDRT	WDIT	WDT	WDY	WE					WDZ
X	Unclassified																			
Y	Event/State/Presence		YIC	YC		YR	YI	YSH	YSL				YT	YY	YE					YZ
Z	Position/Dimension	ZRC	ZIC	ZC	ZCV	ZR	ZI	ZSH	ZSL	ZSHL	ZRT	ZIT	ZT	ZY	ZE					ZV
ZD	Gauging/Deviation	ZDRC	ZDIC	ZDC	ZDCV	ZDR	ZDI	ZDSH	ZDSL		ZDRT	ZDIT	ZDT	ZDY	ZDE					ZDV

Note: This table is not all-inclusive.

*A, alarm, the annunciating device, may be used in the same fashion as S, switch, the actuating device.

**The letters H and L may be omitted in the undefined case.

Other Possible Combinations:

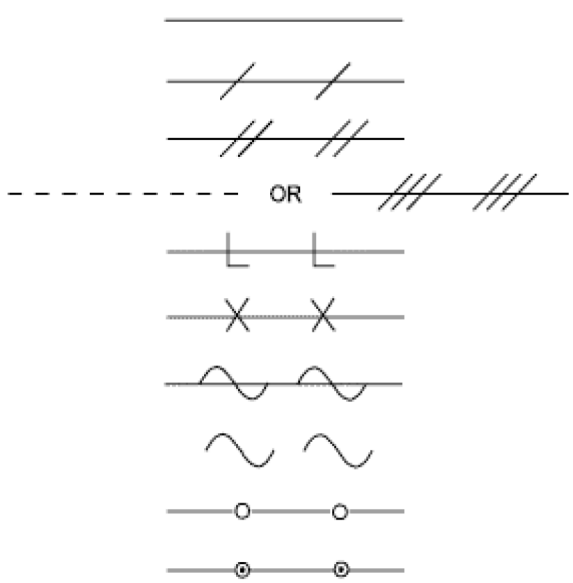
FO	(Restriction Orifice)	PFR	(Ratio)
FRK, HIK	(Control Stations)	KQI	(Running Time Indicator)
FX	(Accessories)	QQI	(Indicating Counter)
TJR	(Scanning Recorder)	WKIC	(Rate-of-Weight-Loss Controller)
LLH	(Pilot Light)	HMS	(Hand Momentary Switch)

General Instrument or Function Symbol

	PRIMARY LOCATION NORMALLY ACCESSIBLE TO OPERATOR	FIELD MOUNTED	AUXILIARY LOCATION NORMALLY ACCESSIBLE TO OPERATOR	BEHIND THE PANEL NORMALLY INACCESSIBLE TO OPERATOR
DISCRETE INSTRUMENT				
SHARED DISPLAY, SHARED CONTROL				
COMPUTER FUNCTION				
PROGRAMMABLE LOGIC CONTROL				
	INSTRUMENTS SHARING COMMON HOUSING	INSTRUMENT WITH LONG TAG NUMBER	INTERLOCK LOGIC	CONVERT SUCH AS CURRENT TO PRESSURE
MORE COMMON SYMBOLS				

Signal Lines

1. INSTRUMENT SUPPLY OR CONNECTED TO PROCESS
2. UNDEFINED SIGNAL
3. PNEUMATIC SIGNAL
4. ELECTRIC SIGNAL
5. HYDRAULIC SIGNAL
6. CAPILLARY SIGNAL
7. ELECTROMAGNETIC OR SONIC SIGNAL (GUIDED)
8. ELECTROMAGNETIC OR SONIC SIGNAL (NOT GUIDED)
9. INTERNAL SYSTEMS LINK (SOFTWARE OR DATA LINK)
10. MECHANICAL LINK



Worked Examples

Selection and Sizing of Pressure Relief Valves

Introduction

The function of a **pressure relief valve** is to protect pressure vessels, piping systems, and other equipment from pressures exceeding their **design pressure** by more than a fixed predetermined amount. The permissible amount of **overpressure** is covered by various codes and is a function of the type of equipment and the conditions causing the overpressure.

It is not the purpose of a pressure relief valve to control or regulate the pressure in the vessel or system that the valve protects, and it does not take the place of a control or regulating valve. There are modulating type proportional valves available for the purpose of regulating over pressure such as in the application of positive displacement pumps, but the back pressure will have to be known for proper sizing.

The aim of safety systems in processing plants is to prevent damage to equipment, avoid injury to personnel and to eliminate any risks of compromising the welfare of the community at large and the environment. Proper sizing, selection, manufacture, assembly, test, installation, and maintenance of a pressure relief valve are critical to obtaining maximum protection.

List of Code Sections Pertaining to Pressure Relief Valves

Section I	Power Boilers
Section III, Division 1	Nuclear Power Plant Components
Section IV	Heating Boilers
Section VI	Recommended Rules for the Care and Operation of Heating Boilers
Section VII	Recommended Rules for the Care of Power Boilers
Section VIII, Division 1	Pressure Vessels
Appendix 11	Capacity Conversions for Safety Valves
Appendix M	Installation and Operation
Section VIII, Division 2	Pressure Vessels - Alternative Rules
B31.3, Chapter II, Part 3	Power Piping - Safety and Relief Valves
B31.3, Chapter II, Part 6	Power Piping - Pressure Relief Piping

Some of the code excerpts pertaining to the CSE examination are listed in the book in the section on pressure relief.

Unless otherwise noted, all symbols used are defined as follows:

A = Valve effective orifice area, in²

C = Flow constant determined by the ratio of specific heats, see Table 2 (use *C* = 315 if *k* is unknown)

G = Specific gravity referred to water = 1.0 at 70°F

K = Coefficient of discharge obtainable from valve manufacture (*K* = 0.975 for many nozzle-type valves)

K_b = Correction factor due to back pressure. This is valve specific; refer to manufacturer's literature

K_n = Correction factor for saturated steam at set pressures > 1,500 psia, see Equation 6

K_p = Correction factor for relieving capacity vs. lift for relief valves in liquid service, see Equations 1 and 2

K_{sh} = Correction factor due to the degree of superheat in steam (K_{sh} = 1.0 for saturated steam)

K_u = Correction factor for viscosity, see Equations 8 and 9 (use K_u =1.0 for all but highly viscous liquids)

K_w = Correction factor due to back pressure for use with balanced bellows valves

M = Molecular weight, see Table 2 for values of some common gases

P_1 = Upstream pressure, psia (set pressure + overpressure + atmospheric pressure)

! ***P*** = Differential pressure (set pressure, psig **!** back pressure, psig)

Q = Flow, gpm

T = Inlet vapor temperature, °R

R_{ne} = Reynolds numbers

W = Flow, lb/hr

Z = Compressibility factor (use *Z* = 1 for ideal gas)

'' = Liquid dynamic (absolute) viscosity, centipoises

FORMULAS

Vapor or gases,

$$A = \frac{W\sqrt{TZ}}{K_b CKP\sqrt{M}}$$

Steam,

$$A = \frac{W_s}{51.5KPK_bK_{sh}}$$

Liquids,

$$A = \frac{Q_g \sqrt{G}}{27.2\sqrt{P_d} K_p K_u K_w}$$

Table A11. Standard Nozzle Orifice Data

NOZZLE ORIFICE AREAS	
Size Designation	Orifice Area, in^2
E	0.196
F	0.307
G	0.503
H	0.785
J	1.280
K	1.840
L	2.850
M	3.600
N	4.340
P	6.380
Q	11.050
R	16.000
T	26.000

Manufacturer's customized versions of the LIQUIDS Equation should be used when available. These typically modify the equation presented to reflect actual coefficients of discharge (K) based on required ASME capacity certification testing. In some cases, the variable K may be absent. The gas and vapor formula presented is based on perfect gas laws. Many real gases and vapors, however, deviate from a perfect gas. The compressibility factor Z is used to compensate for the deviations of real gases from the ideal gas. In the event the compressibility factor for a gas or vapor cannot be determined, a conservative value of $Z = 1$ is commonly used. Values of Z based on temperature and pressure considerations are available in the open literature.

The standard equations listed above may not fully take into consideration the effect of back pressure on the valve capacity. The capacity of pressure relief valves of conventional design will be markedly reduced if the back pressure is greater than 10% of the set pressure. For example, a back pressure of 15% of the set pressure may reduce the capacity as much as 40%. The capacities of bellows valves with balanced discs are not affected by back pressure until it reaches 40 to 50% of the set pressure. If back pressure on valves in gas and vapor service exceeds the critical pressure (generally taken as 55% of accumulated inlet pressure, absolute), the flow correction factor K_b must be applied. If the back pressure is less than critical pressure, no correction factor is generally required.

Overpressure Considerations

Back pressure correction factors should not be confused with the correction factor K_p that accounts for the variation in relieving capacity of relief valves in liquid service that occurs with the change in the amount of overpressure or accumulation. Typical values of K_p range from 0.3 for an overpressure of 0%, 1.0 for 25%, and up to 1.1 for an overpressure of 50%. A regression analysis on a typical manufacturer's performance data produced the following correlation equations for K_p:

For % overpressure < 25,

1. K_p = -0.0014 (% overpressure)^2 + 0.073 (% overpressure) + 0.016

For 25% overpressure < 50,

2. K_p = 0.00335 (% overpressure)^2 + 0.918

Equation 4 is based on the empirical Napier formula for steam flow. Correction factors are included to account for the effects of superheat, back pressure and subcritical flow. An additional correction factor K_n is required by ASME when relieving pressure (P_1) is above 1,500 psia:

EXAMPLE 1 - Steam Application

Given:

Fluid:	Saturated steam
Required Capacity:	40,000 lb/hr Set
Pressure:	140 psig Overpressure: 10% (or 14 psig)
Back Pressure:	Atmospheric
Inlet relieving Temperature:	Saturation temperature
Molecular Weight:	18

Find: XYZ Valve Company's standard orifice for this application

Solution: Refer to Figure 3 and find that a "P" orifice is required, which will have a capacity of 53,290 lb/hr. Find the Required pressure of 140 in the left column, follow the line to the right. Column N = 36,610 lb/hr - this is too small. Column P = 53,290 lb/hr. - this is the size to select.

The problem may also be solved by using the formula and finding the required area for the proper orifice form the above chart, **Table A11. Standard Nozzle Orifice Data. Use Manufacturer's data and formulas to receive exact results.**

See the orifice chart on the next page.

THE XYZ VALVE COMPANY

Approved: API-ASME and ASME Certified: National Board of Boiler Pressure Vessel Codes and Pressure Vessel Inspectors

HYPOTHETICAL TYPICAL CAPACITY TABLE

Capacity in Pounds per Hour of Saturated Steam at Set Pressure Plus 10% Overpressure

Set Press (psig)	ORIFICE DESIGNA													
	D	E	F	G	H	J	K	L	M	N	P	Q	R	T
10	141	252	395	646	1009	165	10	3666	4626	5577	8198	14200	20550	33410
20	202	360	563	923	1440	2362	3373	5235	6606	7964	11710	20280	29350	47710
30	262	467	732	1200	1872	3069	4384	6804	8586	10350	15220	26350	38200	62010
40	323	575	901	1476	2304	3777	5395	8374	10570	12740	18730	32430	47000	76310
50	383	683	1070	1753	2736	4485	6405	9943	12550	15120	22230	38510	55800	90610
60	444	791	1939	9030	3167	5193	7416	11510	14530	17510	25740	44590	64550	104900
70	504	899	1408	2306	3599	5901	8427	13080	16510	19900	29250	50660	73400	119200
80	565	1005	1576	2583	4031	6609	9438	14650	18490	22290	32760	56740	82100	133500
90	625	1115	1745	2860	4463	7317	10450	16220	20470	24670	36270	69890	90900	147800
100	686	1220	1914	3136	4894	8024	11460	17790	22450	27060	39780	68900	99700	162110
120	807	1440	2252	2690	5758	9440	13480	20930	26410	318300	46800	81050	117000	190710
140	998	1655	2590	4943	6621	10860	15550	24070	30370	36610	53290	93210	135000	
160	1050	1870	2927	4796	7485	12270	17530	27200	34330	41380	60830	105400	152500	
180	1170	2085	3265	5349	8348	136900	19550	30340	38290	46160	67850	117500	170000	
200	1290	2300	36030	5903	9212	15100	21570	33480	42250	50930	74870	129700	188000	
220	1410	2515	3940	6456	10080	16520	23590	36620	46210	55700	81890	141800	205500	
240	1535	2730	4278	7009	10940	17930	25610	39760	50170	60480	88910	154000	223000	
260	1655	2945	4616	7563	11800	19350	27630	49890	54130	65250	95920	166100	240500	
280	1775	3160	4953	8116	12670	20770	29660	46030	58090	70030	102900	178300	258000	
300	1895	3380	5291	8669	13530	22180	31680	49170	62050	74800	110000	190400	276000	
320	2015	3595	5629	9223	14390	23600	33700	52310	66010	79570	117000	202600		
340	2140	3810	5967	9776	15260	25010	35720	55450	69970	84350	124000	214800		
360	2260	4025	6304	10330	16120	26430	37740	58590	73930	89120	131000	226900		
380	2380	4240	6642	10880	16980	27840	39770	61720	77890	93900	138000	239100		
400	2500	4455	6980	1440	17850	29260	41790	64860	81850	98670	145100	251200		
420	2620	4670	7317	11990	18710	30680	43810	68000	85810	103400	152100	263400		
440	2745	4885	7655	12400	19570	32090	45830	71140	89770	108200	159100	275500		
460	2865	5105	7993	13100	20440	33510	47850	74280	93730	113000	166100	287700		
480	2985	5320	8330	13650	21300	34920	49870	77420	97690	117800	173100	299800		

EXAMPLE 2 - Manual calculation verification of Example 1

Given: Same conditions and fluid properties as Example 1

Find: The correct size standard orifice to meet the given requirements.

Solution:

1. Because the steam is saturated and the set pressure < 1,500 psia, K_{sh} = 1.0 and K_n = 1.0

2. Calculate an orifice effective area using Equation 4:

$$A = \frac{W_s}{51.5KPK_bK_{sh}} = \frac{40{,}000}{51.5(0.975)(140+14+14.7)(1)(1)} = 4.72in^2$$

3. From Table 1, find the smallest standard orifice designation that has an area equal to or greater than *A*.

4. Select a "P" orifice with an actual area equal to 6.38 in^2.

EXAMPLE 3 - Gas/Vapor Application (see Table 2 on the next page)

Given:

Fluid:	Saturated ammonia vapor
Required Capacity:	15,000 lb/hr
Set Pressure:	325 psig (constant back pressure of 15 psig deducted)
Overpressure:	10%
Back Pressure:	15 psig (constant)
Inlet relieving Temperature:	NH_3 saturation temperature @ P_1 (138°F) Molecular Weight:17

Find: The correct size standard orifice to meet the given requirements.

Solution:

1. Determine from Table 2 below that NH_3 has a nozzle constant of $C = 347$.

2. Because the back pressure is < 40% of set pressure, assume $K_b = 1.0$

3. Assume that NH_3 is an ideal gas, # $Z = 1.0$

4. Calculate an orifice effective area using Equation 3:

$$A = \frac{W\sqrt{TZ}}{K_b CKP\sqrt{M}} = \frac{15{,}000\sqrt{(138+460)(1)}}{(0.975)(347)(1)(325+32.5+14.7)\sqrt{17}} = 0.707 in^2$$

5. From Table 1, find the smallest standard orifice designation that is equal to or greater than A.

6. Select an "H" orifice with an actual area equal to 0.785 in^2.

Table A11. Typical Properties of Gases

Gas or Vapor	Molecular Weight M	Ratio of Specific Heats k (14.7 psia)	Coefficient C*	Specific Gravity	Critical Pressure psia	Critical Temp.(°R) (°F+460)
Acetylene	26.04	1.25	342	0.899	890	555
Air	28.97	1.40	356	1.000	547	240
Ammonia	17.03	1.30	347	0.588	1638	730
Argon	39.94	1.66	377	1.379	706	272
Benzene	78.11	1.12	329	2.696	700	1011
N-Butane	58.12	1.18	335	2.006	551	766
Iso-Butane	58.12	1.19	336	2.006	529	735
Carbon Dioxide	44.01	1.29	346	1.519	1072	548
Carbon Disulphide	76.13	1.21	338	2.628	1147	994
Carbon Monoxide	28.01	1.40	356	0.967	507	240
Chlorine	70.90	1.35	352	2.447	1118	751
Cyclohexane	84.16	1.08	325	2.905	591	997
Ethane	30.07	1.19	336	1.038	708	550
Ethyl Alcohol	46.07	1.13	330	1.590	926	925
Ethyl Chloride	64.52	1.19	336	2.227	766	829
Ethylene	28.03	1.24	341	0.968	731	509
Freon 11	137.37	1.14	331	4.742	654	848
Freon 12	120.92	1.14	331	4.174	612	694
Freon 22	86.48	1.18	335	2.985	737	665
Freon 114	170.93	1.09	326	5.900	495	754
Helium	4.02	1.66	377	0.139	33	10
N-Heptane	100.20	1.05	321	3.459	397	973
Hexane	86.17	1.06	322	2.974	437	914
Hydrochloric Acid	36.47	1.41	357	1.259	1198	584
Hydrogen	2.02	1.41	357	0.070	188	60
Hydrogen Chloride	36.47	1.41	357	1.259	1205	585
Hydrogen Sulphide	34.08	1.32	349	1.176	1306	672
Methane	16.04	1.31	348	0.554	673	344
Methyl Alcohol	32.04	1.20	337	1.106	1154	924
Methyl Butane	72.15	1.08	325	2.491	490	829
Methyl Chloride	50.49	1.20	337	1.743	968	749
Natural Gas (Typical)	19.00	1.27	344	0.656	671	375
Nitric Oxide	30.00	1.40	356	1.036	956	323
Nitrogen	28.02	1.40	356	0.967	493	227
Nitrous Oxide	44.02	1.31	348	1.520	1054	557
N-Octane	114.22	1.05	321	3.943	362	1025
Oxygen	32.00	1.40	356	1.105	737	279
N-Pentane	72.15	1.08	325	2.491	490	846
Iso-Pentane	72.15	1.08	325	2.491	490	829
Propane	44.09	1.13	330	1.522	617	666
Sulfur Dioxide	64.04	1.27	344	2.211	1141	775
Toluene	92.13	1.09	326	3.180	611	1069

Viscous Fluid Considerations

The procedure to follow to correct for a viscous fluid, *i.e.,* a fluid whose viscosity is greater than 150 centipoise (cP) is to:

1. Determine a preliminary required pressure relief valve orifice size (effective area) discounting any effects for viscosity. This is done by using the standard liquid sizing formula and setting the viscosity correction factor K_u= 1.0. Select the standard orifice size letter designation that has an actual area equal to or greater than this effective area.
2. Use the actual area of the viscous trial size orifice to calculate a Reynolds number (R_{NE}) using the following formula:

   (equation 7) $$R_{NE} = \frac{2800GQ}{\mu A}$$

3. Use the Reynolds number calculated in Step 2 to calculate a viscosity correction factor K_u from the following equations:

   (equation 8)

   For $R_{NE} < 200$, $$K_u = 0.27 \ln R_{NE} - 0.65$$

   (equation 8)

   For $R_{NE} >= 200$ and $R_{NE} < 10{,}000$,

   $$Ku = -0.00777 \left(\ln R_{NE} \right)^2 + 0.165 \ln R_{NE} + 0.128$$

4. Determine a corrected required effective area of the pressure relief valve orifice using the standard liquid sizing formula and the value of K_u determined in Step 3.
5. Compare the corrected effective area determined in Step 4 with the chosen actual orifice area in Step 1. If the corrected effective area is less than the actual trial area assumed in Step 1, then the initial viscous trial size assumed in Step 1 is acceptable. Repeat this iterative process until an acceptable size is found.

EXAMPLE 4 - Viscous Liquid Application

Given:

Fluid:	No. 6 Fuel Oil Required Capacity: 1,200 gal/min
Set Pressure:	150 psig Overpressure:10%
Back Pressure:	Atmospheric Inlet relieving
Temperature:	60°F
Dynamic Viscosity:	850 cP
Specific Gravity:	0.993

Find: The correct size standard orifice to meet the given requirements.

Solution:

1. Since the overpressure is < 25%, determine the correction factor K_p from Equation 1:
   For % overpressure < 25,

   K_p = -0.0014 (% overpressure)^2 + 0.073 (% overpressure) + 0.016
   K_p = -0.0014 (10)^2 + 0.073 (10) + 0.016 = 0.606 = 0.61

2. Select an orifice trial size by setting K_u = 1.0 and using Equation 5. Since the back pressure = 0, then K_w = 1.0:

$$A = \frac{Q_g\sqrt{G}}{27.2\sqrt{P_d}K_pK_uK_w} = \frac{1,200\sqrt{0.993}}{(27.2)\sqrt{(150-0)}(0.61)(1)(1)} = 589in^2$$

3. From Table 1, it can be seen that an orifice size designation "P" with an actual area of 6.38 in^2 must be used.
4. Using the "P" orifice area, calculate the Reynolds number using Equation 7:

$$R_{NE} = \frac{2800GQ}{\mu A} = \frac{2800(0.993)(1200)}{(850)(6.38)} = 615$$

5. Since R_{NE} > 200, use Equation 9 and compute a viscosity correction factor K_u:

$$Ku = -0.00777(\ln R_{NE})^2 + 0.165\ln R_{NE} + 0.128$$

$$= -0.00777(6.422)^2 + 0.165(6.422) + 0.128 = 0.87$$

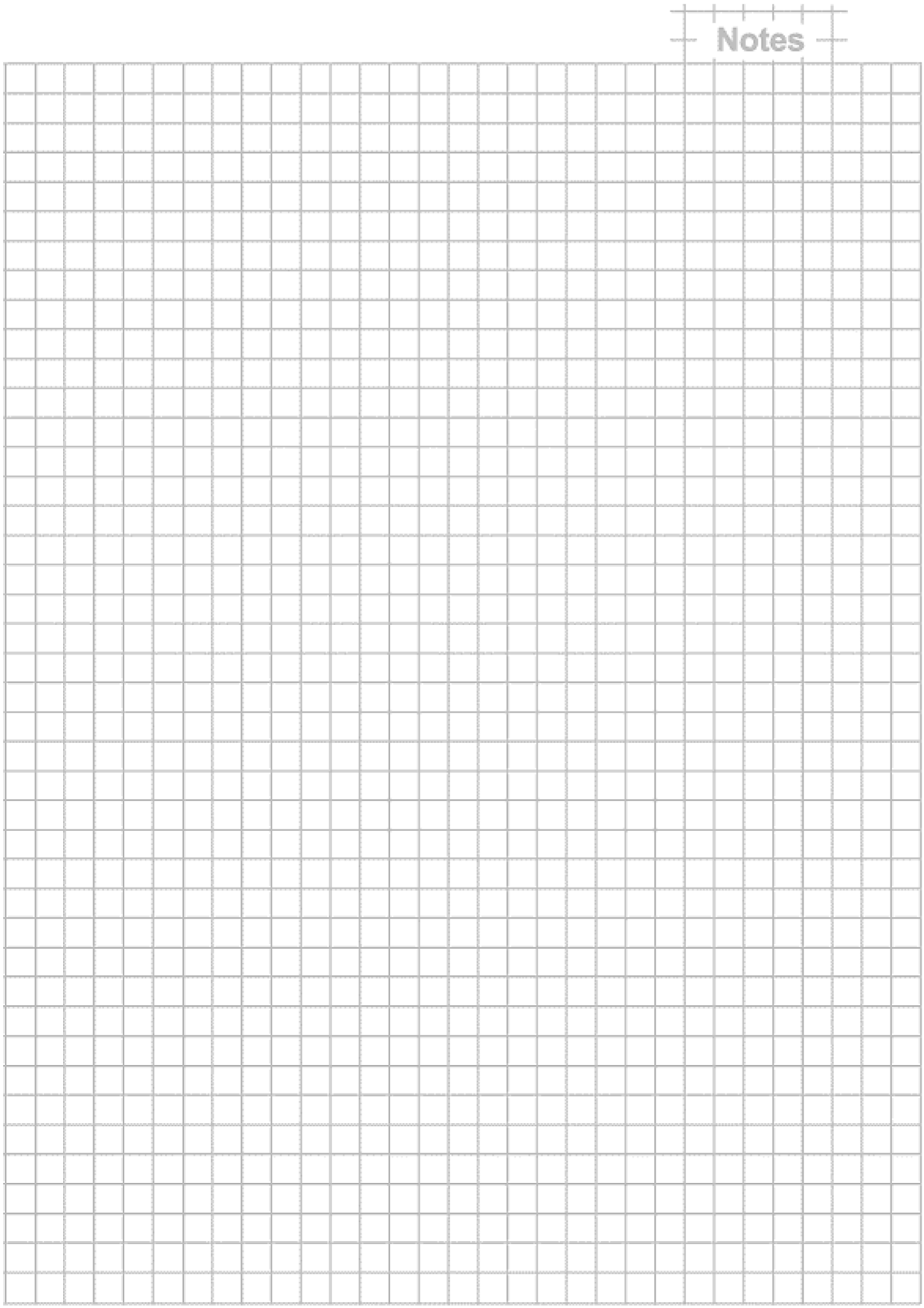
6. Compute a corrected orifice effective area based on the now known value of K_u:

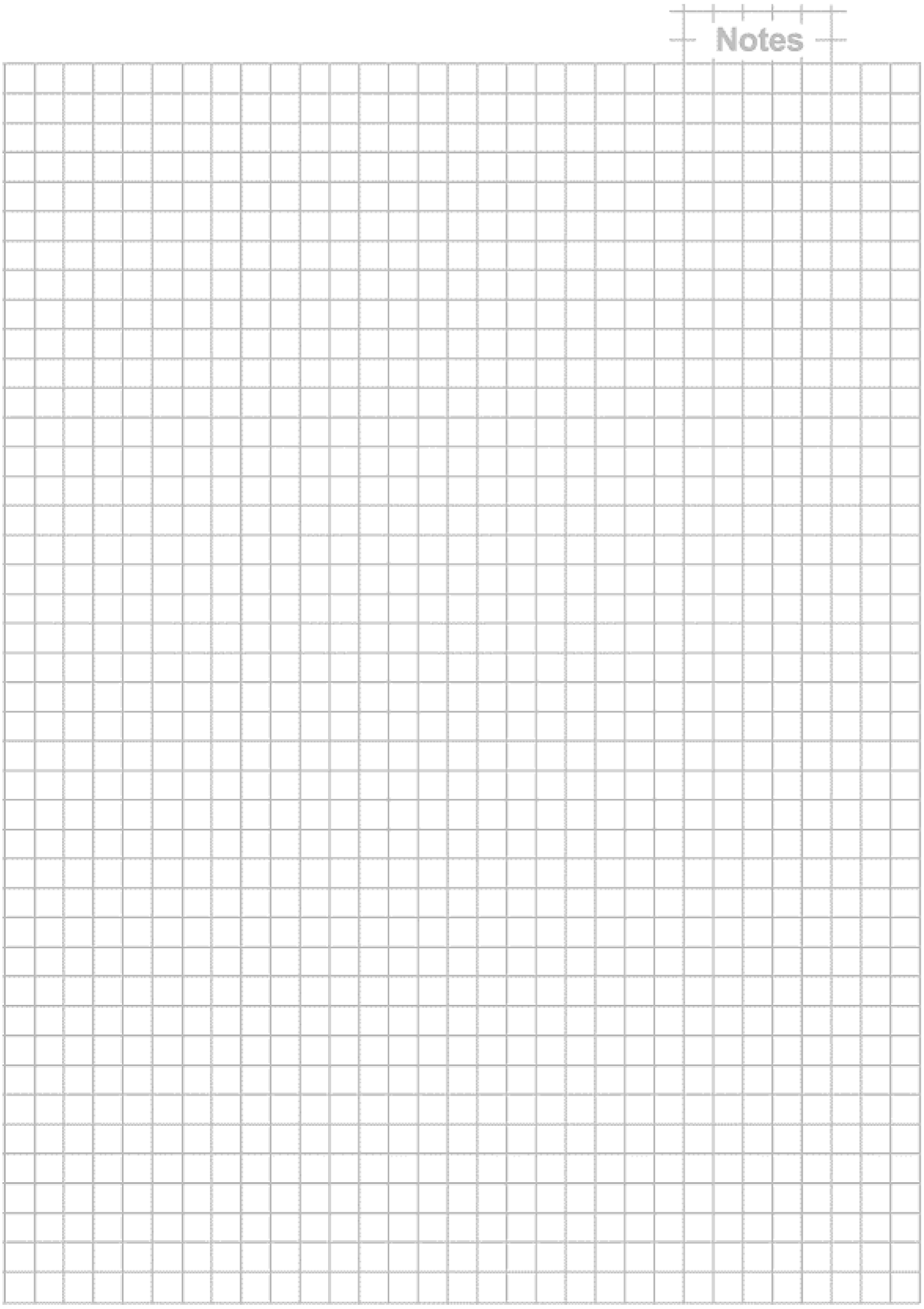
$$A = \frac{Q_g\sqrt{G}}{27.2\sqrt{P_d}K_pK_uK_w} = \frac{1,200\sqrt{0.993}}{(27.2)\sqrt{(150-0)}(0.61)(0.87)(1)} = 676in^2$$

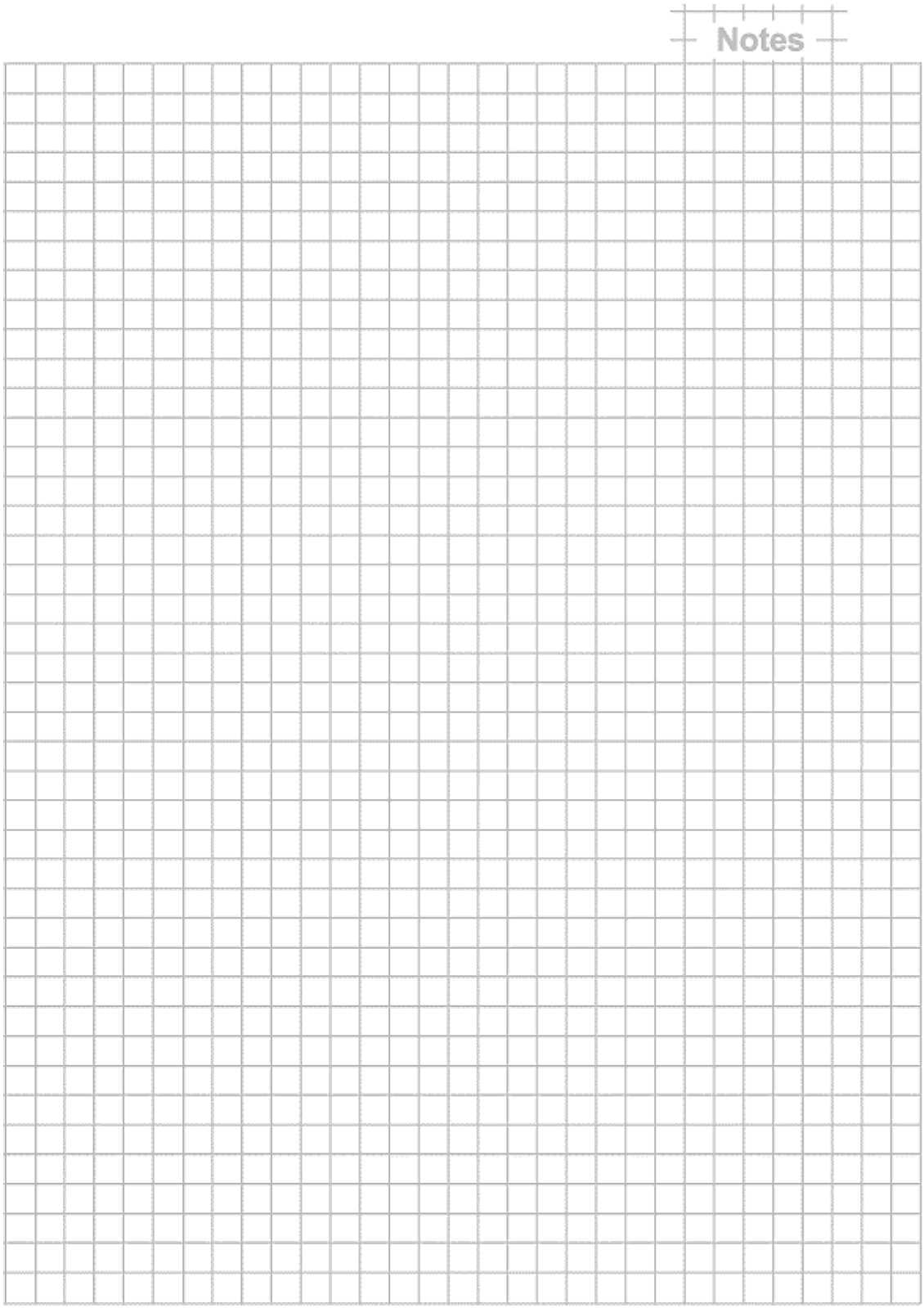
7. Since the corrected orifice effective area (6.76 in^2) is greater than the selected trial orifice area (6.38 in^2), the "P" orifice is unacceptable. Select the next larger size orifice (Q) with an area of 11.05 in^2 for this viscous application.

References

1. Fisher Controls, *Fisher Control Valve Handbook*, 3rd ed
2. Norman A. Anderson, *Instrumentation for Process Measurement and Control*, 3rd ed
3. Robert N. Bateson, *Introduction to Control System Technology*, 6th ed
4. ISA, ANSI/ISA–75.01.01–2002, Flow Equations for Sizing Control Valves
5. Richard Dorf, Robert Bishop, *Modern Control Systems*, 8th ed
6. Dale Seborg, Thomas Edgar, Mellichamp, *Process Dynamics and Control*, 1st ed
7. ASME Boiler and Pressure Vessel Code Section VIII, Div. 1, UG-125 through UG-136;
8. Miller, Richard W., *Flow Measurement Engineering Handbook*, 3rd ed., p. 13.4.
9. B. G. Lipták, *Instrument Engineer's Handbook - Process Control, 2nd Edition (Revised)*
10. ISA, *Control Systems Engineer Study Guide*, 4th ed
11. ANSI/ISA-51.1-1979(R1993) Process Instrumentation Terminology
12. ANSI/ISA-75.01.01-2002 Flow Equations for Sizing Control Valves
13. ANSI/ISA-5.1-1984(R1992) Instrumentation Symbols and Identification
14. NFPA 70 National Electrical Code
15. NFPA 77 Static Electricity
16. NFPA 78 Lightning Protection
17. NFPA 79 Industrial Machinery
18. NFPA 496 Purged and Pressurized Systems
19. http://www.EngineeringToolbox.com
20. http://virtual.cvut.cz/dynlabmodules/syscontrol/


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